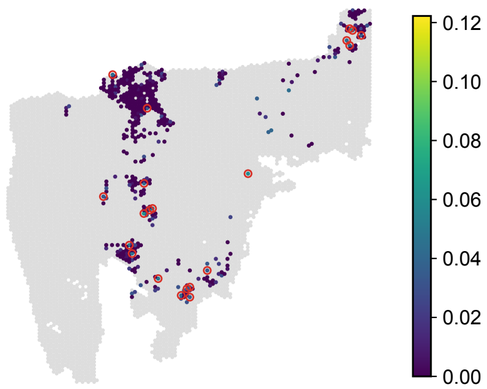
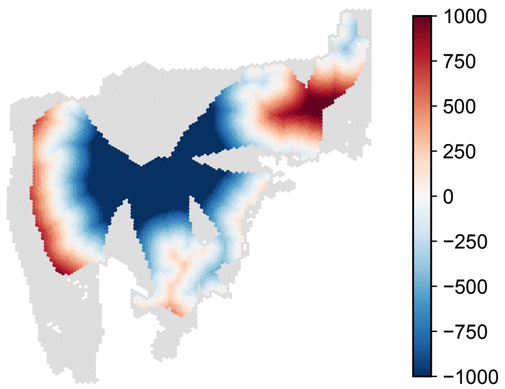
GSE238145_B10 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

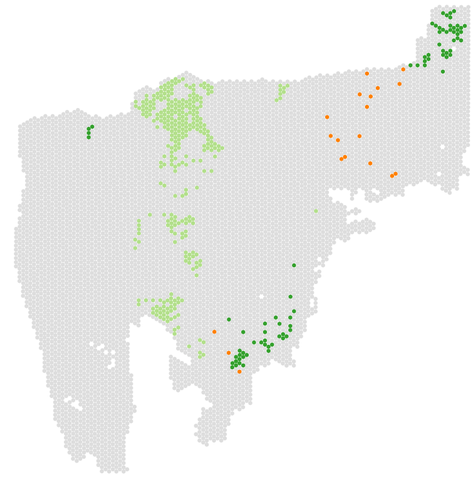
MP10 AUCell and frozen P95



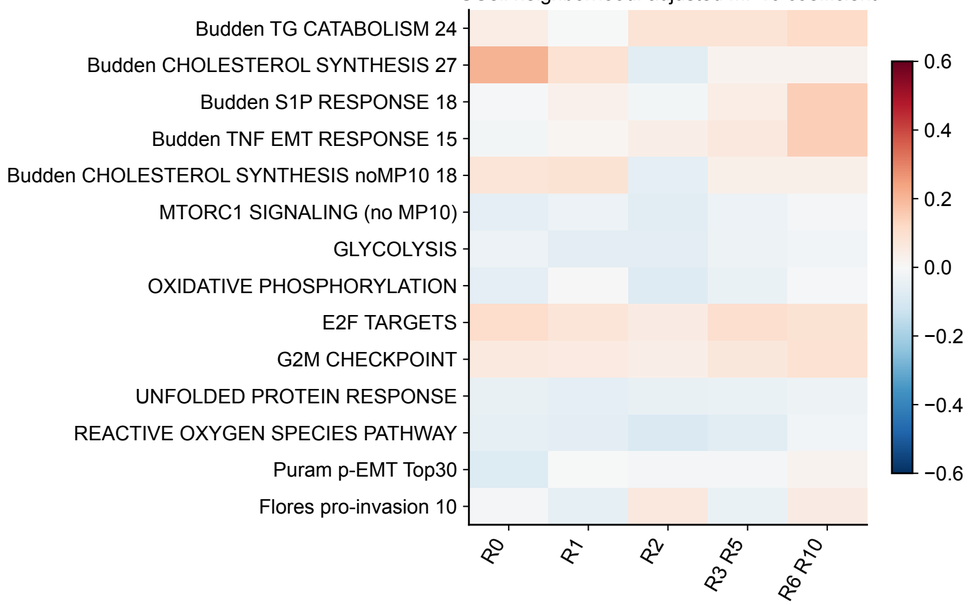
Strict v3.4 signed distance (µm)



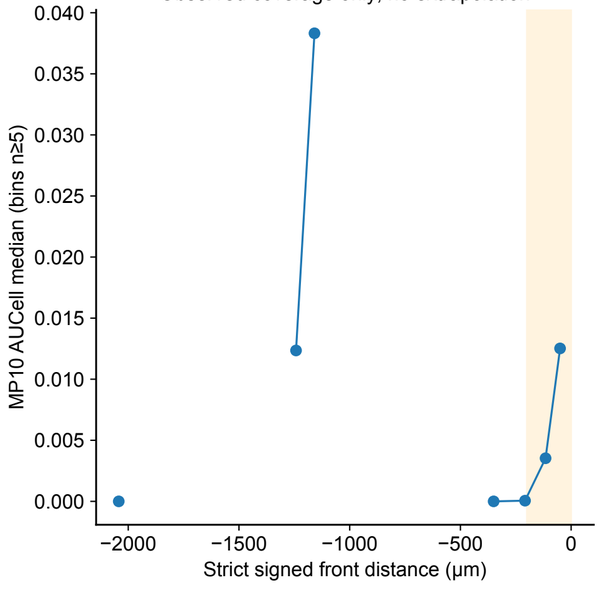
Formal R12 0–8; grey = unlabelled



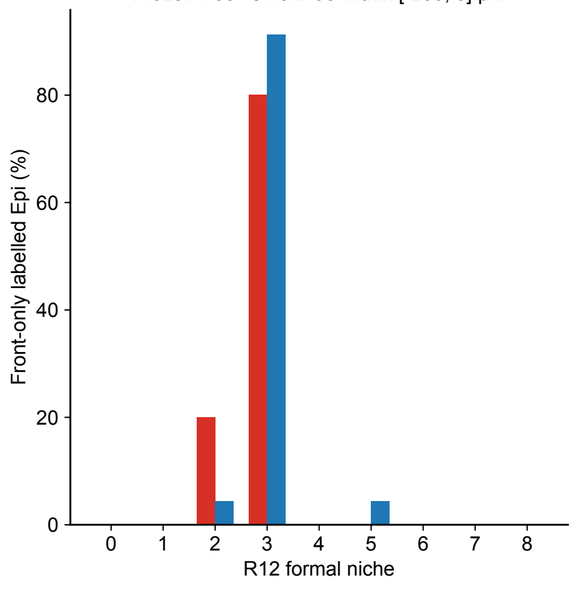
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

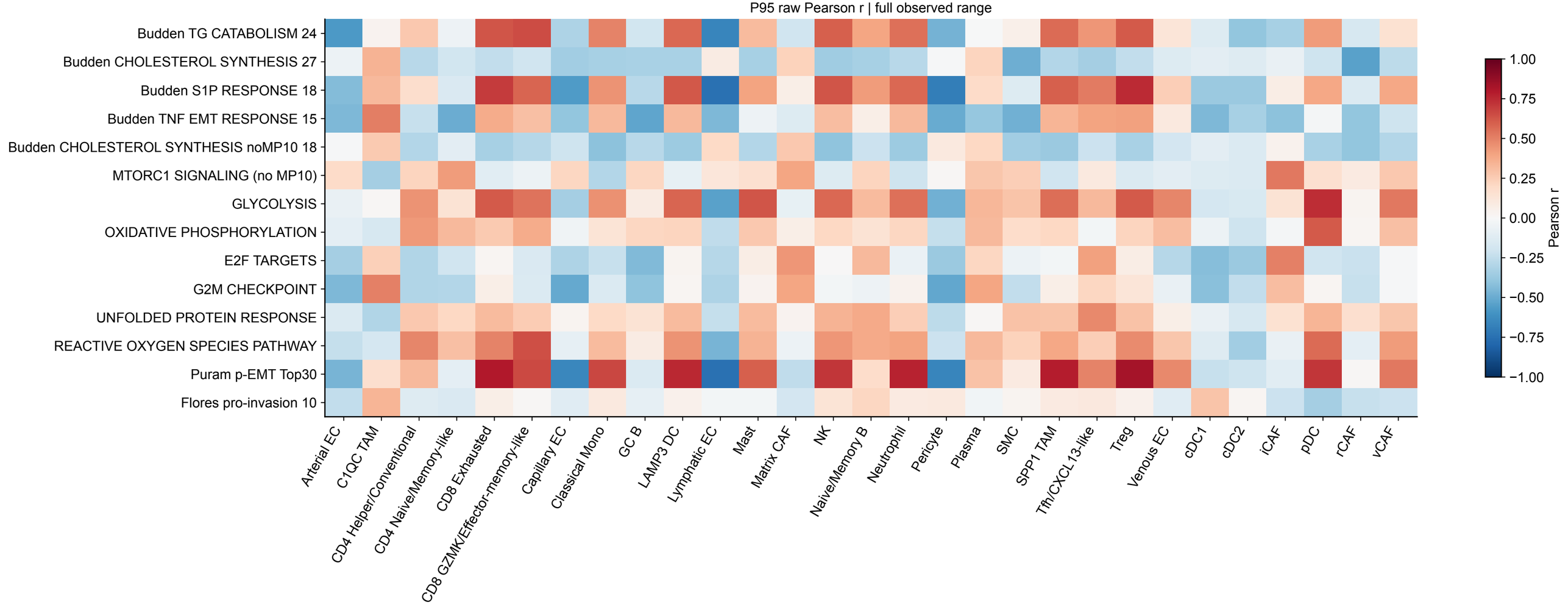


Tissue 7399 | Epi 437 | internal 415 | P95 21
Exact front in Epi 128 | R12 labelled 407

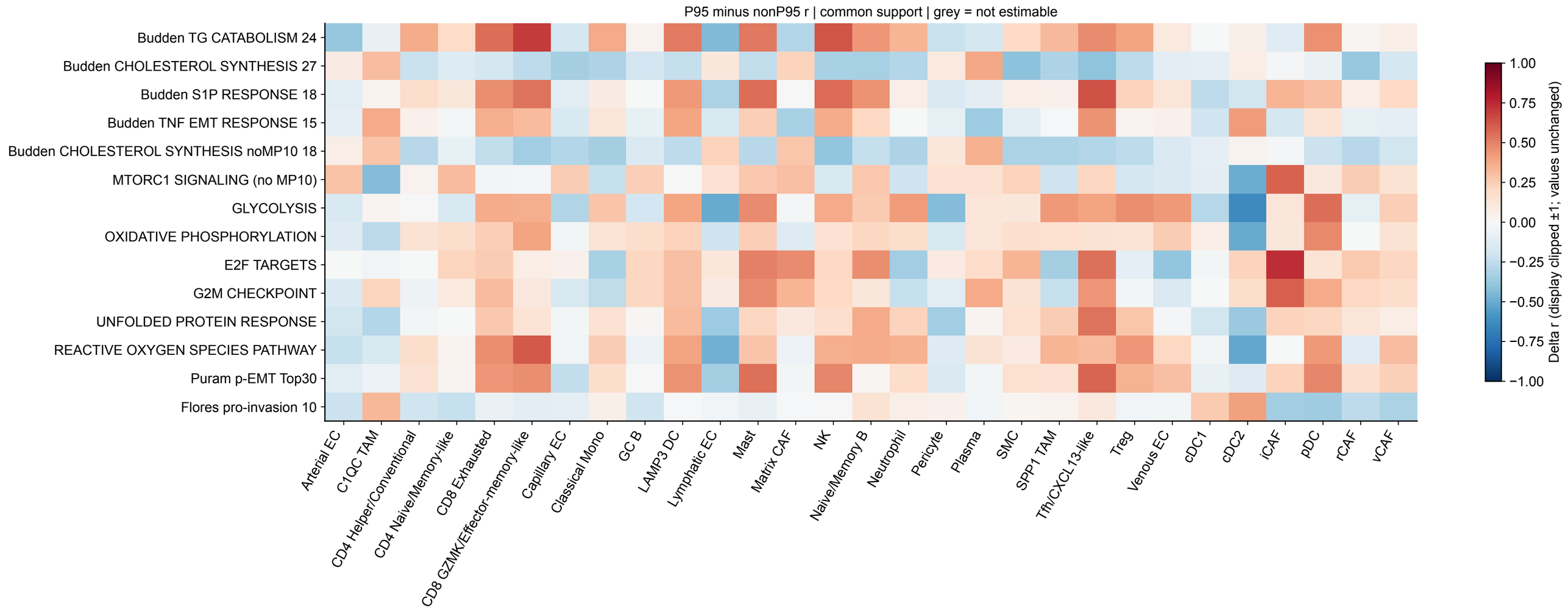
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B10 | all 29 candidates; fixed core programs

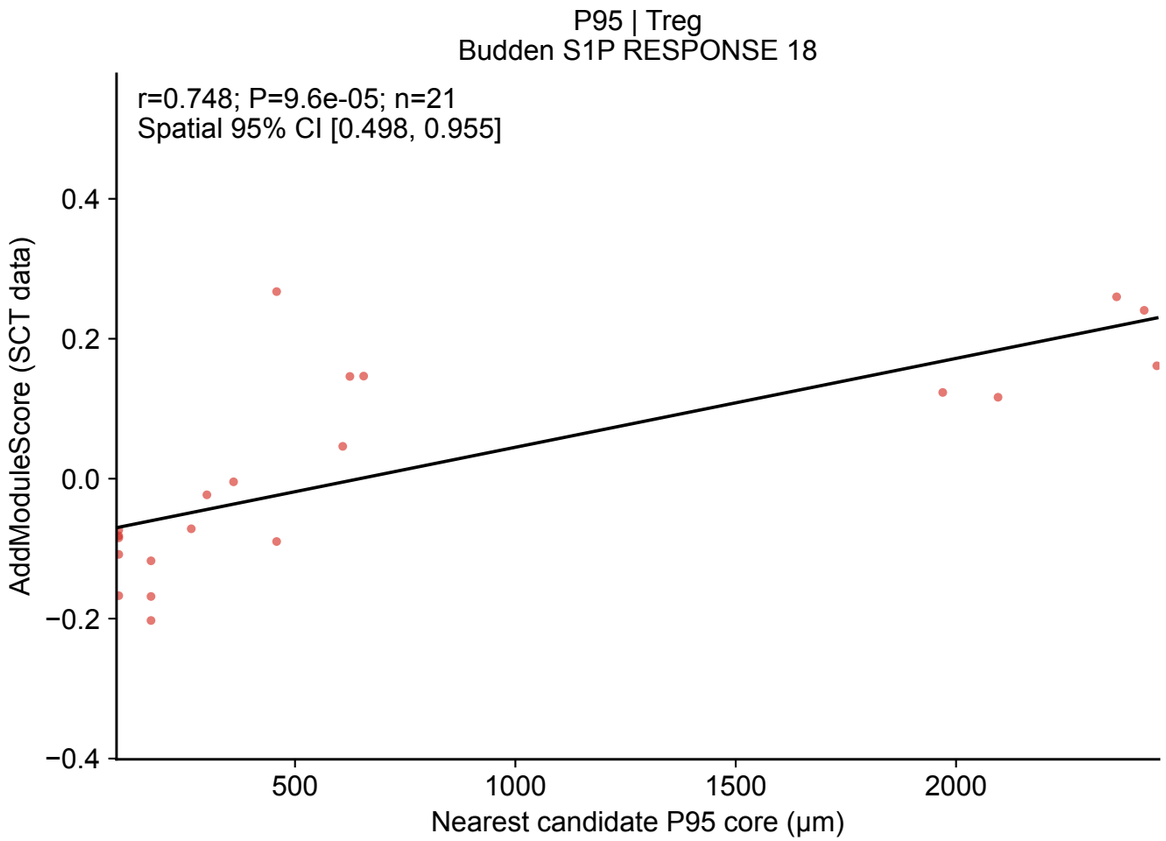
P95 raw Pearson r | full observed range

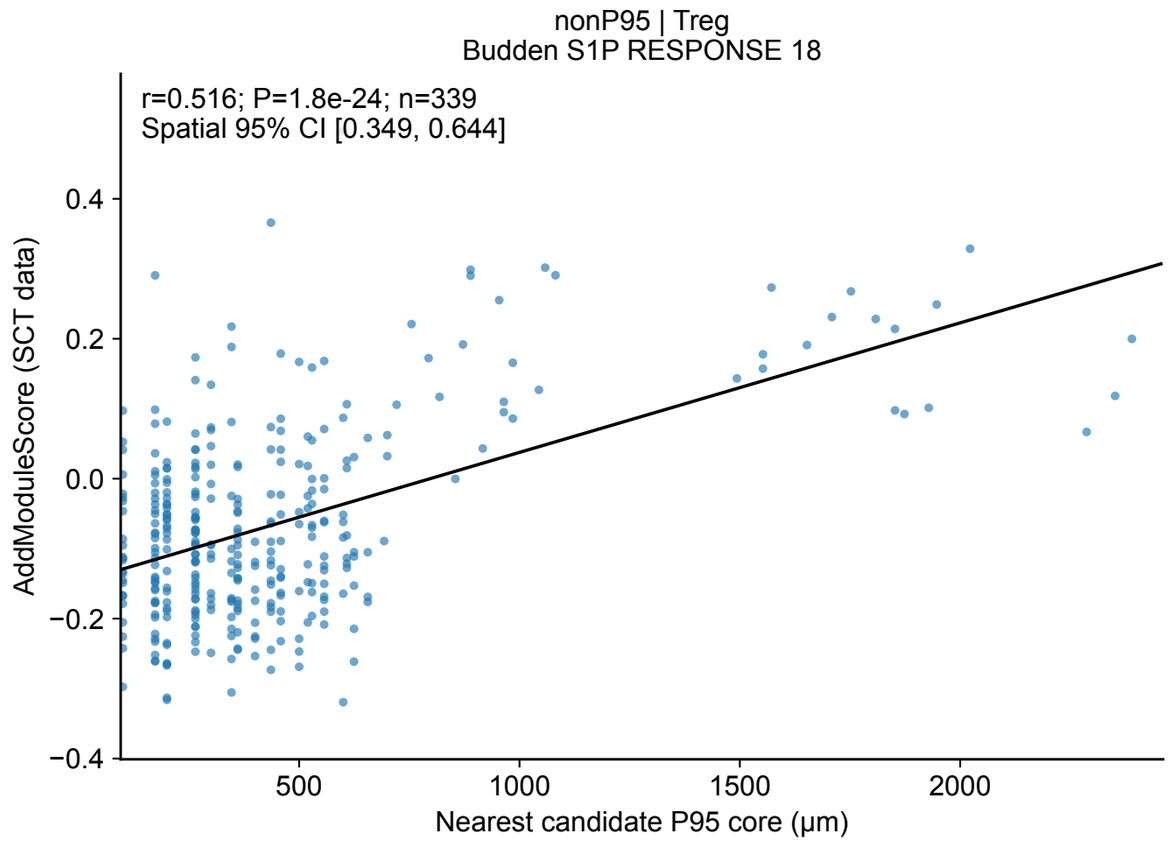


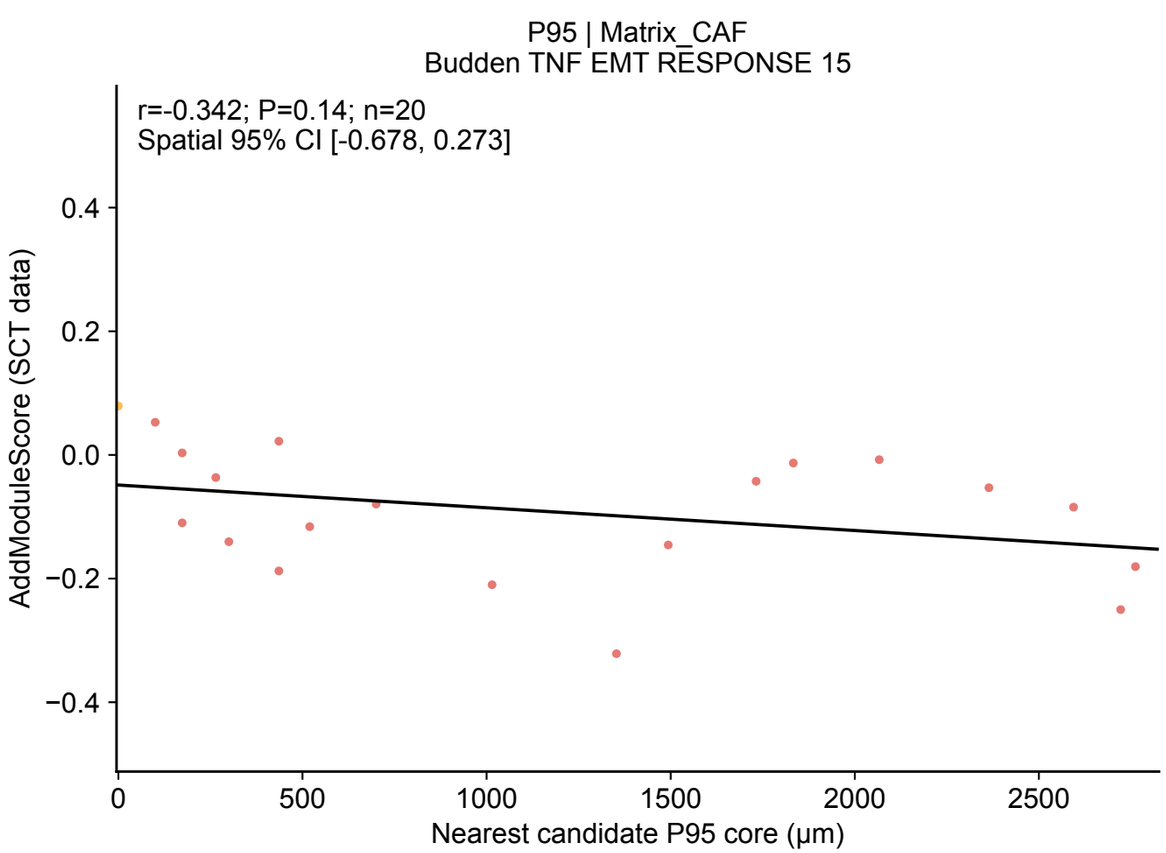
P95 minus nonP95 r | common support | grey = not estimable

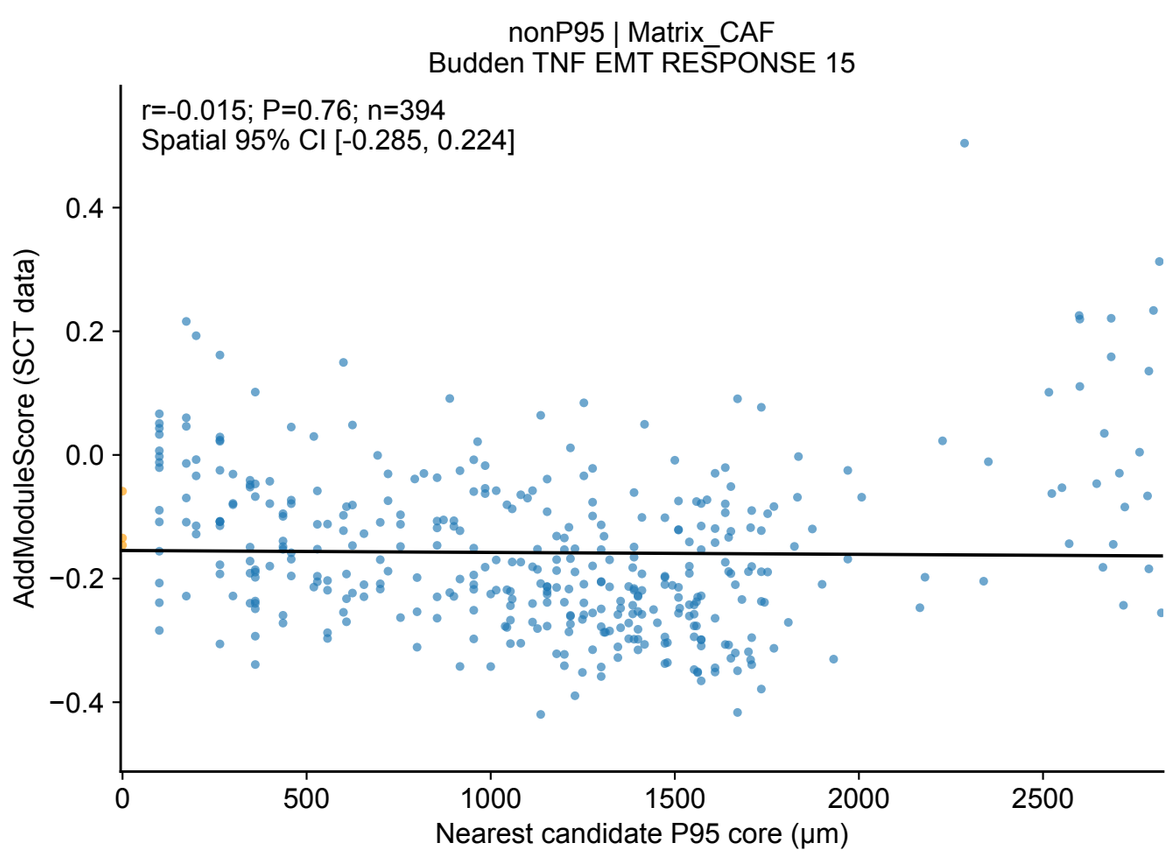


GSE238145_B10 | exploratory scatter display selected by coverage and |delta r|, not P







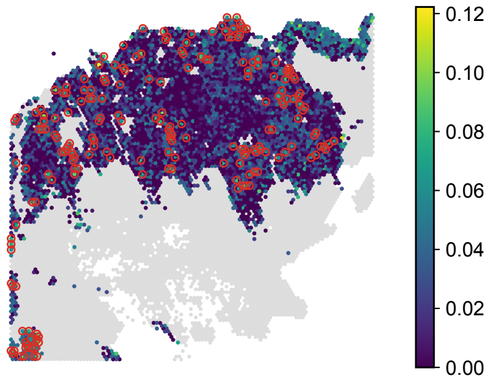


Treg / Budden S1P RESPONSE 18: delta r=0.232 [-0.026, 0.508]; common 100–2456 µm; spatial P=0.17
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=-0.326 [-0.731, 0.283]; common 0–2821 µm; spatial P=0.18
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

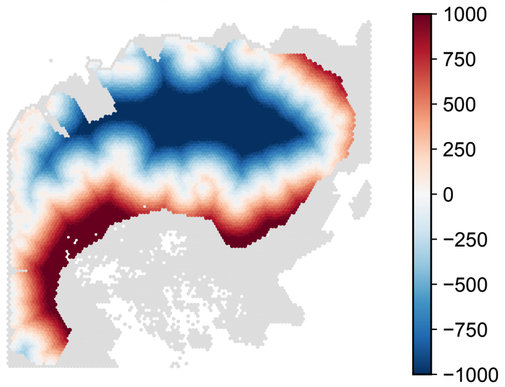
P95 nonP95 distance 0

GSE238145_B11 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

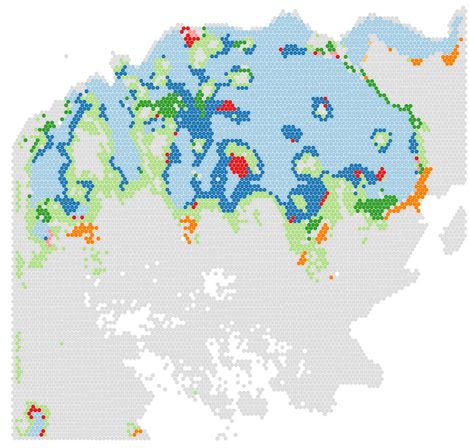
MP10 AUCell and frozen P95



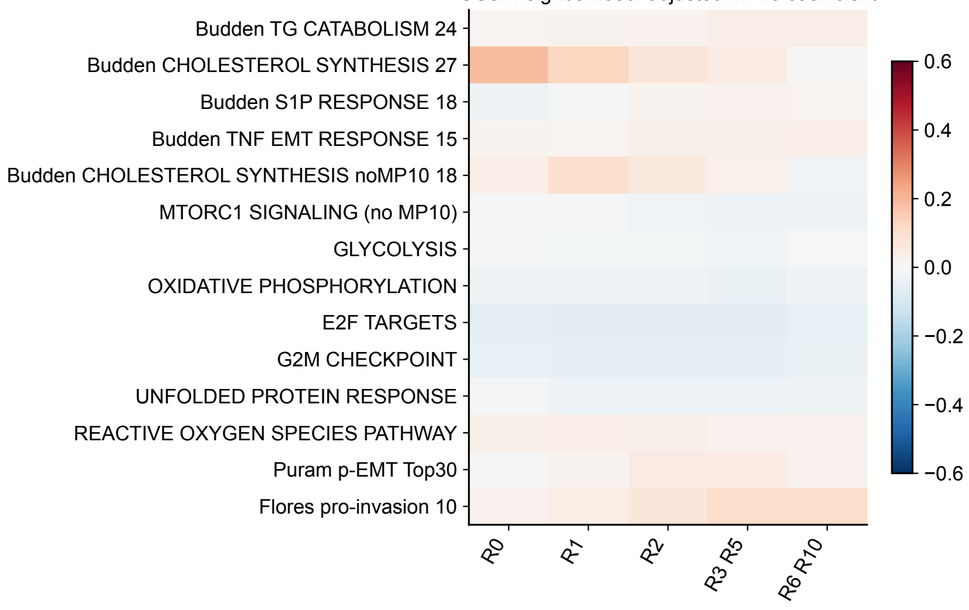
Strict v3.4 signed distance (µm)



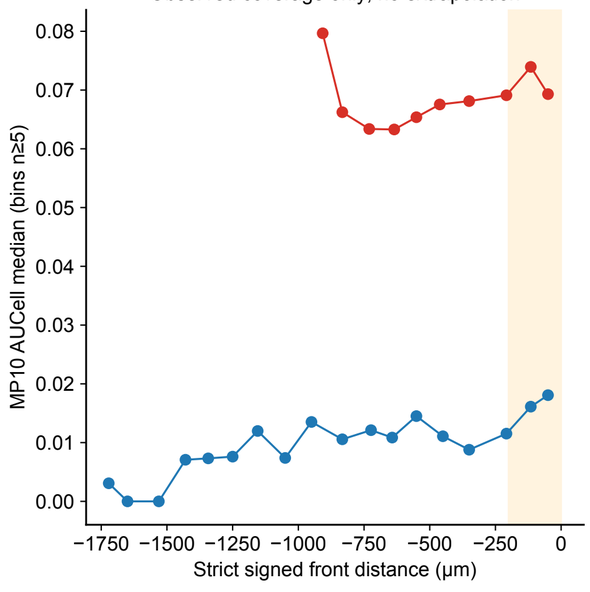
Formal R12 0–8; grey = unlabelled



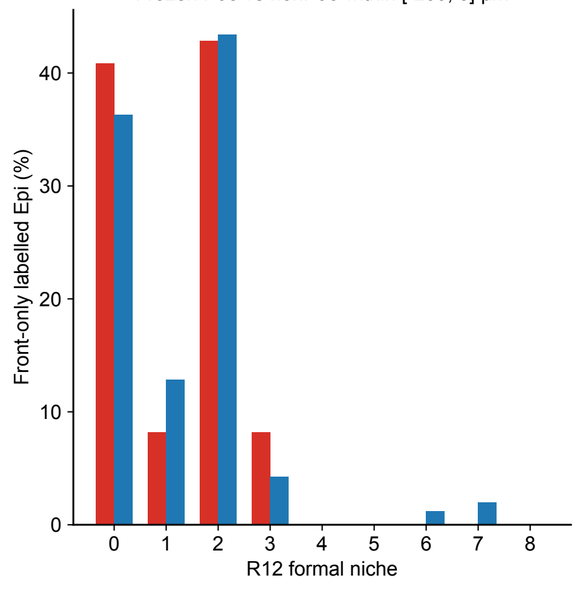
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

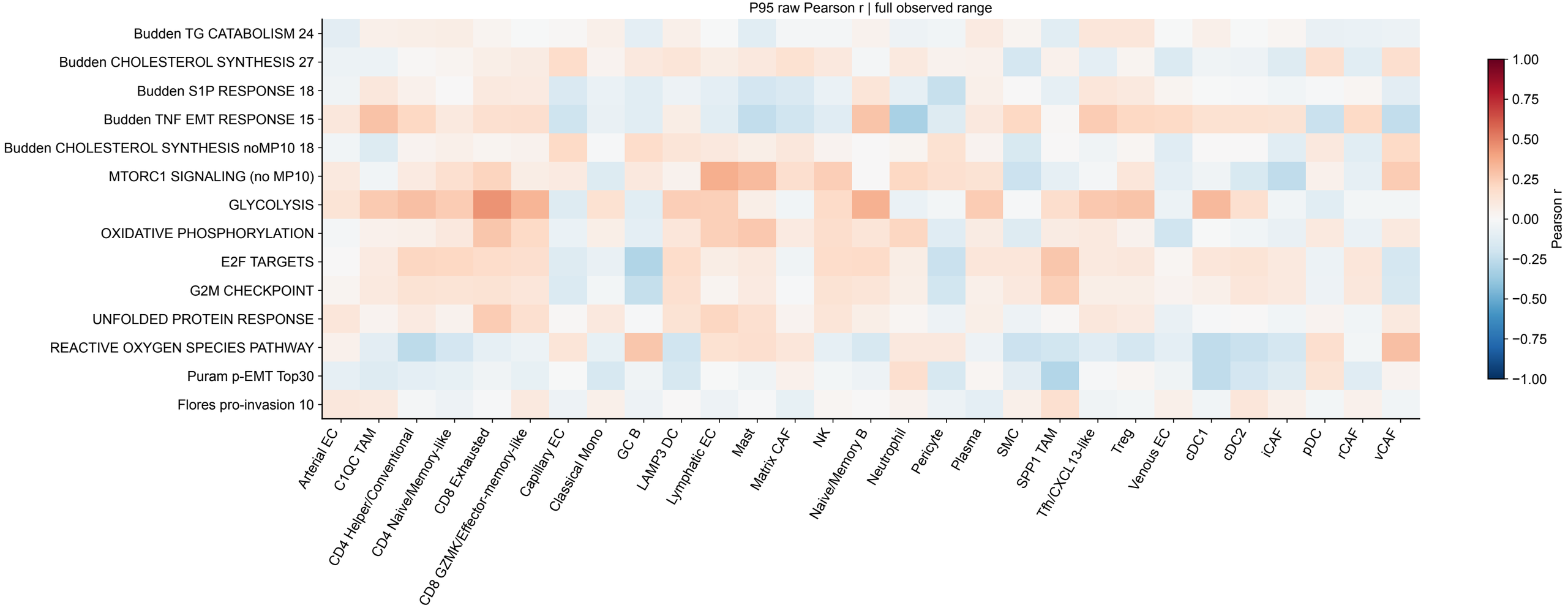


Tissue 10309 | Epi 4887 | internal 3923 | P95 197
Exact front in Epi 4090 | R12 labelled 4542

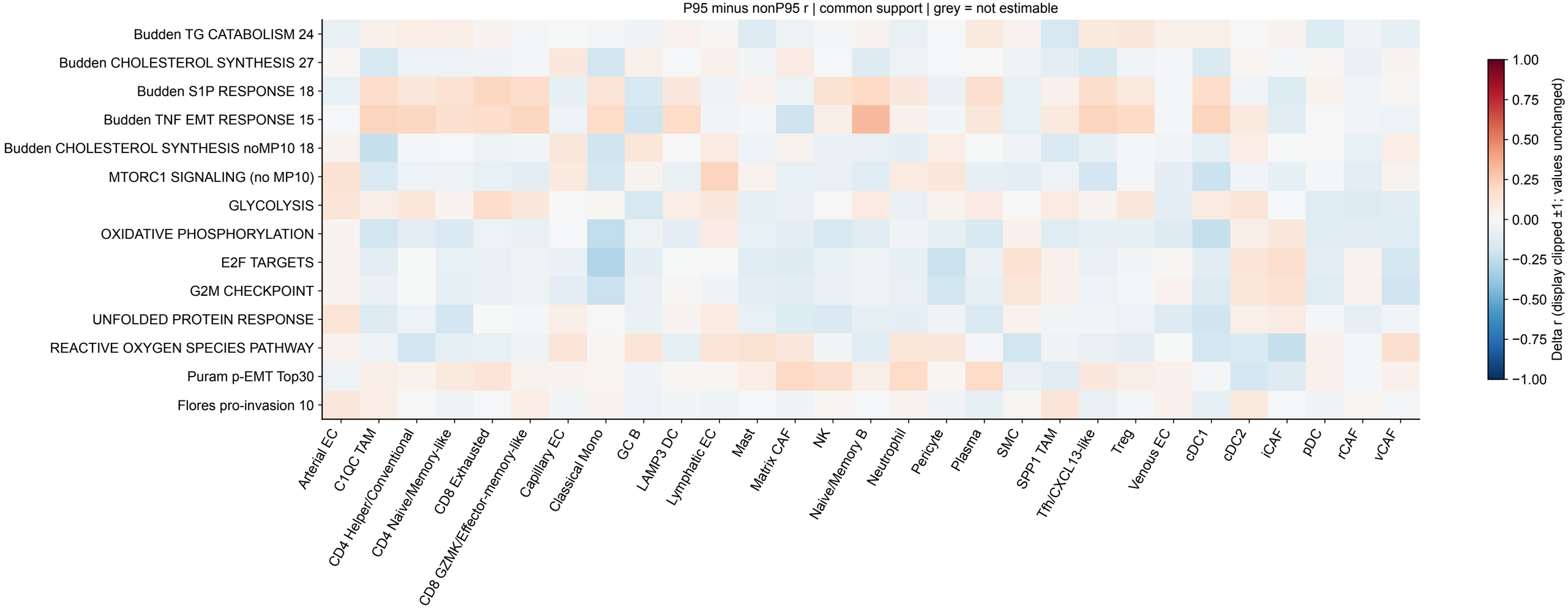
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B11 | all 29 candidates; fixed core programs

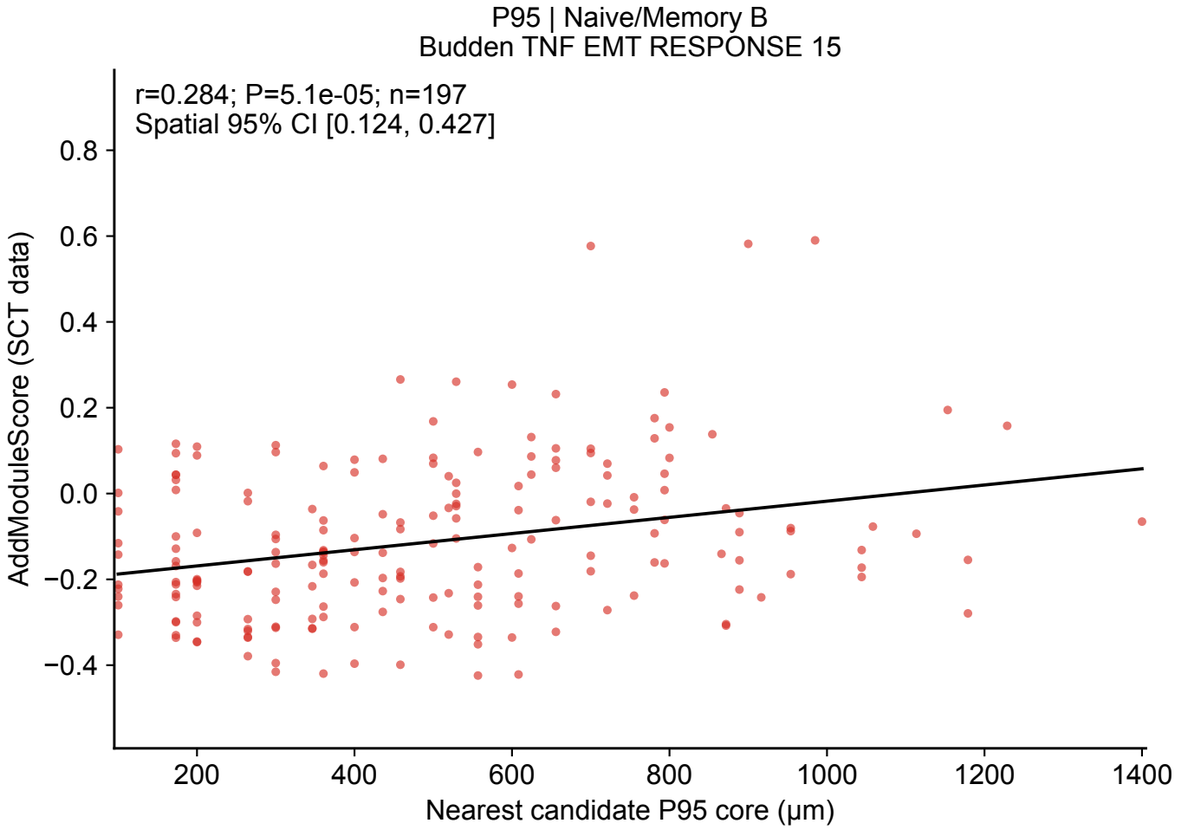
P95 raw Pearson r | full observed range

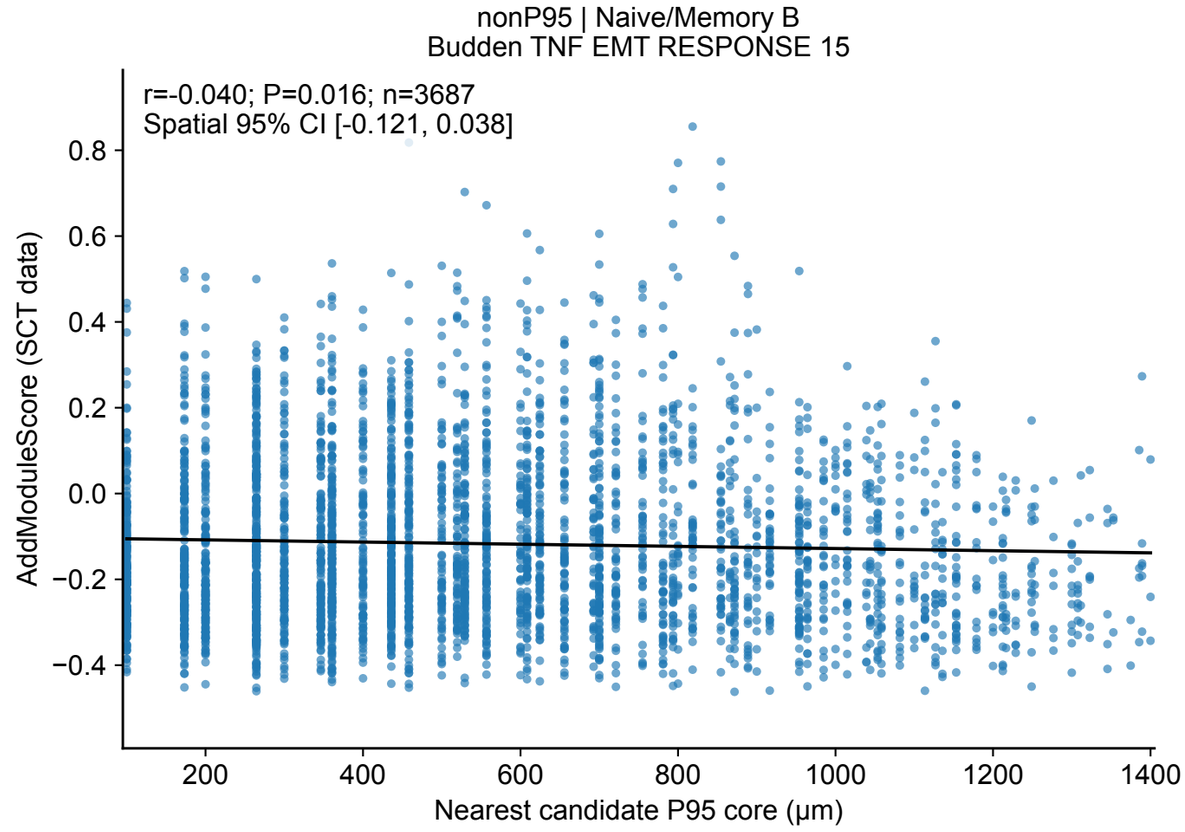


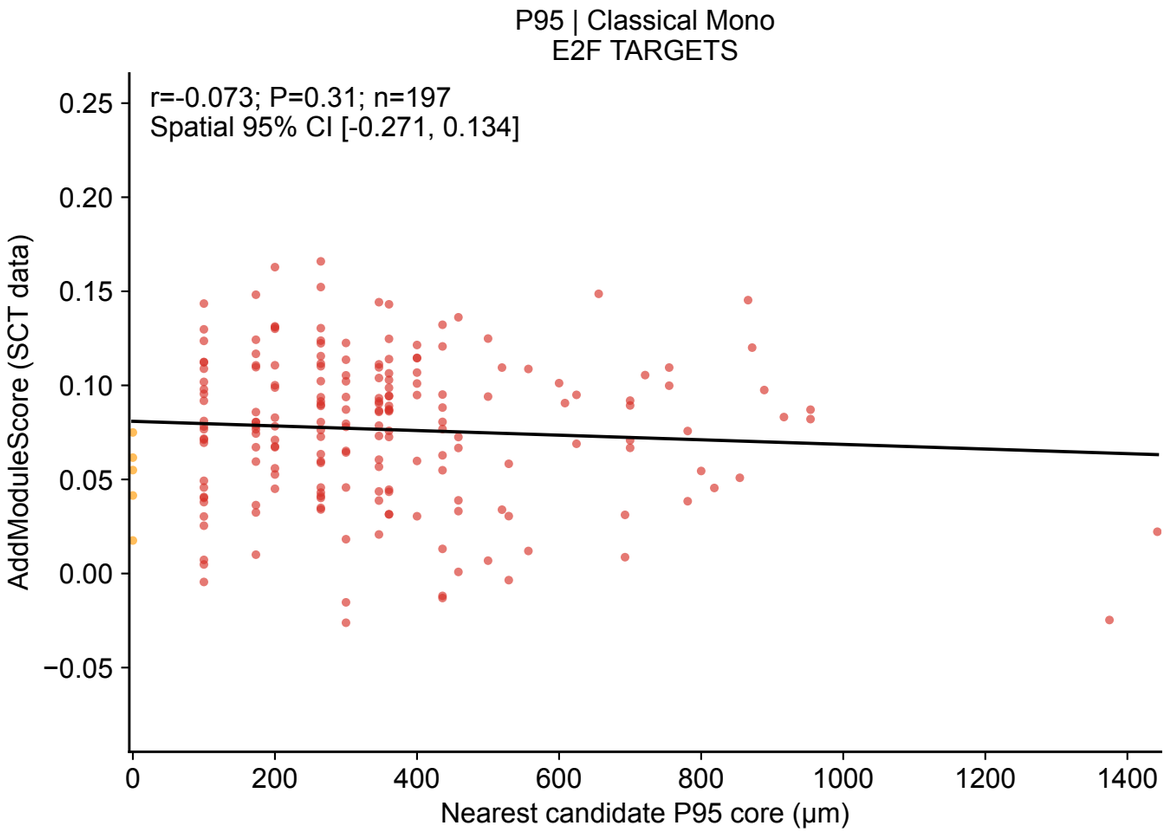
P95 minus nonP95 r | common support | grey = not estimable

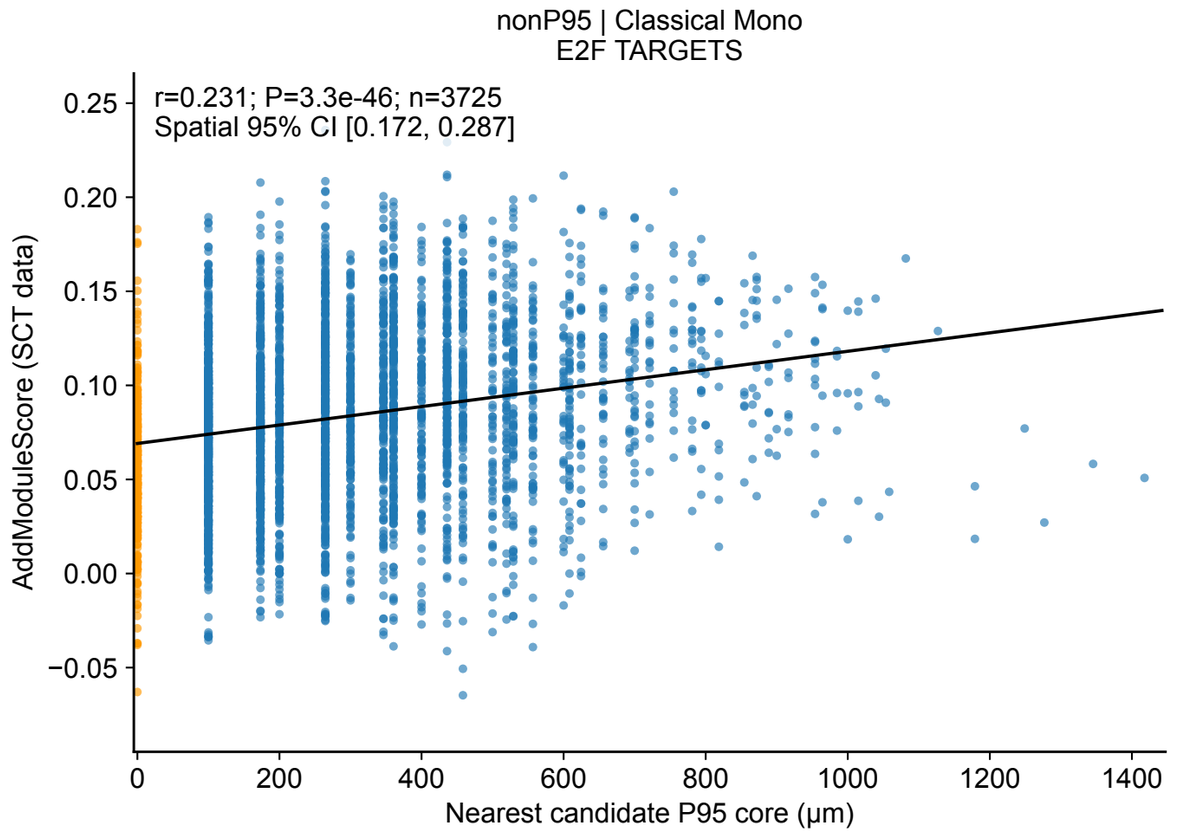


GSE238145_B11 | exploratory scatter display selected by coverage and |delta r|, not P







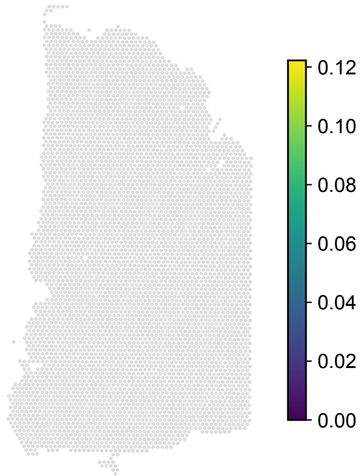


Naive/Memory B / Budden TNF EMT RESPONSE 15: delta r=0.324 [0.169, 0.463]; common 100–1400 μm; spatial P=4.8e-05
Classical Mono / E2F TARGETS: delta r=-0.304 [-0.490, -0.101]; common 0–1442 μm; spatial P=0.0028
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

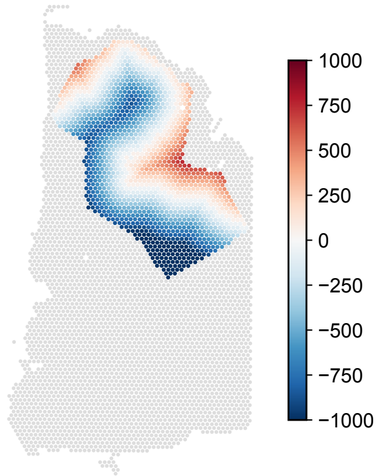
P95 nonP95 distance 0

GSE238145_B13 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

MP10 AUCell and frozen P95

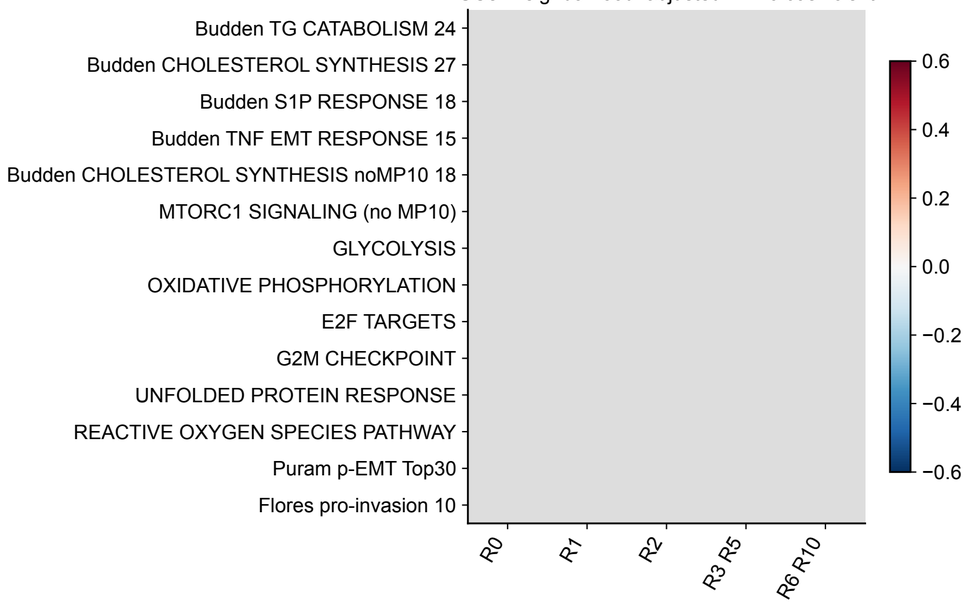


Strict v3.4 signed distance (µm)

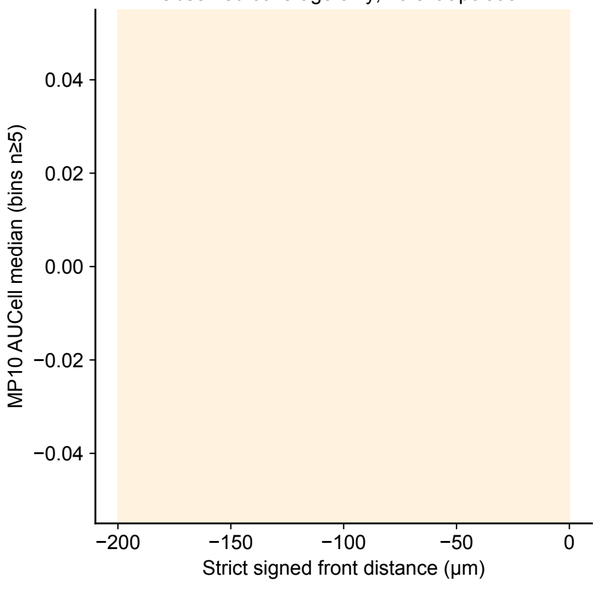


Formal R12 0–8; grey = unlabelled

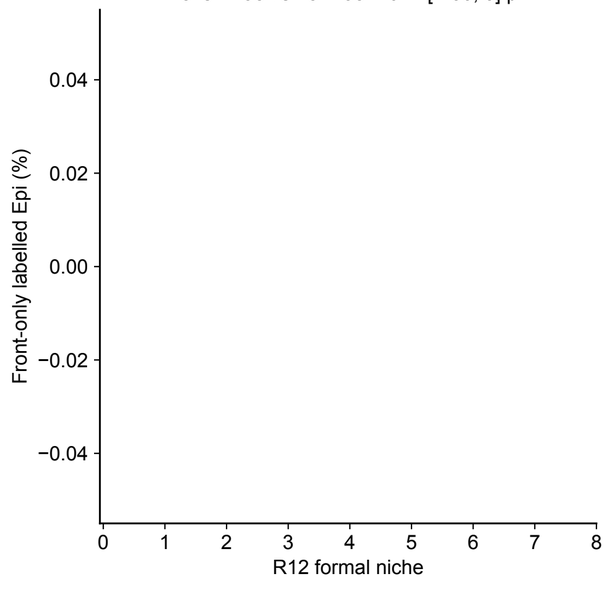
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

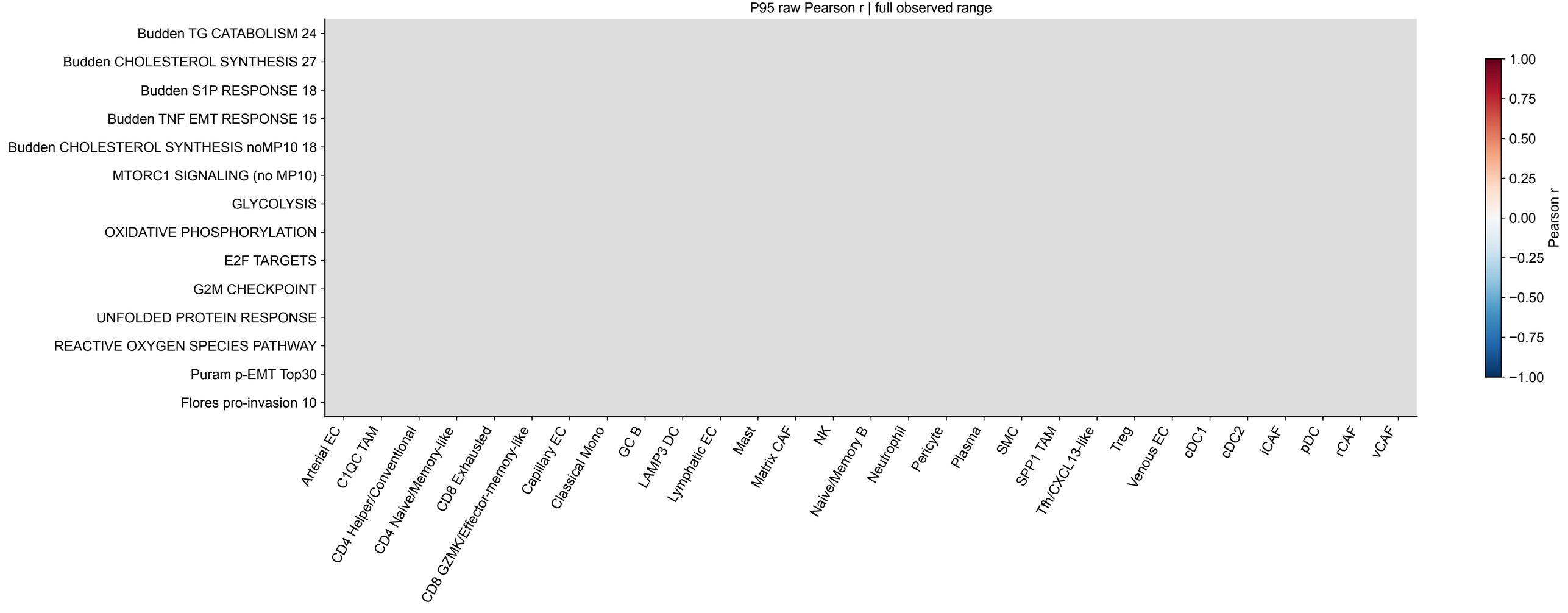


Tissue 4881 | Epi 0 | internal 0 | P95 0
Exact front in Epi 0 | R12 labelled 0

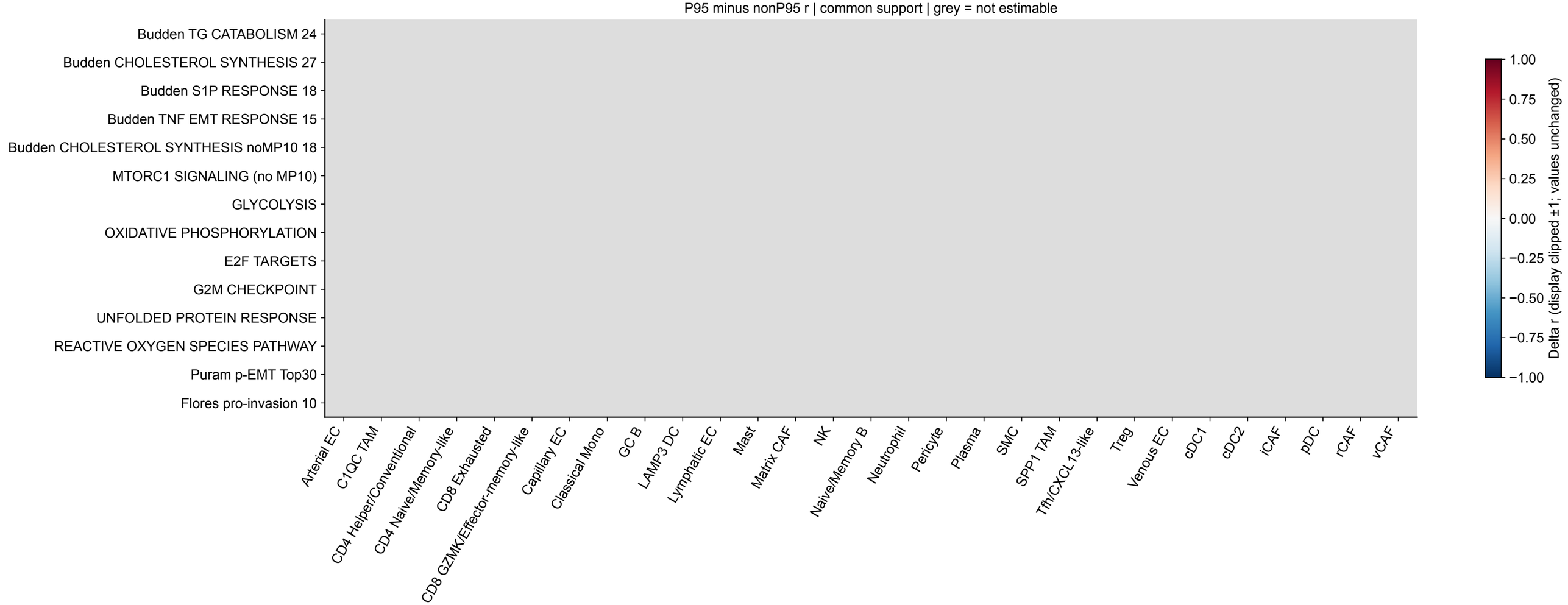


GSE238145_B13 | all 29 candidates; fixed core programs

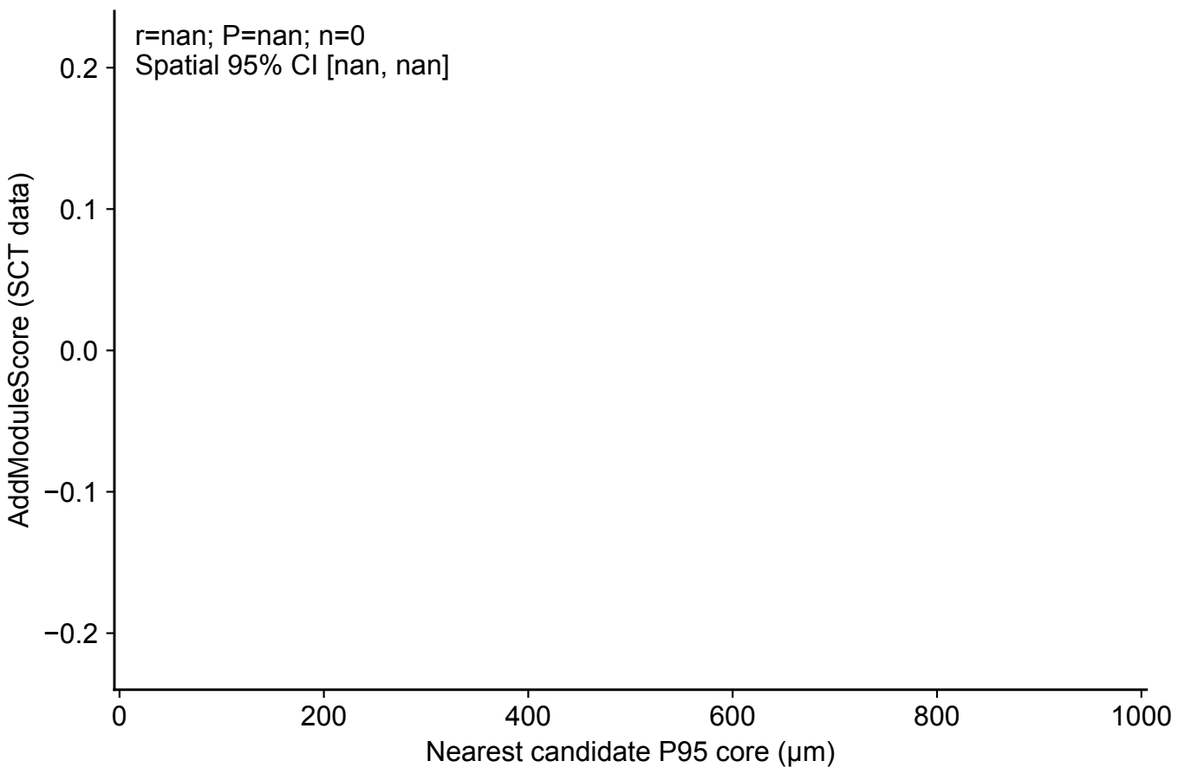
P95 raw Pearson r | full observed range

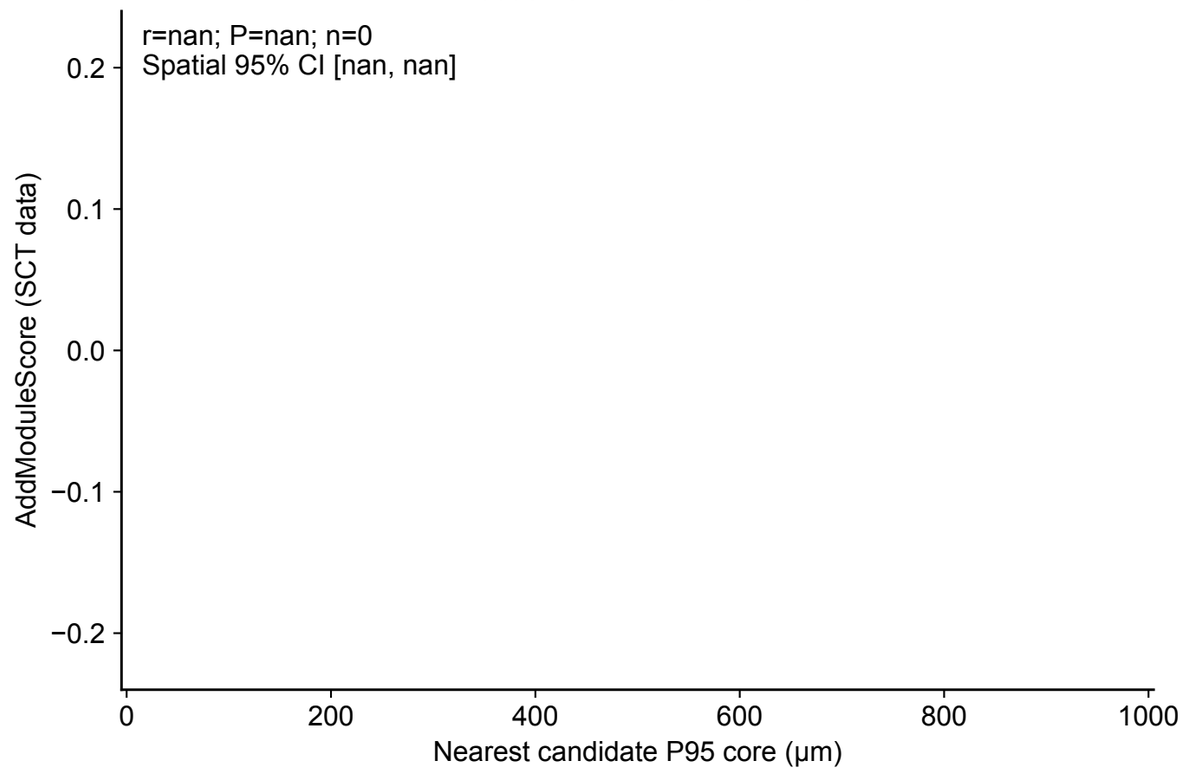


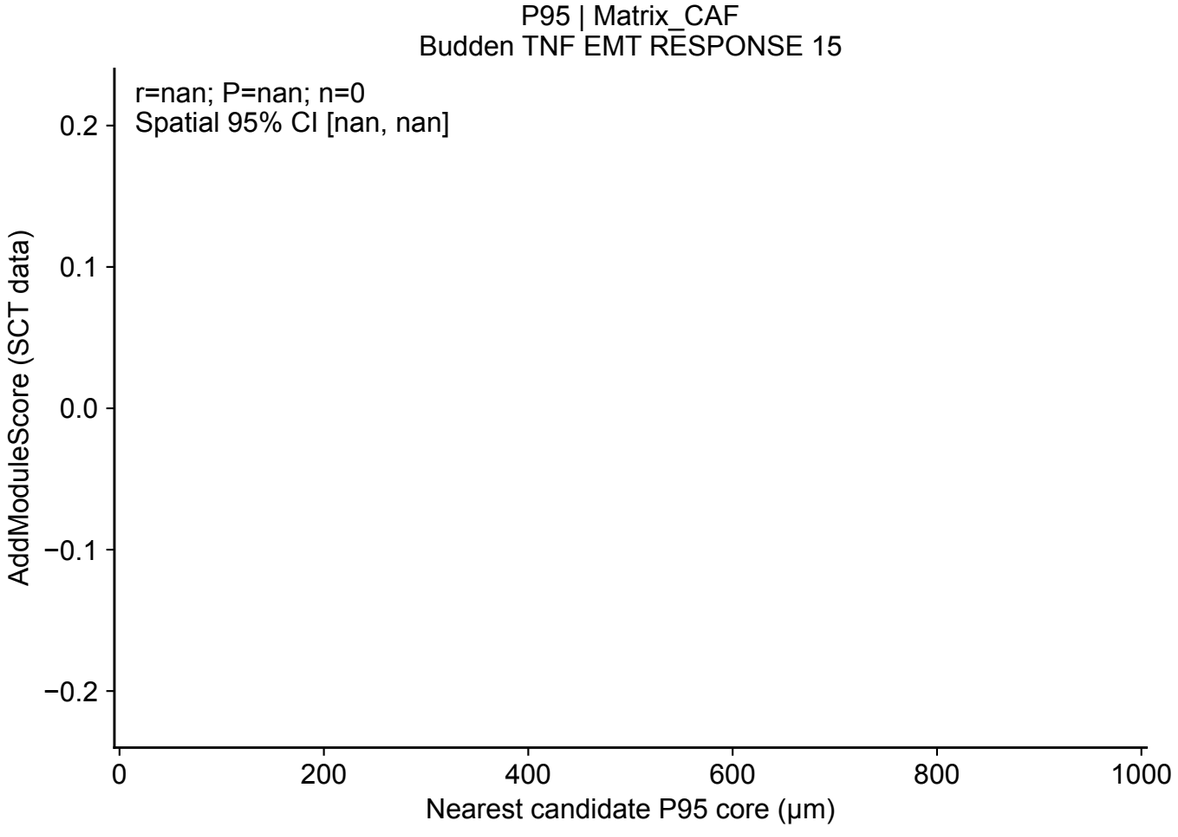
P95 minus nonP95 r | common support | grey = not estimable

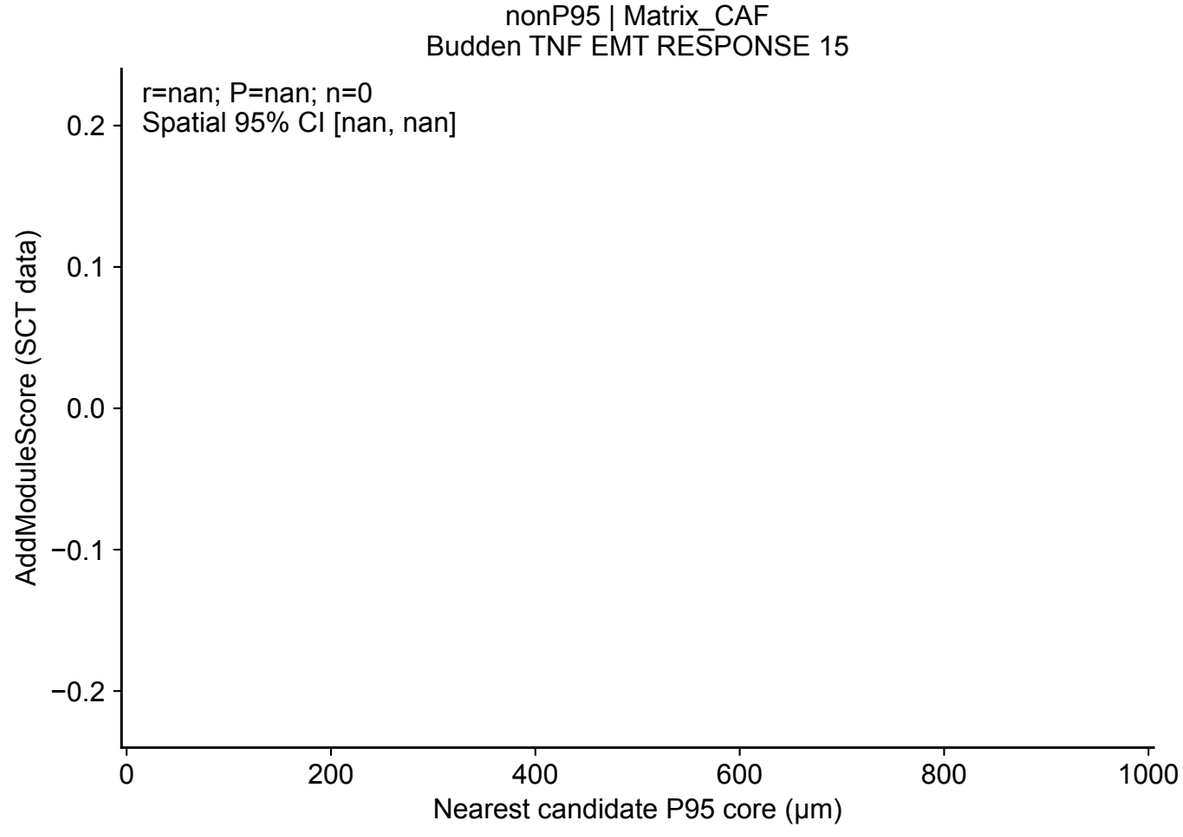


GSE238145_B13 | exploratory scatter display selected by coverage and |delta r|, not P







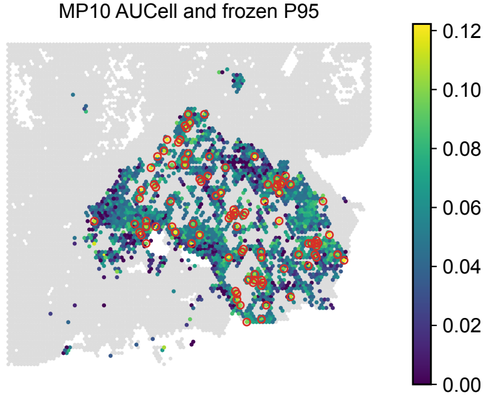


Treg / Budden S1P RESPONSE 18: delta r=nan [nan, nan]; common nan–-inf μm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=nan [nan, nan]; common nan–-inf μm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

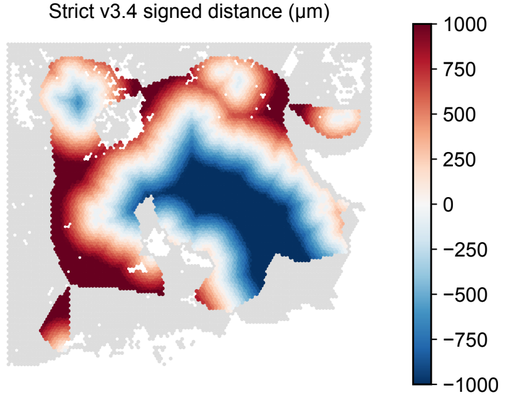
P95 nonP95 distance 0

GSE238145_B14 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

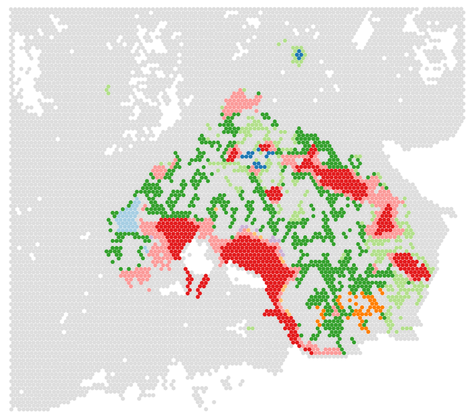
MP10 AUCell and frozen P95



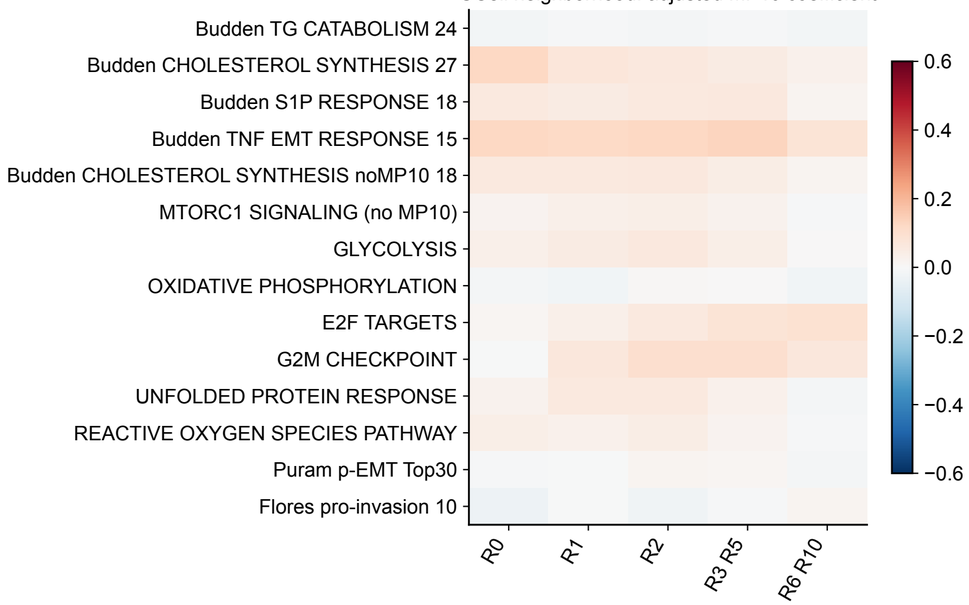
Strict v3.4 signed distance (µm)



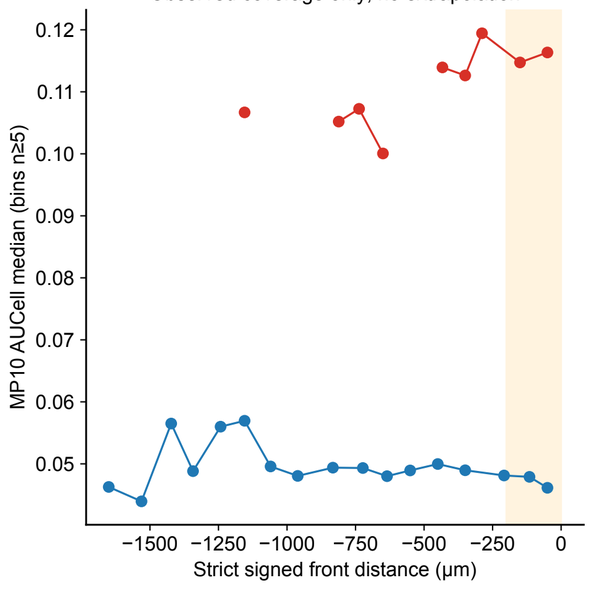
Formal R12 0–8; grey = unlabelled



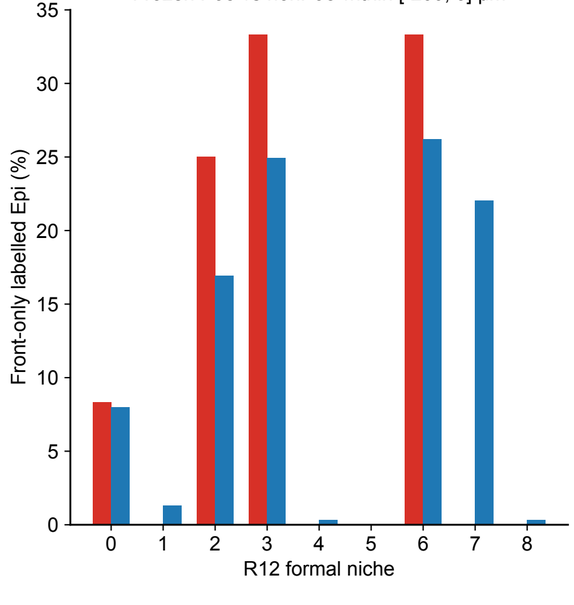
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

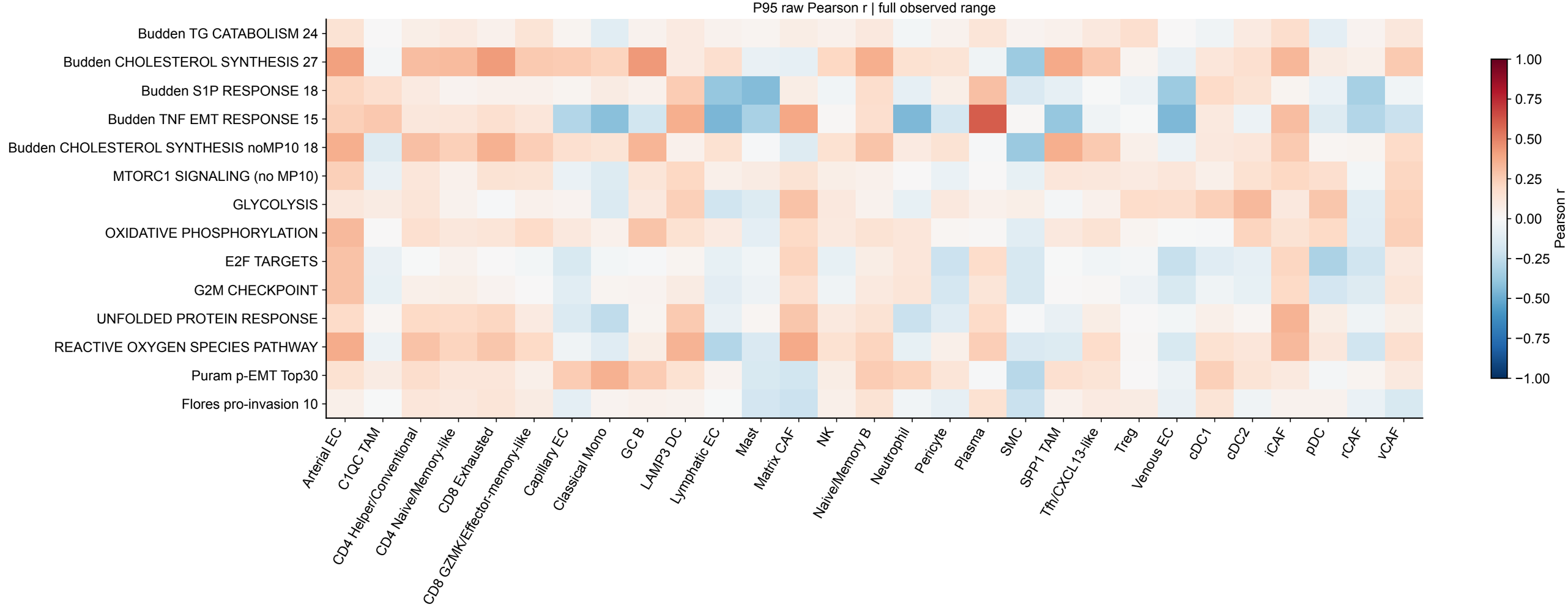


Tissue 10435 | Epi 2138 | internal 1900 | P95 95
Exact front in Epi 1721 | R12 labelled 1940

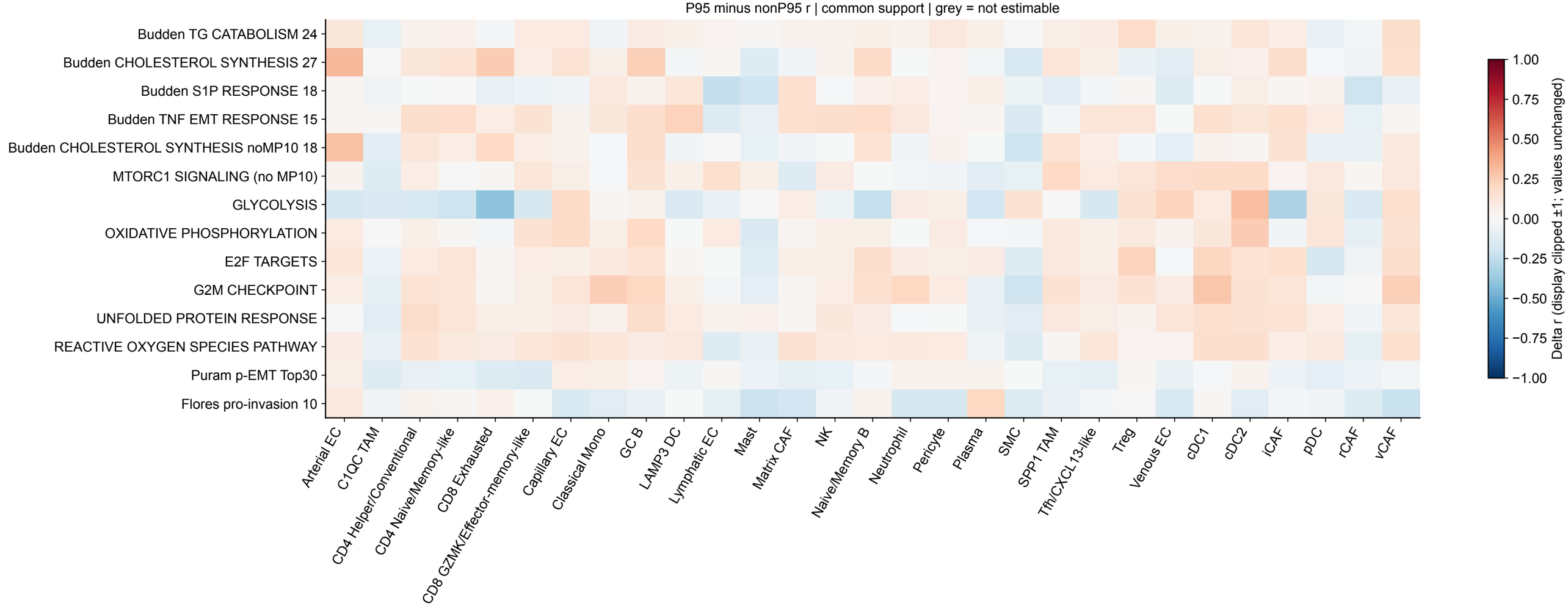
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B14 | all 29 candidates; fixed core programs

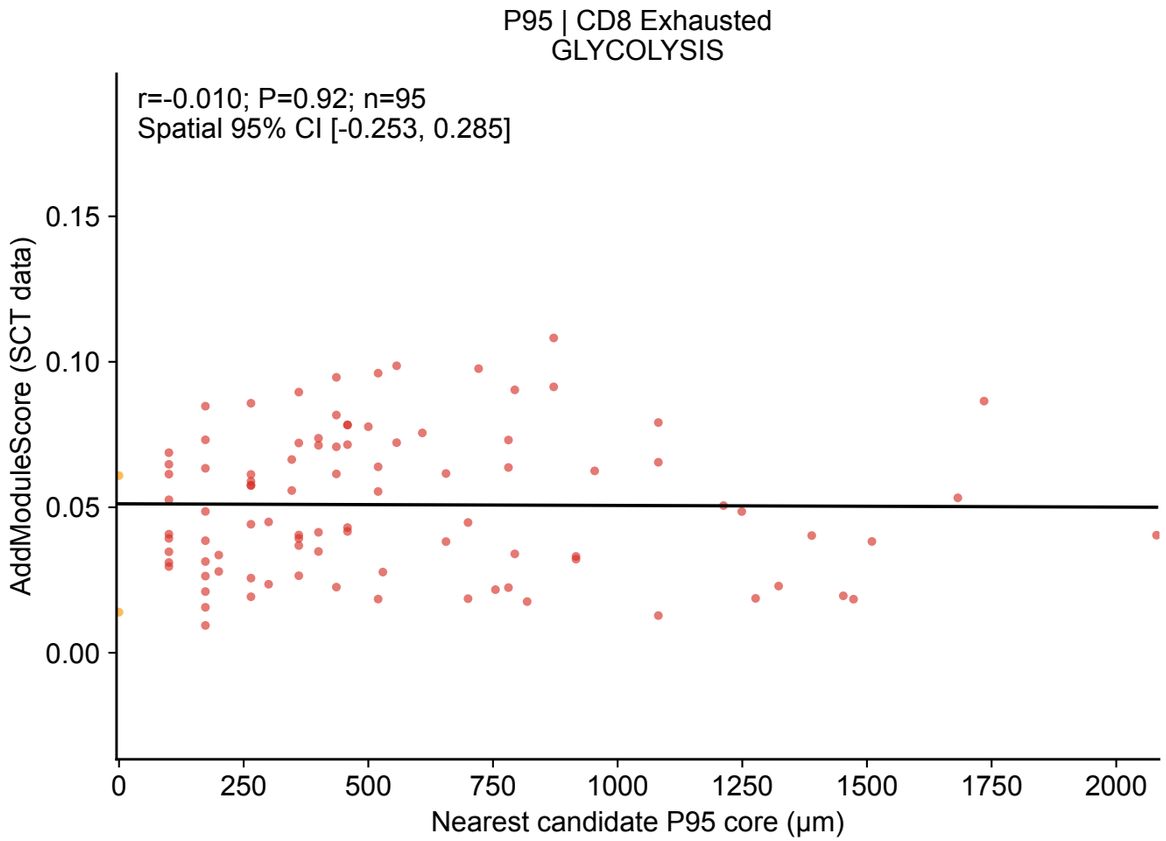
P95 raw Pearson r | full observed range

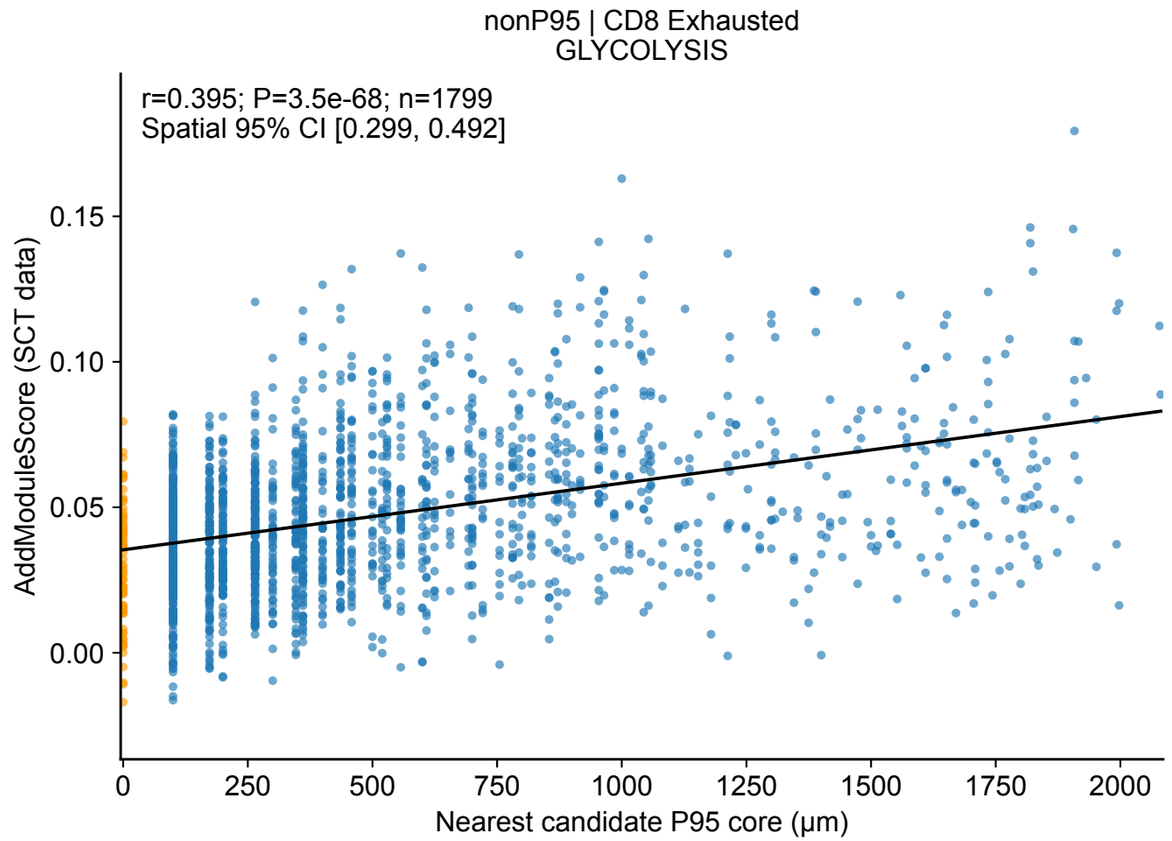


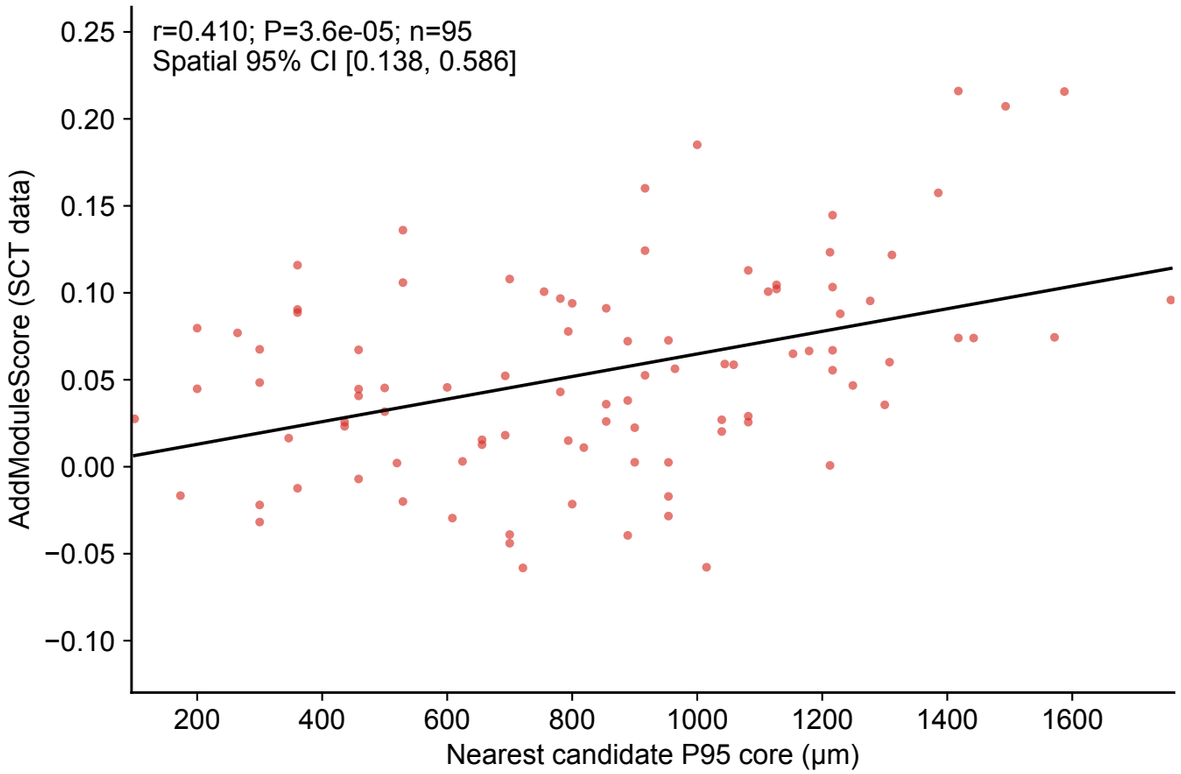
P95 minus nonP95 r | common support | grey = not estimable

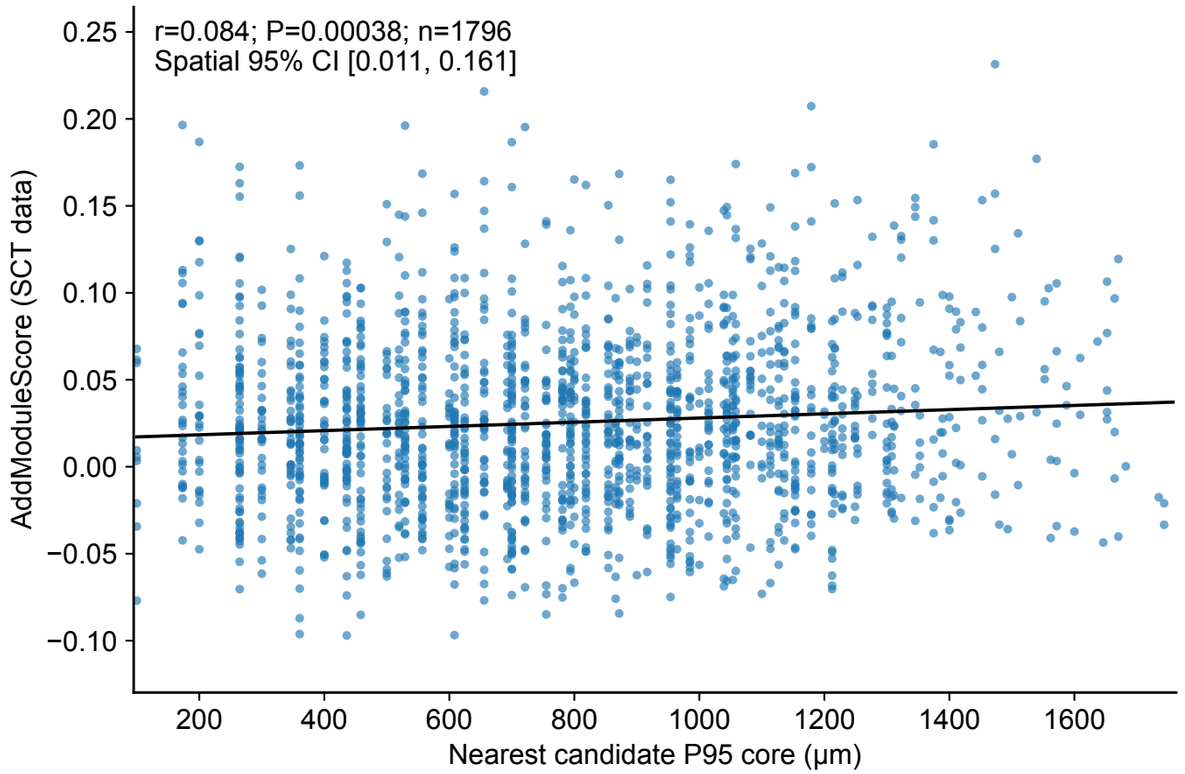


GSE238145_B14 | exploratory scatter display selected by coverage and |delta r|, not P







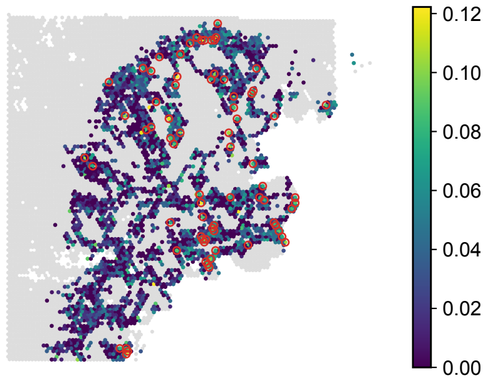


CD8 Exhausted / GLYCOLYSIS: delta r=-0.405 [-0.635, -0.120]; common 0–2081 µm; spatial P=0.0015
Arterial EC / Budden CHOLESTEROL SYNTHESIS 27: delta r=0.327 [0.055, 0.506]; common 100–1758 µm; spatial P=0.0081
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

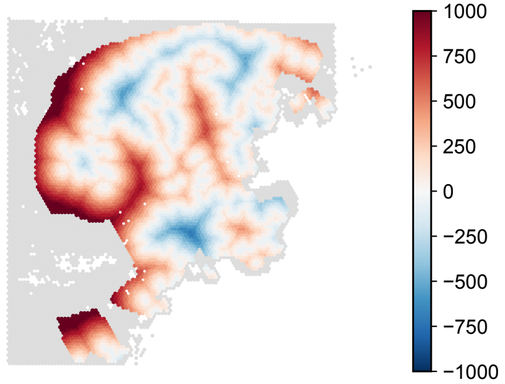
P95 nonP95 distance 0

GSE238145_B15 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

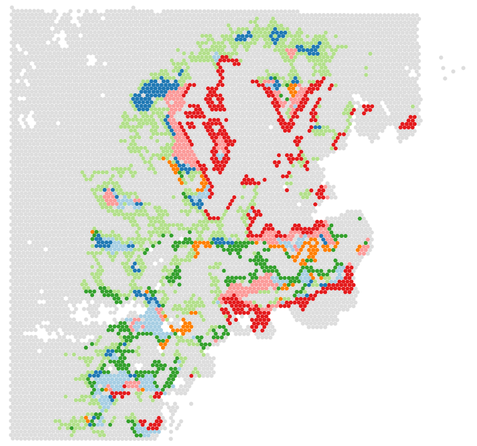
MP10 AUCell and frozen P95



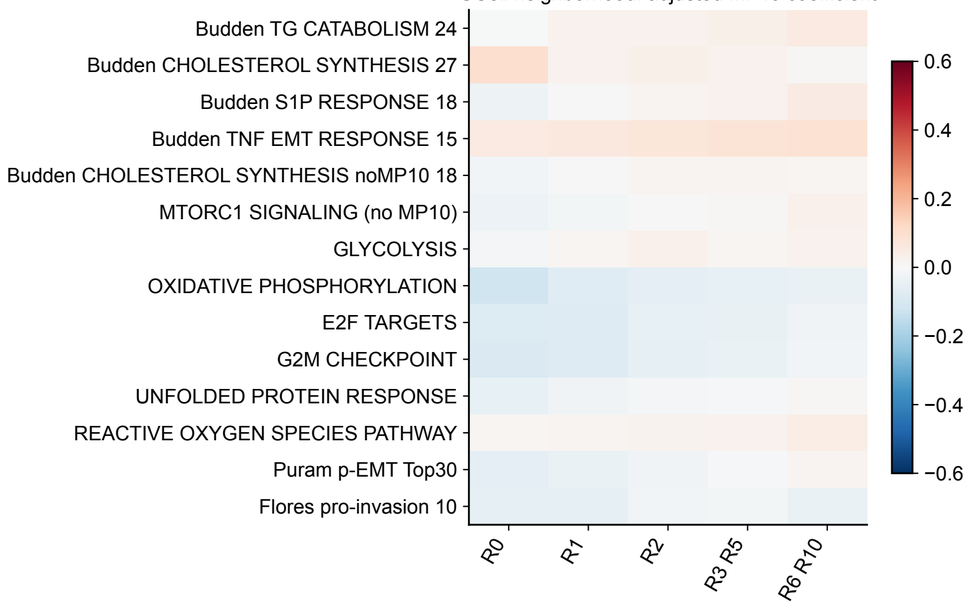
Strict v3.4 signed distance (µm)



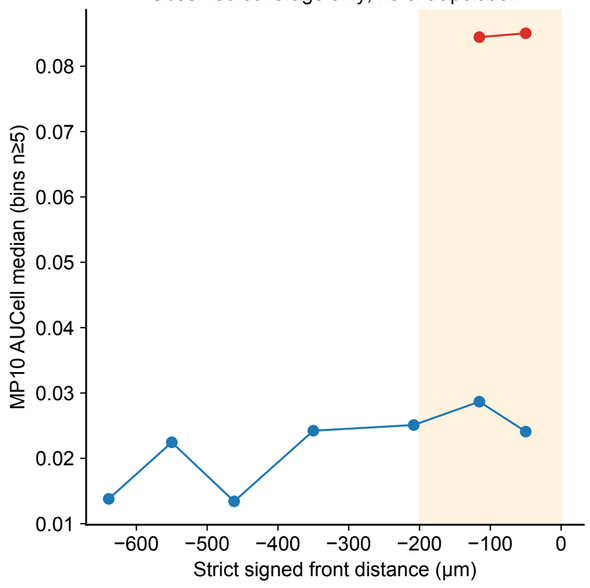
Formal R12 0–8; grey = unlabelled



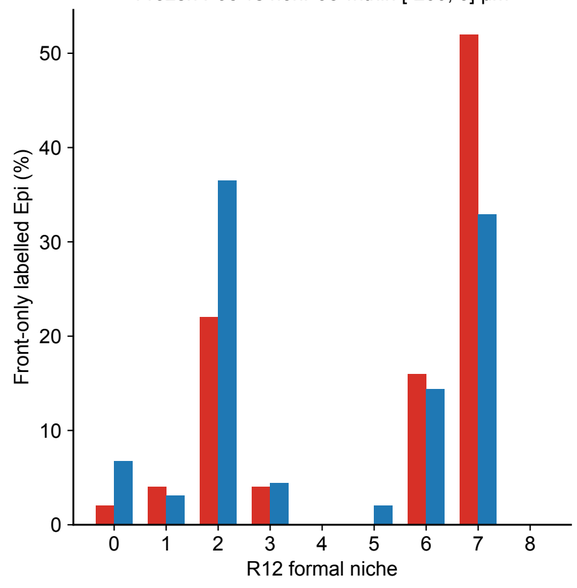
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

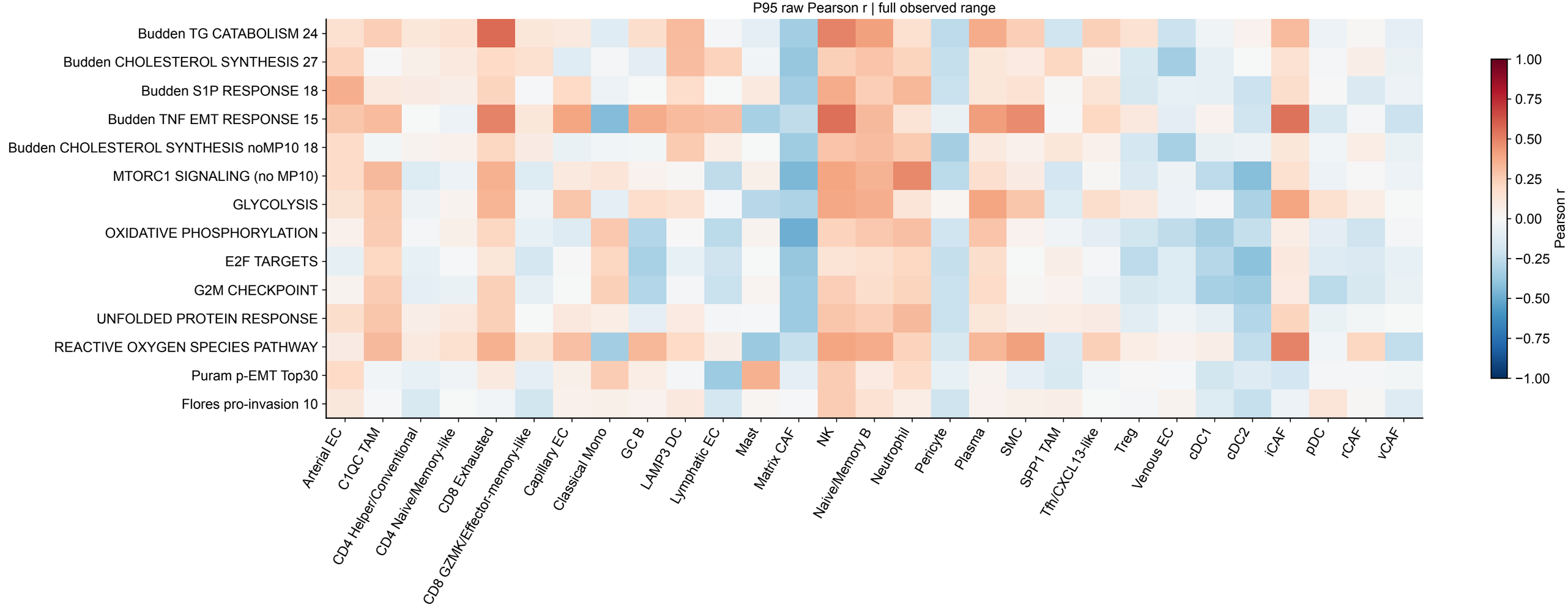


Tissue 9386 | Epi 2782 | internal 1449 | P95 73
Exact front in Epi 2203 | R12 labelled 2525

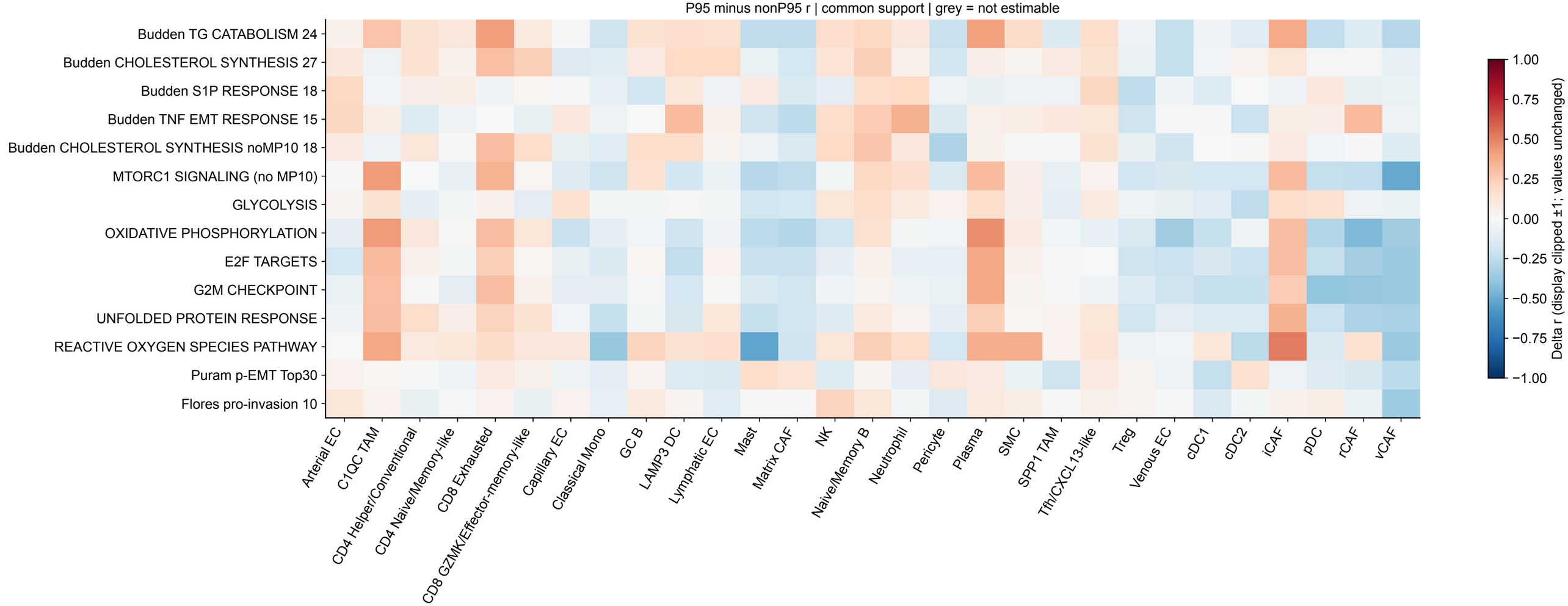
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B15 | all 29 candidates; fixed core programs

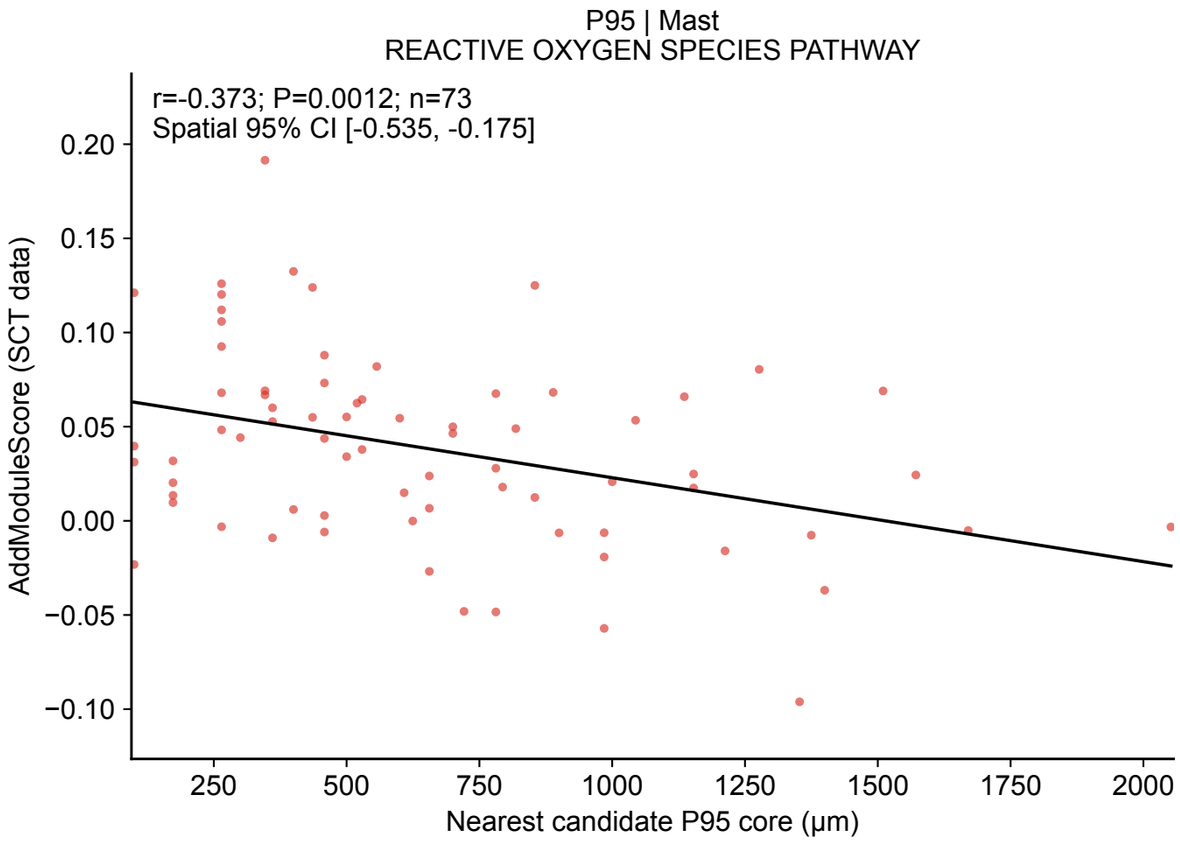
P95 raw Pearson r | full observed range

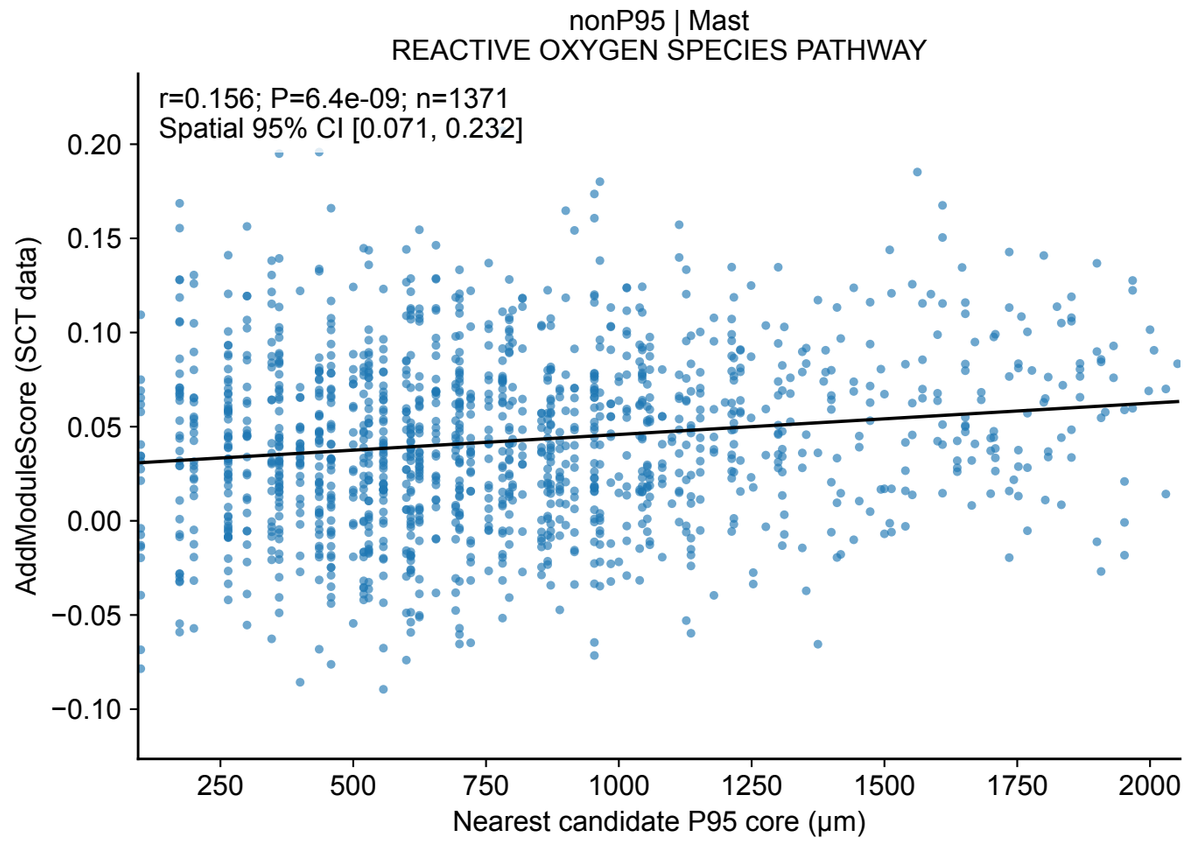


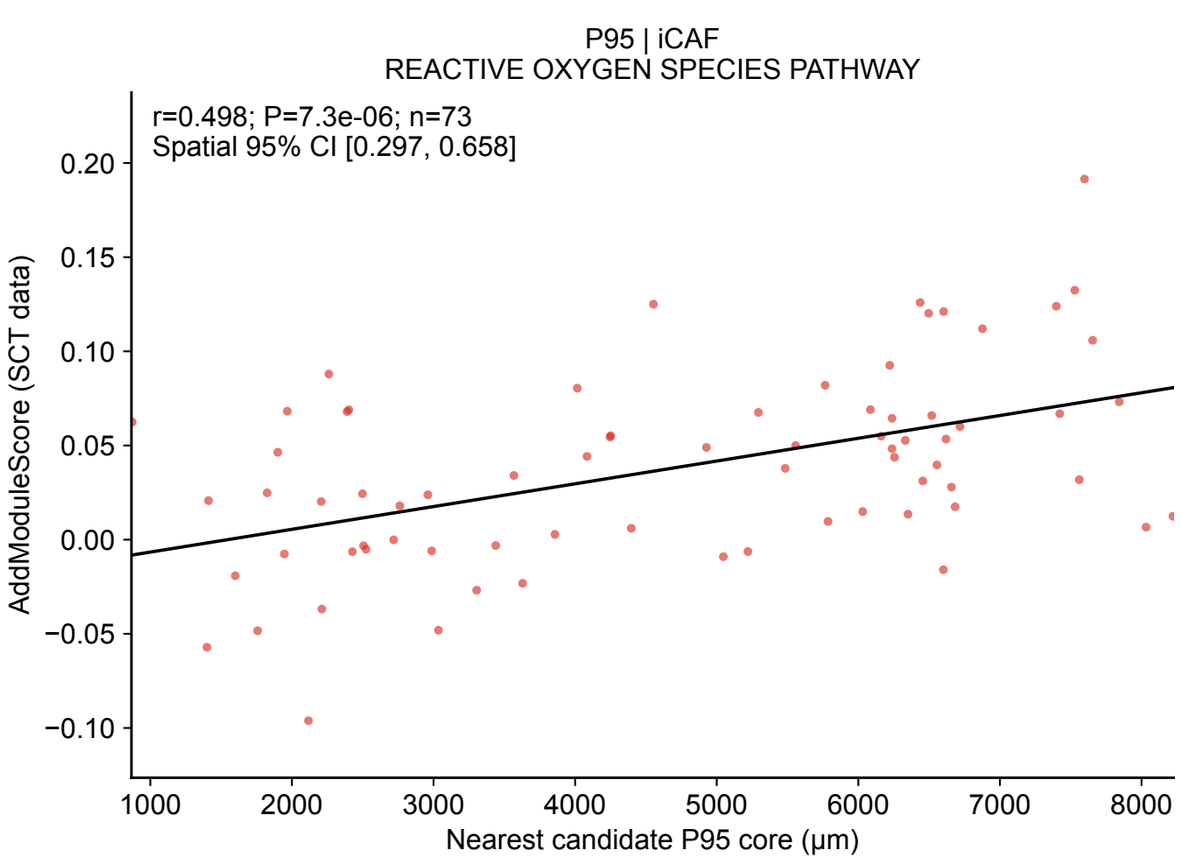
P95 minus nonP95 r | common support | grey = not estimable

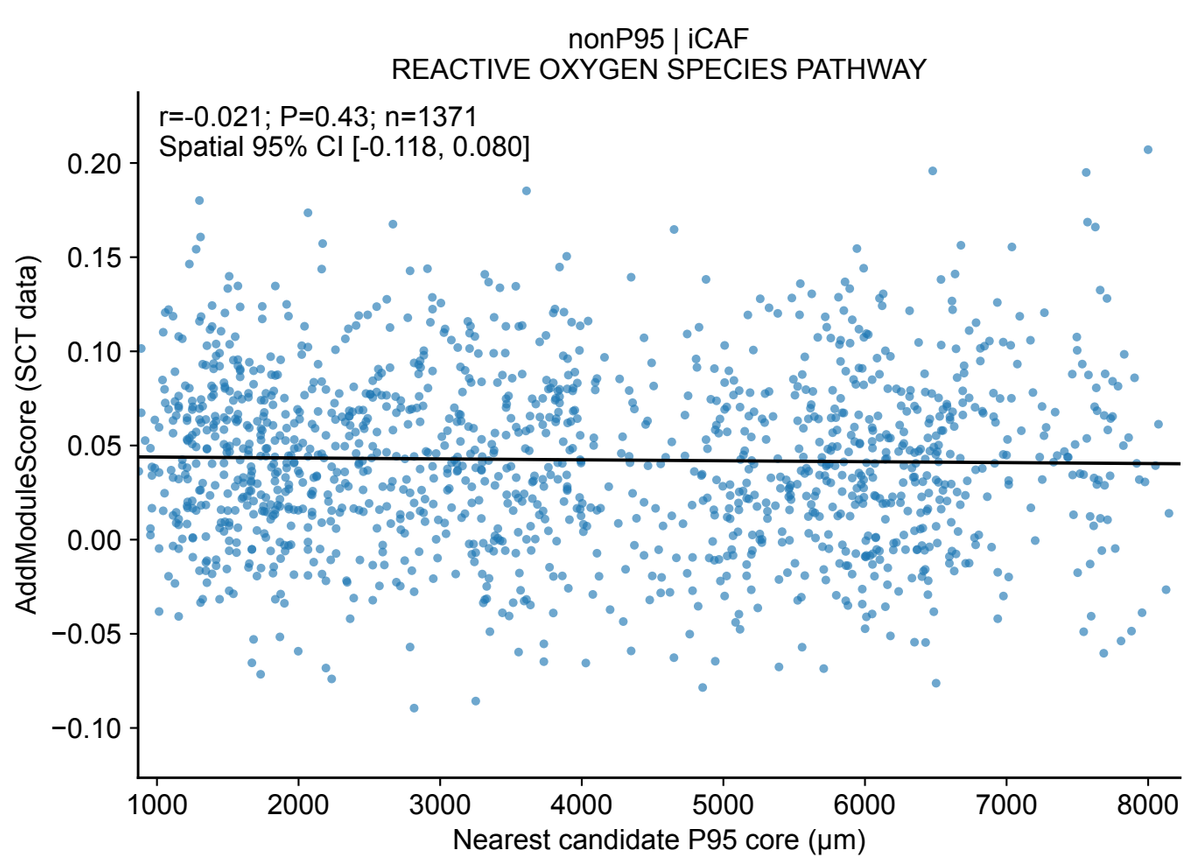


GSE238145_B15 | exploratory scatter display selected by coverage and |delta r|, not P







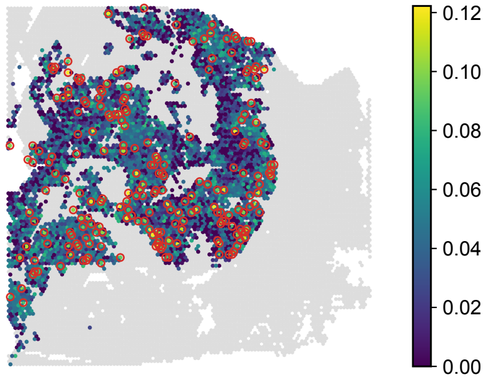


Mast / REACTIVE OXYGEN SPECIES PATHWAY: delta r=-0.529 [-0.698, -0.333]; common 100–2052 µm; spatial P=1.2e-07
iCAF / REACTIVE OXYGEN SPECIES PATHWAY: delta r=0.520 [0.336, 0.669]; common 872–8221 µm; spatial P=2.8e-07
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

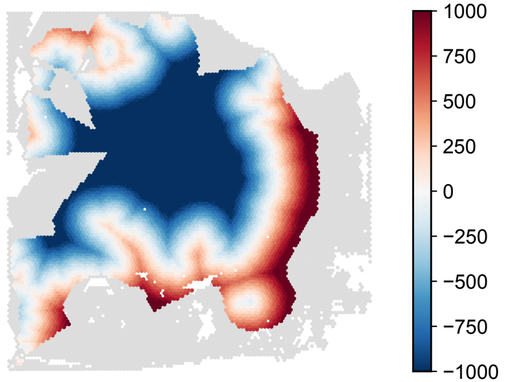
P95 nonP95 distance 0

GSE238145_B16 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

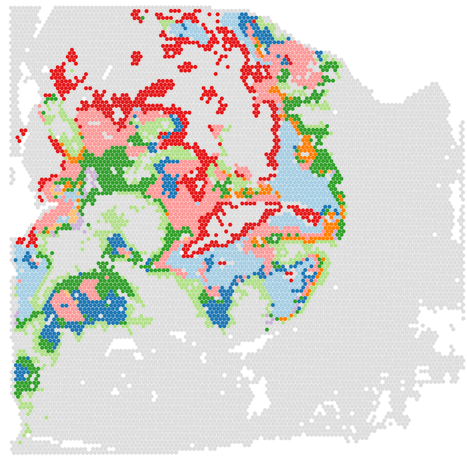
MP10 AUCell and frozen P95



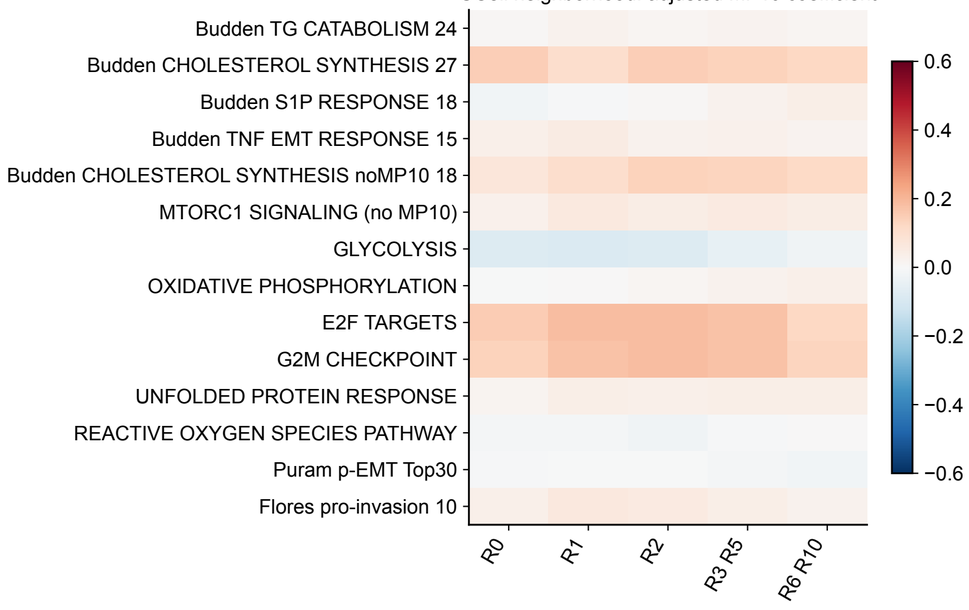
Strict v3.4 signed distance (µm)



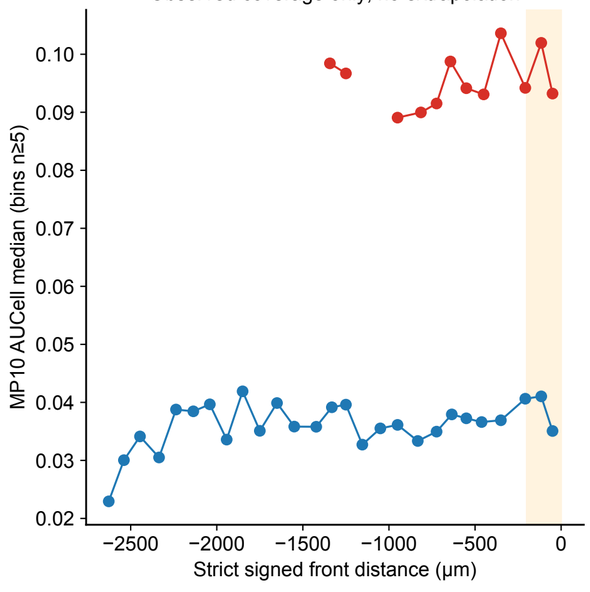
Formal R12 0–8; grey = unlabelled



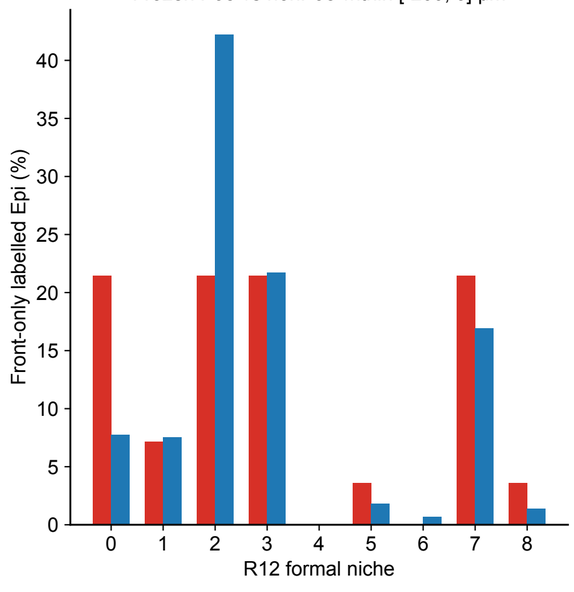
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

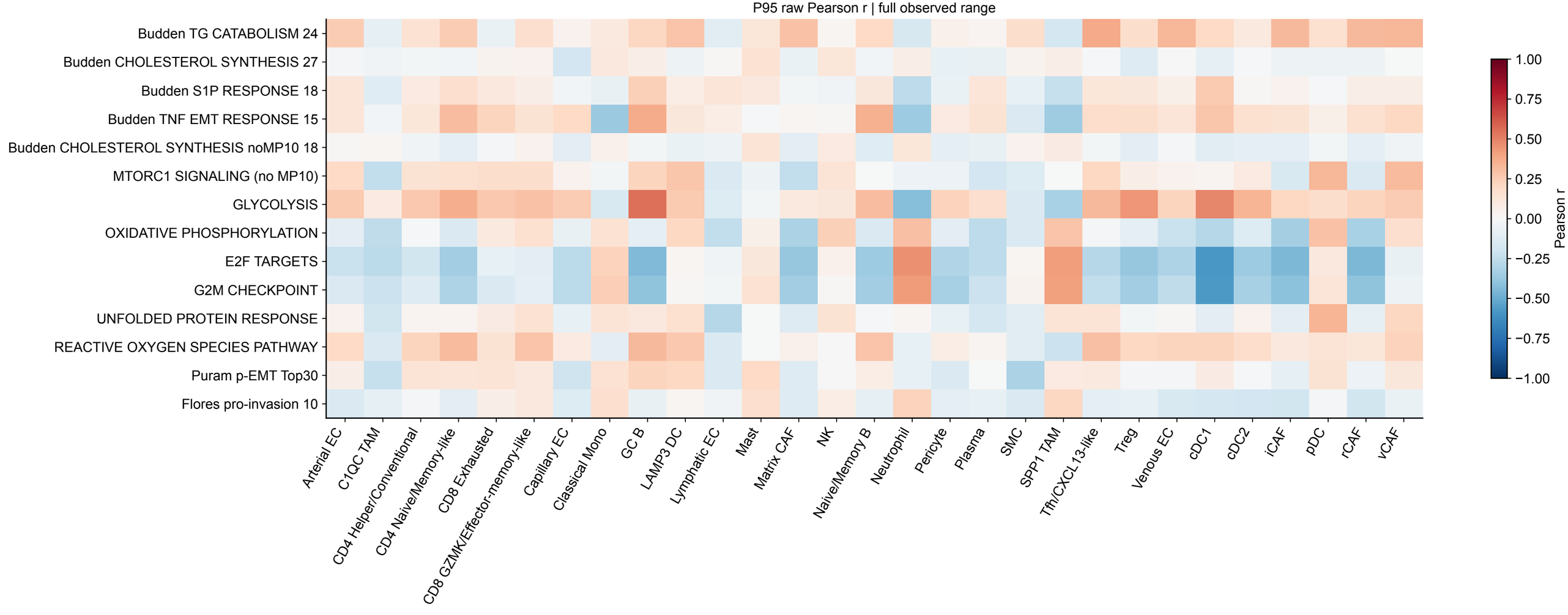


Tissue 12226 | Epi 4440 | internal 4358 | P95 218
Exact front in Epi 3543 | R12 labelled 4150

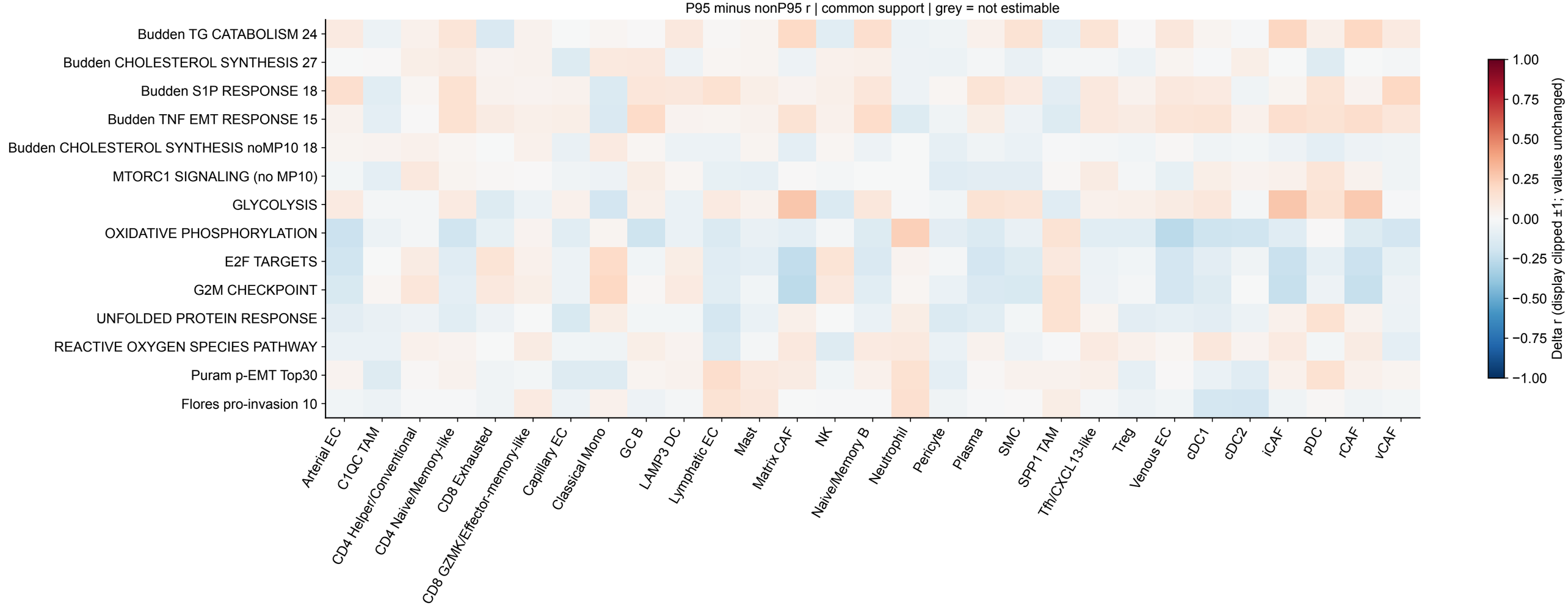
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B16 | all 29 candidates; fixed core programs

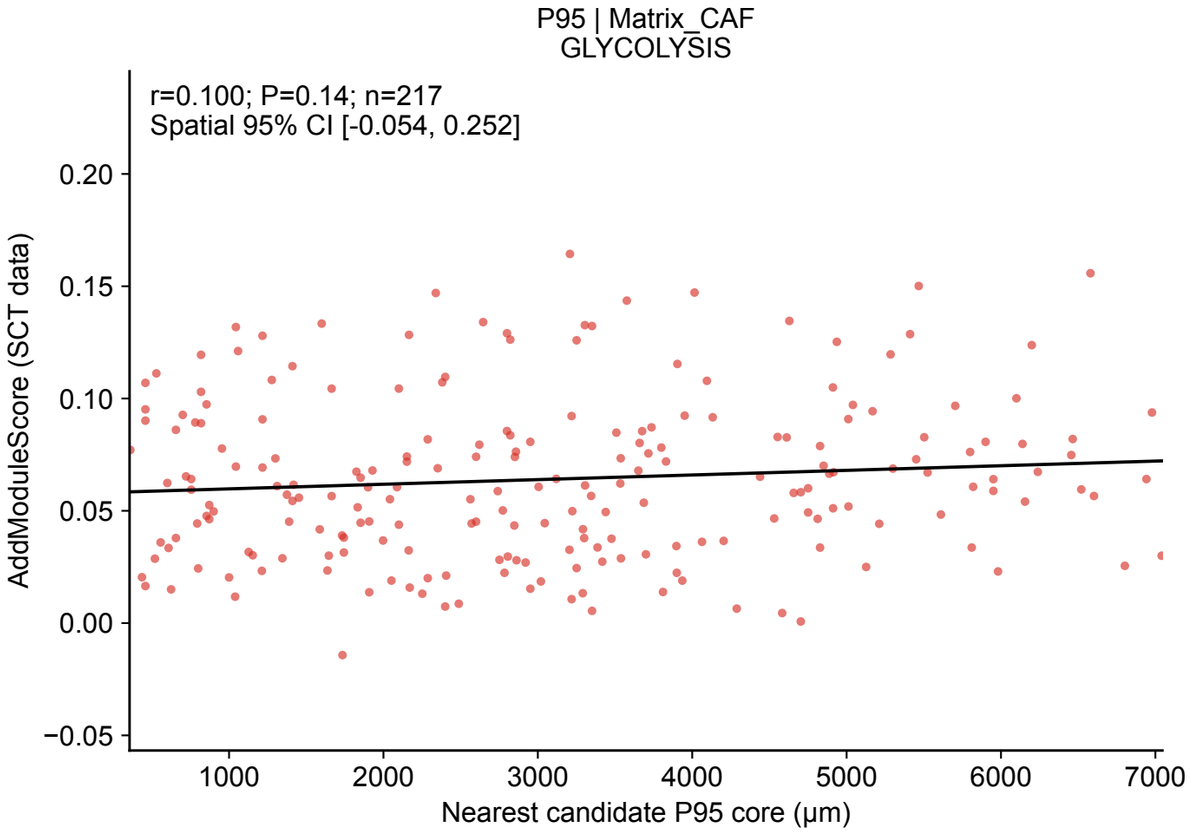
P95 raw Pearson r | full observed range

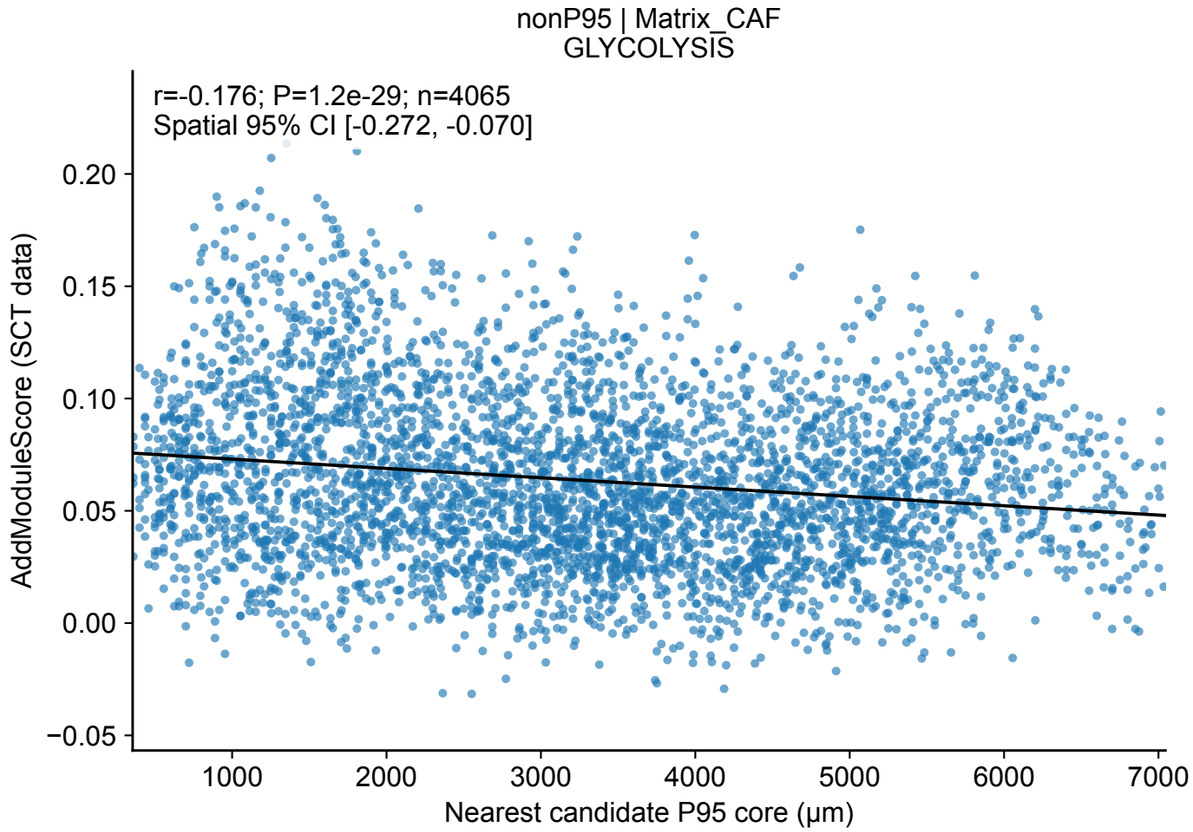


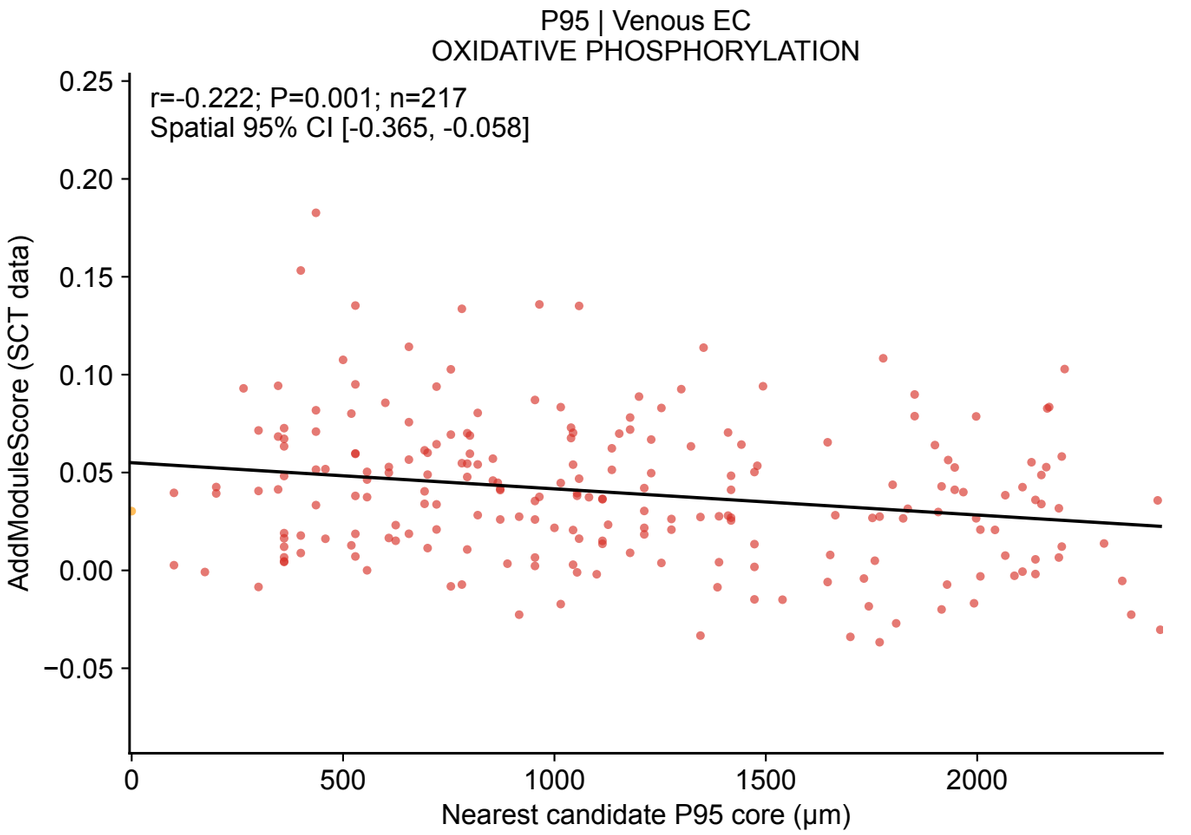
P95 minus nonP95 r | common support | grey = not estimable

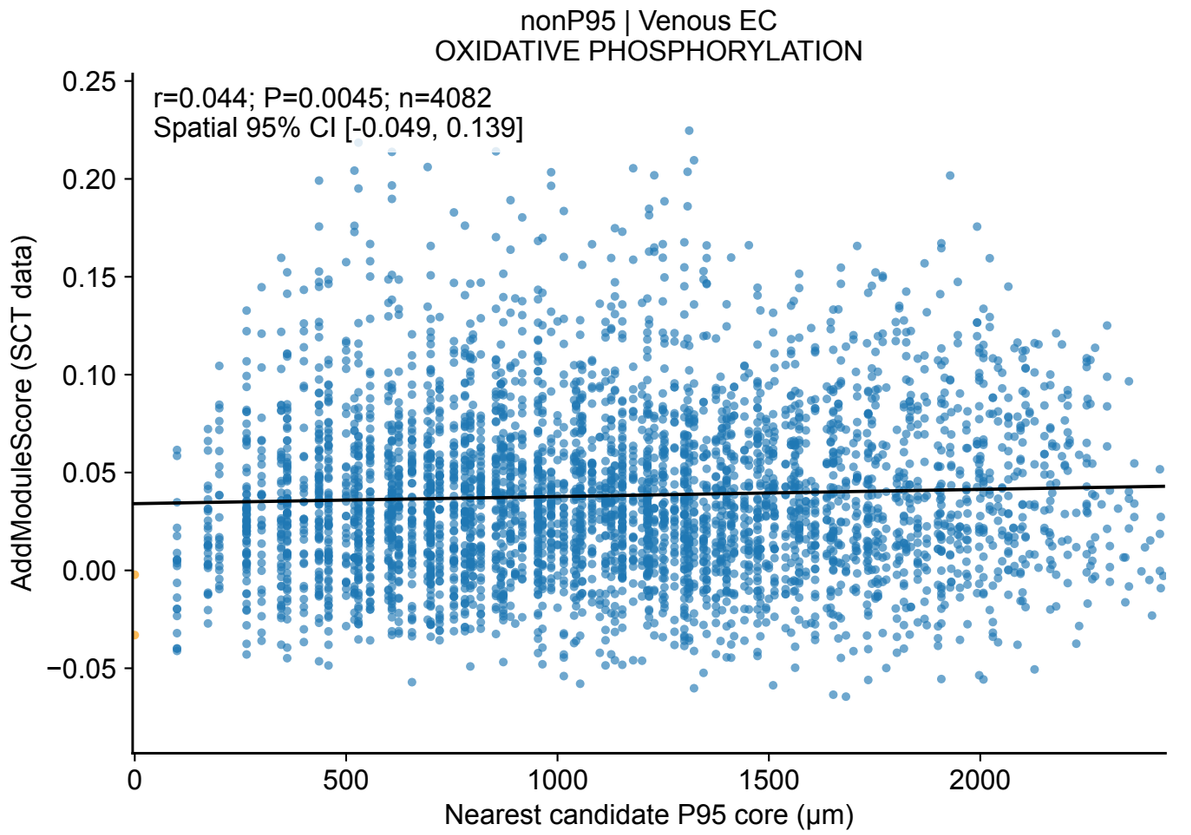


GSE238145_B16 | exploratory scatter display selected by coverage and |delta r|, not P







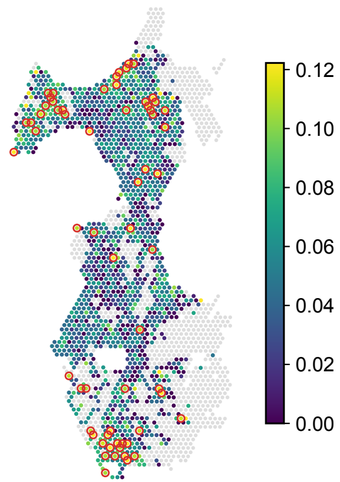


Matrix_CAF / GLYCOLYSIS: delta r=0.276 [0.120, 0.430]; common 361–7041 µm; spatial P=0.00051
Venous EC / OXIDATIVE PHOSPHORYLATION: delta r=-0.266 [-0.401, -0.120]; common 0–2433 µm; spatial P=0.00038
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

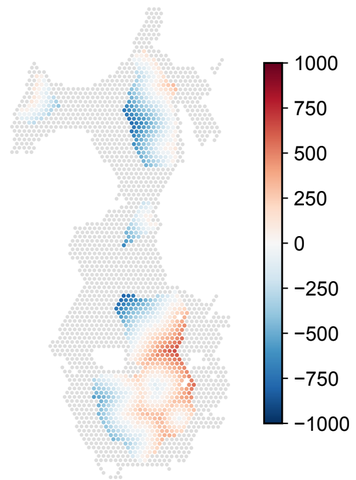
P95 nonP95 distance 0

GSE238145_B17 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

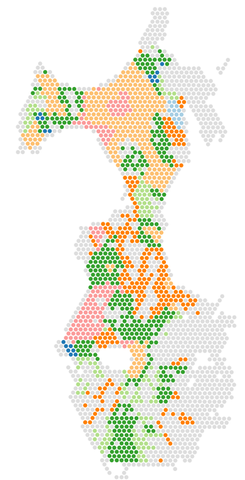
MP10 AUCell and frozen P95



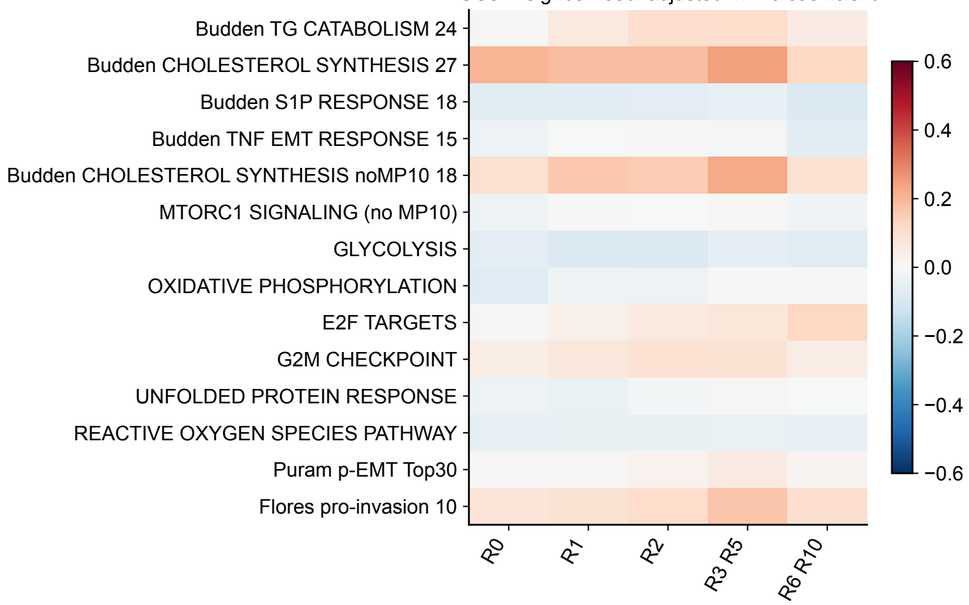
Strict v3.4 signed distance (µm)



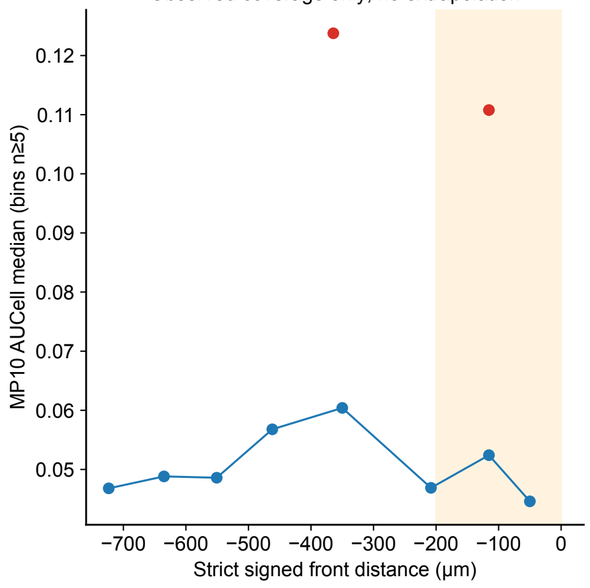
Formal R12 0–8; grey = unlabelled



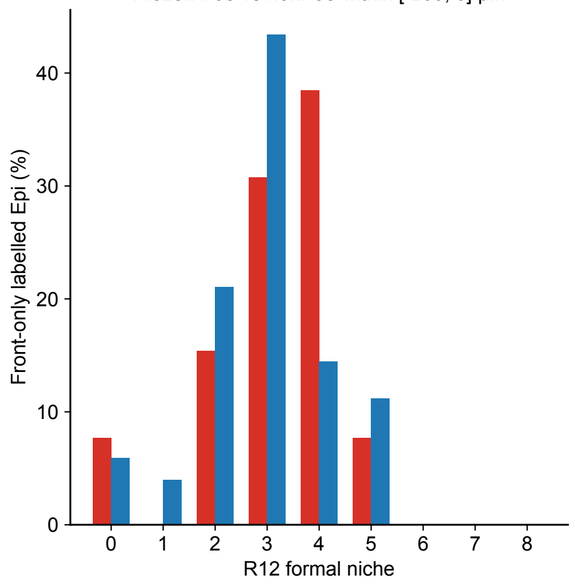
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

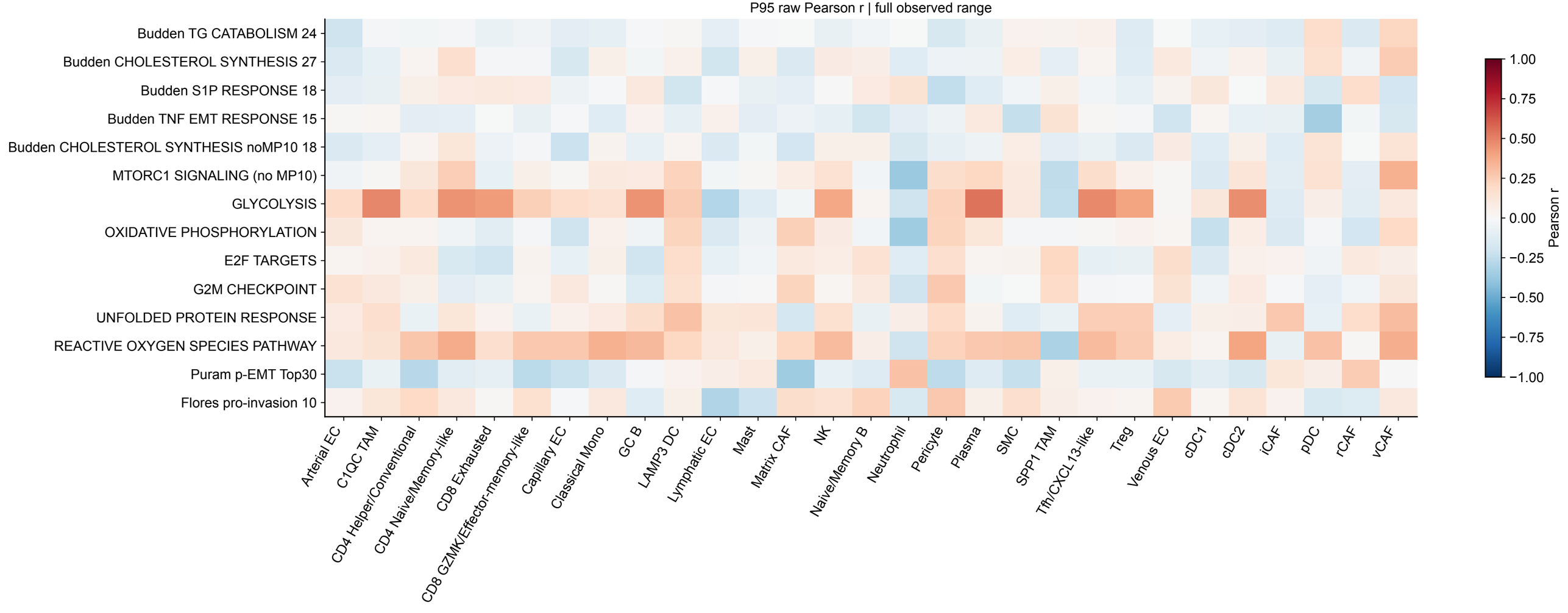


Tissue 2399 | Epi 1494 | internal 1258 | P95 63
Exact front in Epi 563 | R12 labelled 1284

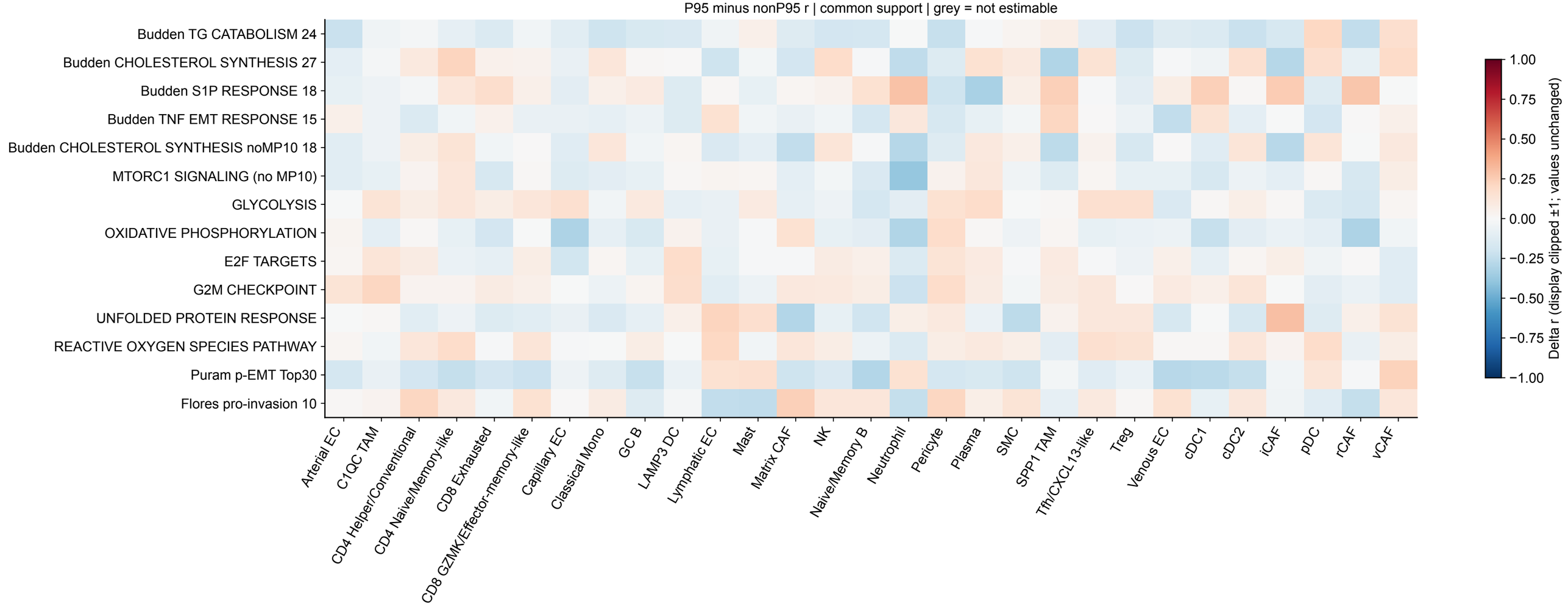
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B17 | all 29 candidates; fixed core programs

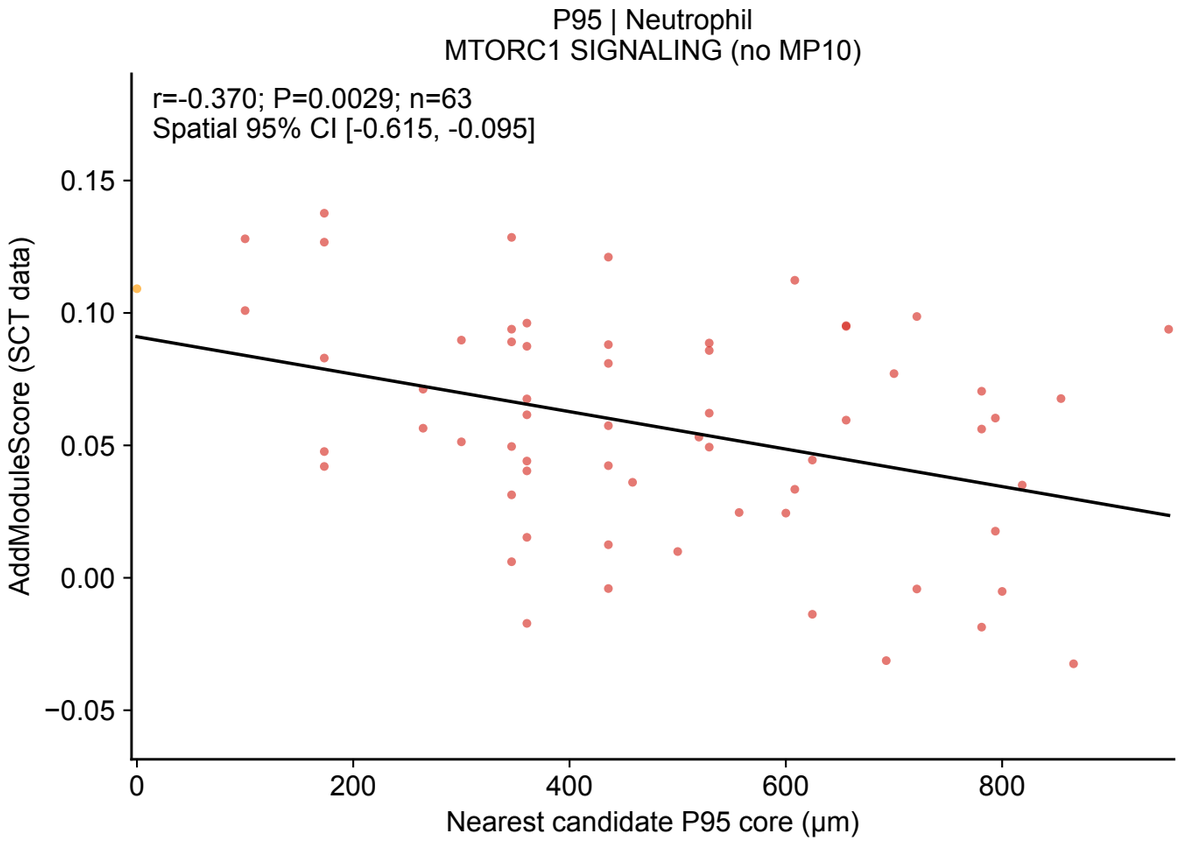
P95 raw Pearson r | full observed range

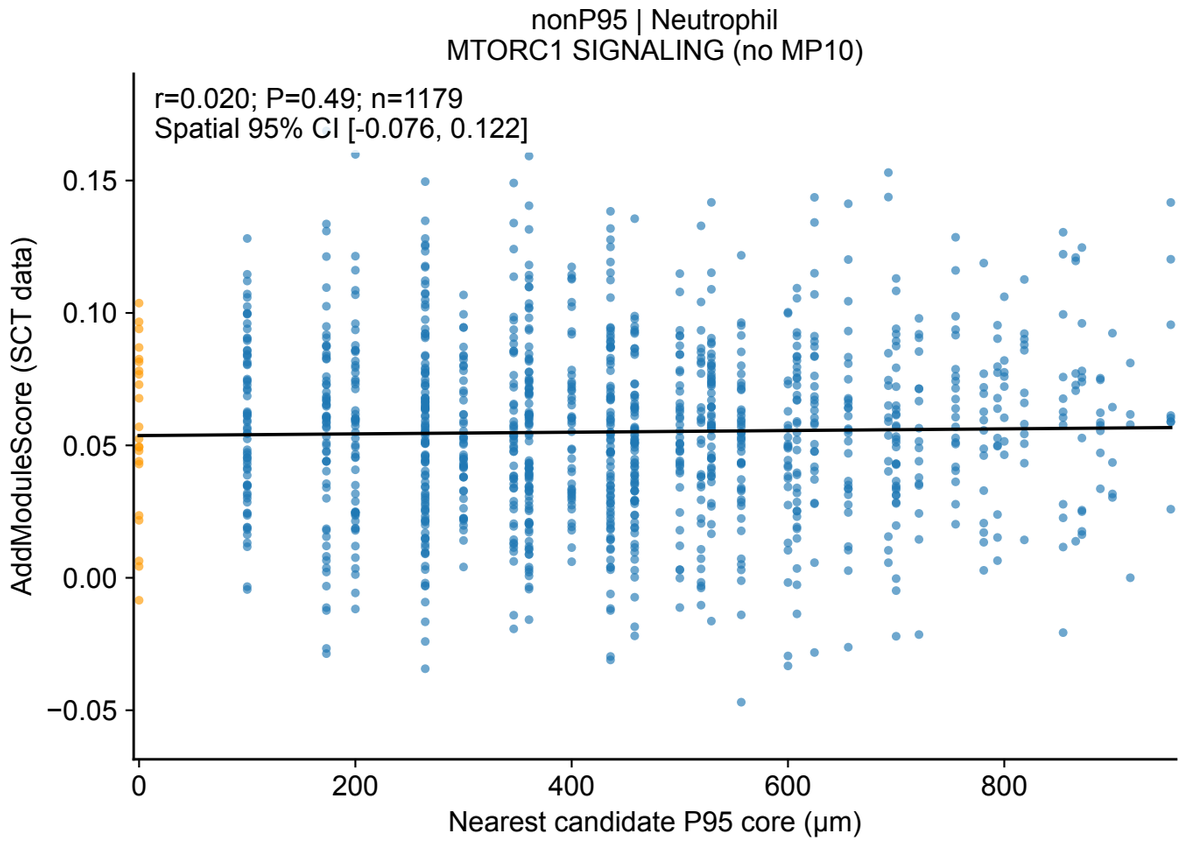


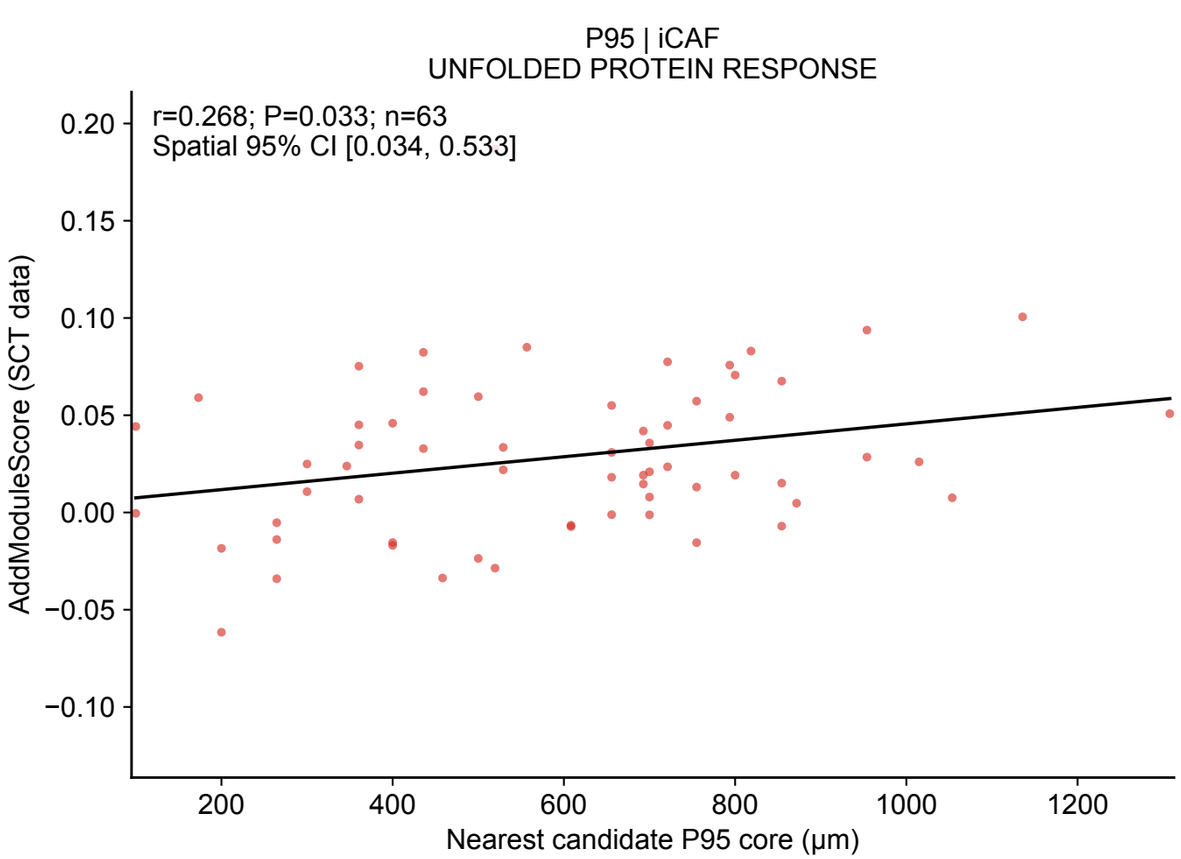
P95 minus nonP95 r | common support | grey = not estimable

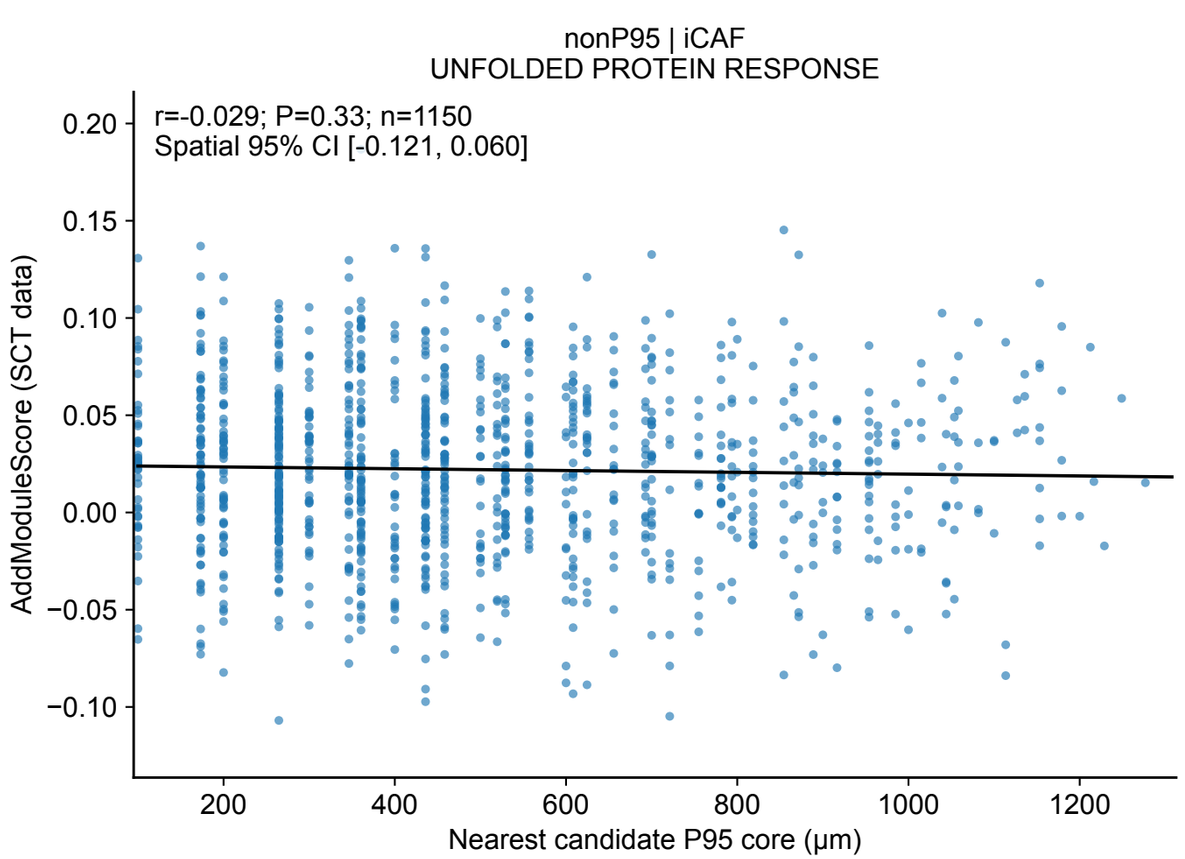


GSE238145_B17 | exploratory scatter display selected by coverage and |delta r|, not P







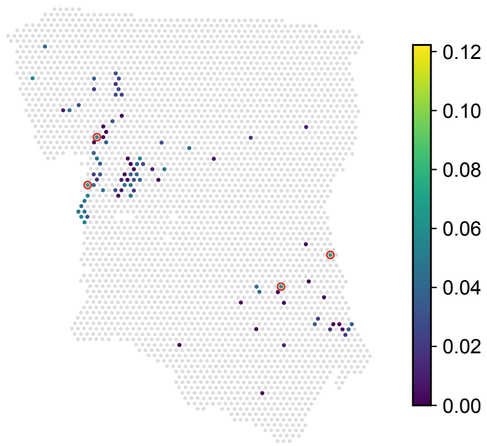


Neutrophil / MTORC1 SIGNALING (no MP10): delta r=-0.390 [-0.616, -0.146]; common 0–954 µm; spatial P=0.0035
iCAF / UNFOLDED PROTEIN RESPONSE: delta r=0.297 [0.061, 0.561]; common 100–1308 µm; spatial P=0.026
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

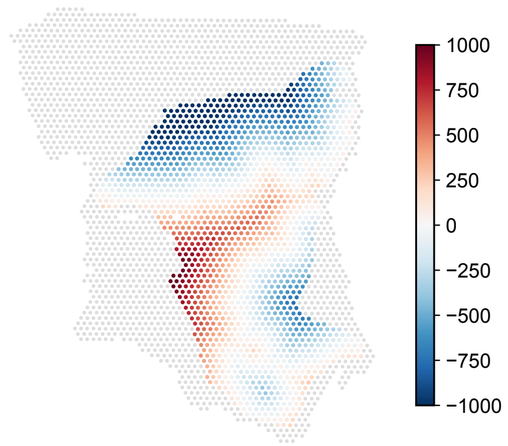
P95 nonP95 distance 0

GSE238145_B18 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

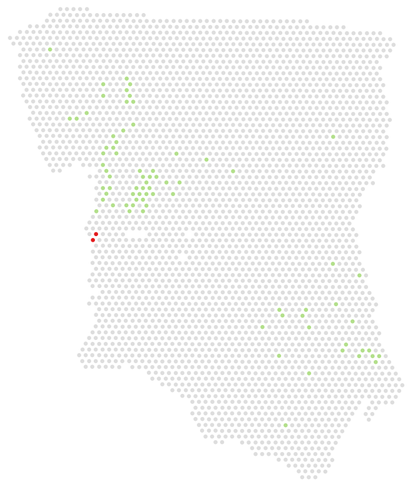
MP10 AUCell and frozen P95



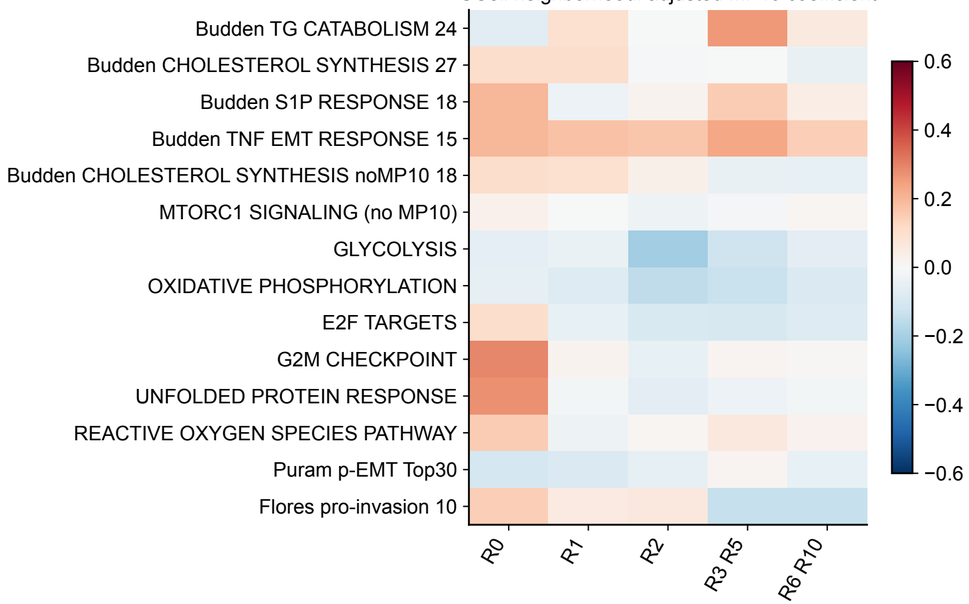
Strict v3.4 signed distance (µm)



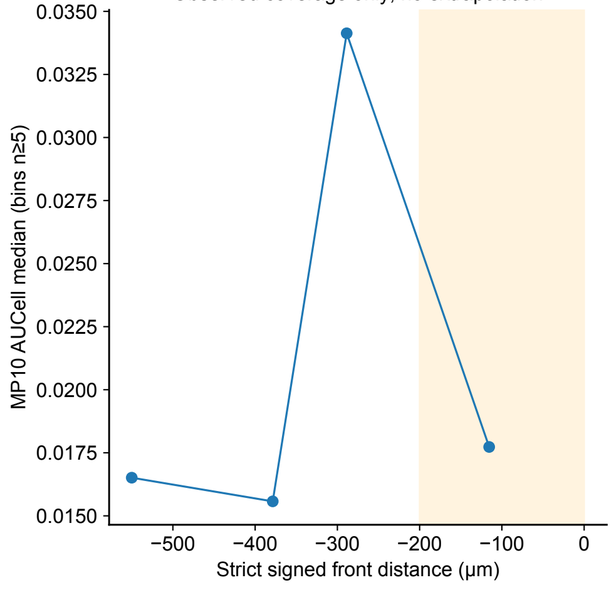
Formal R12 0–8; grey = unlabelled



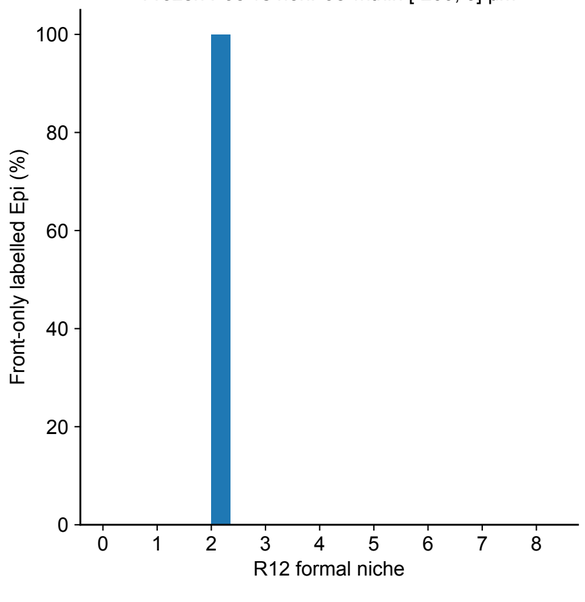
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

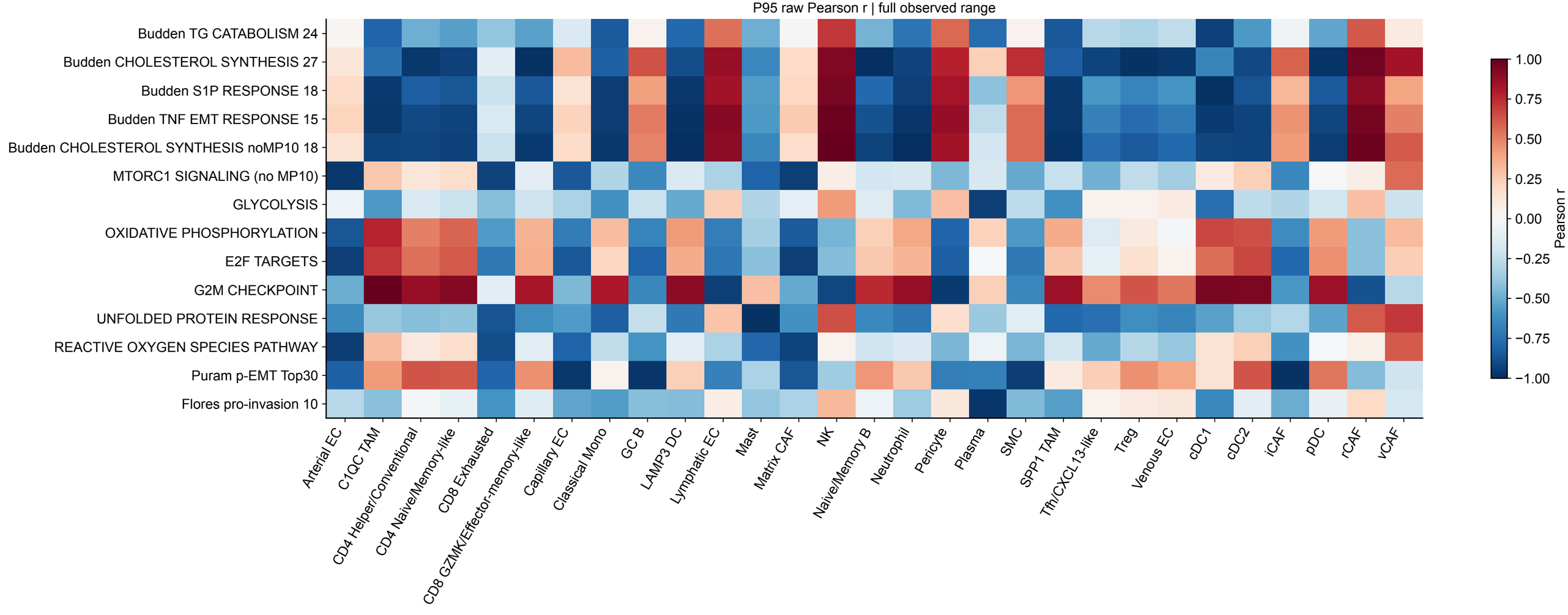


Tissue 3303 | Epi 89 | internal 80 | P95 4
Exact front in Epi 42 | R12 labelled 79

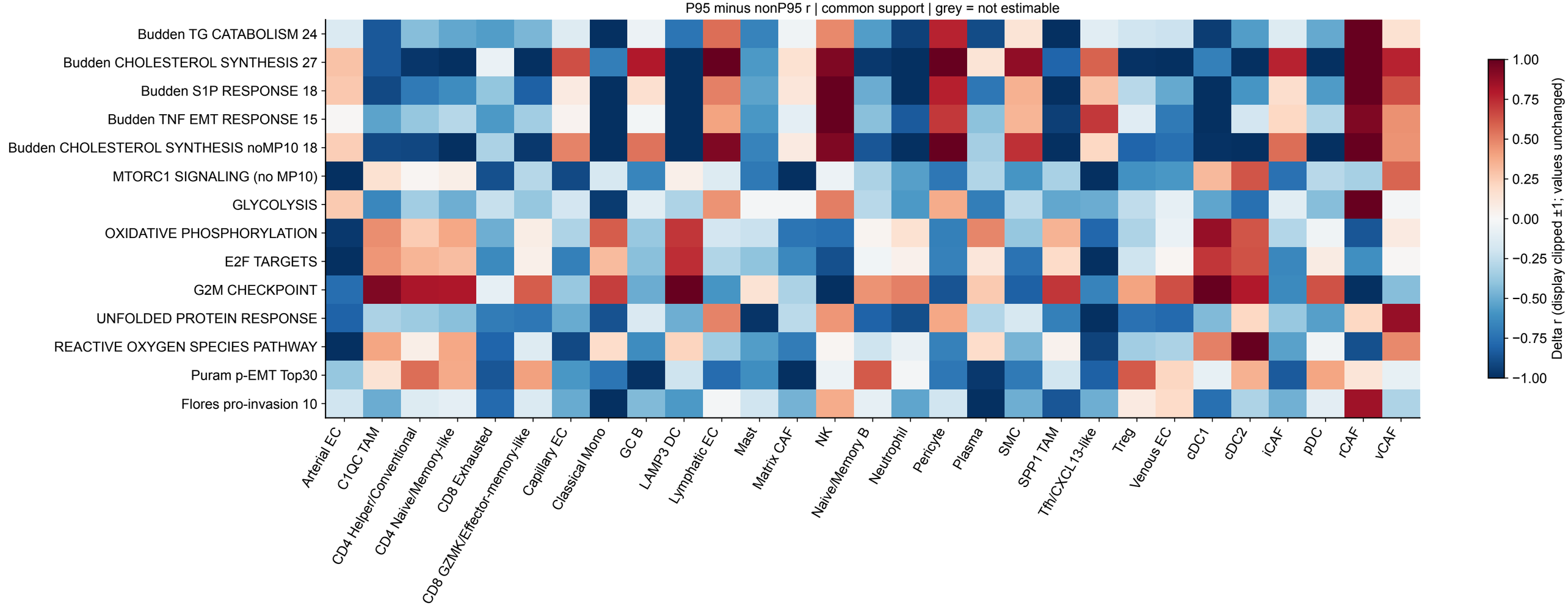
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B18 | all 29 candidates; fixed core programs

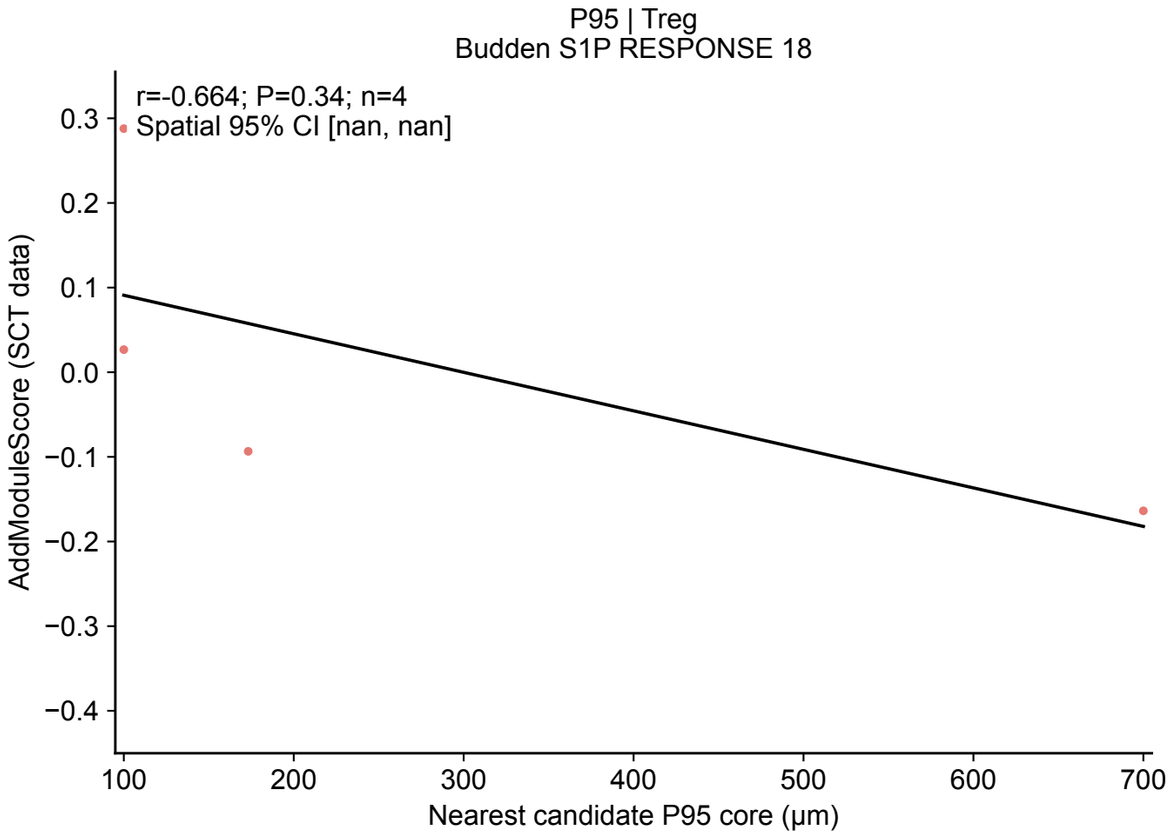
P95 raw Pearson r | full observed range

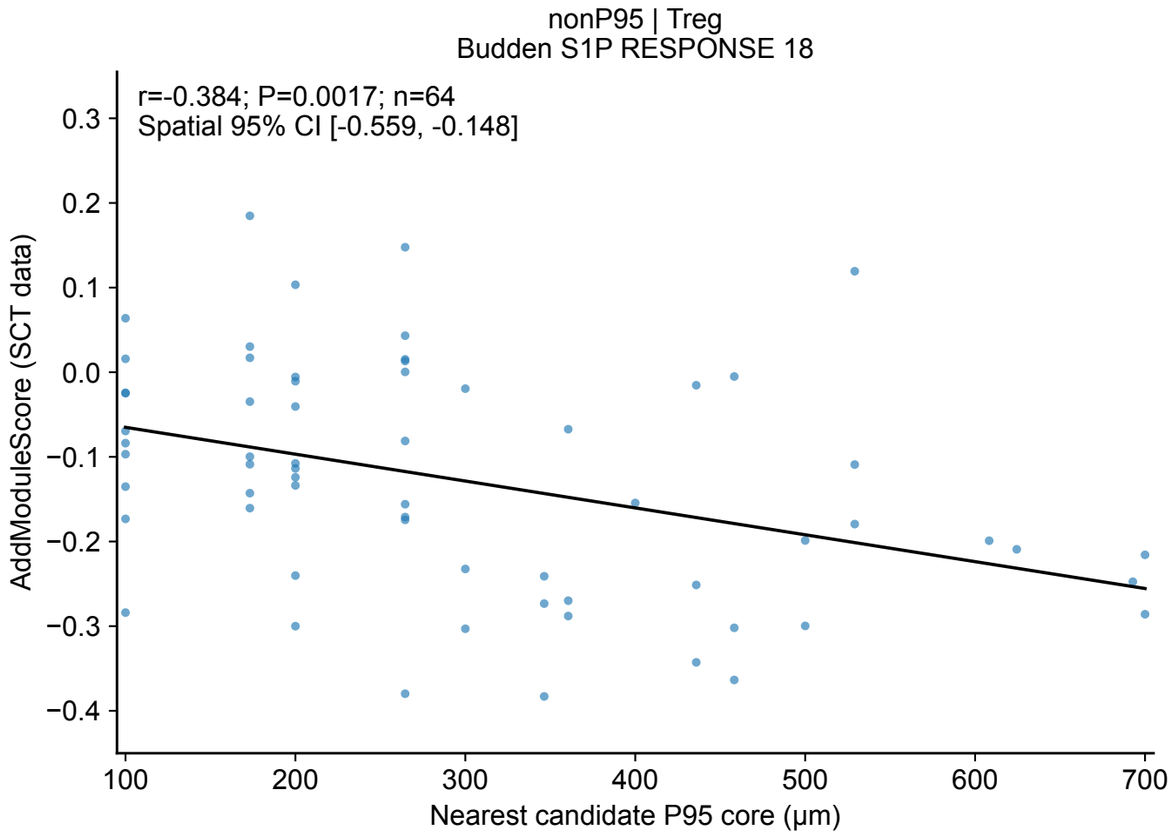


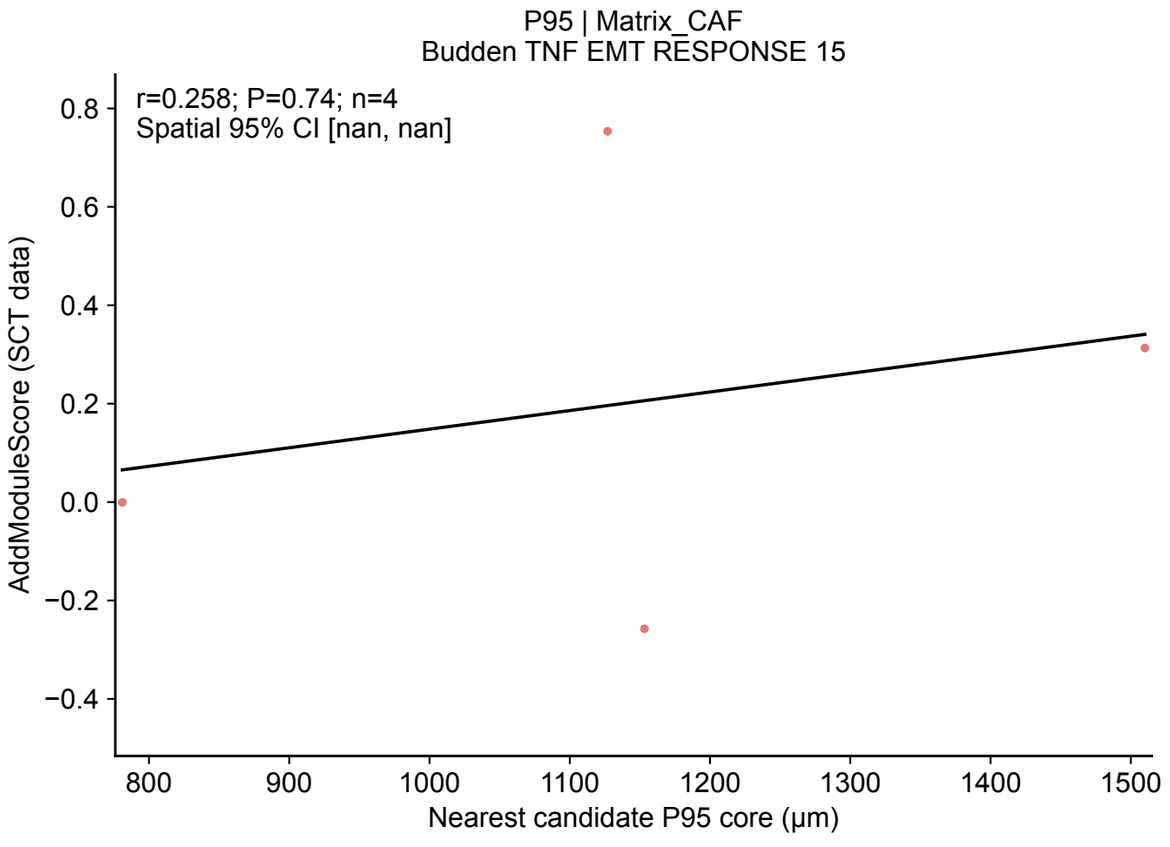
P95 minus nonP95 r | common support | grey = not estimable

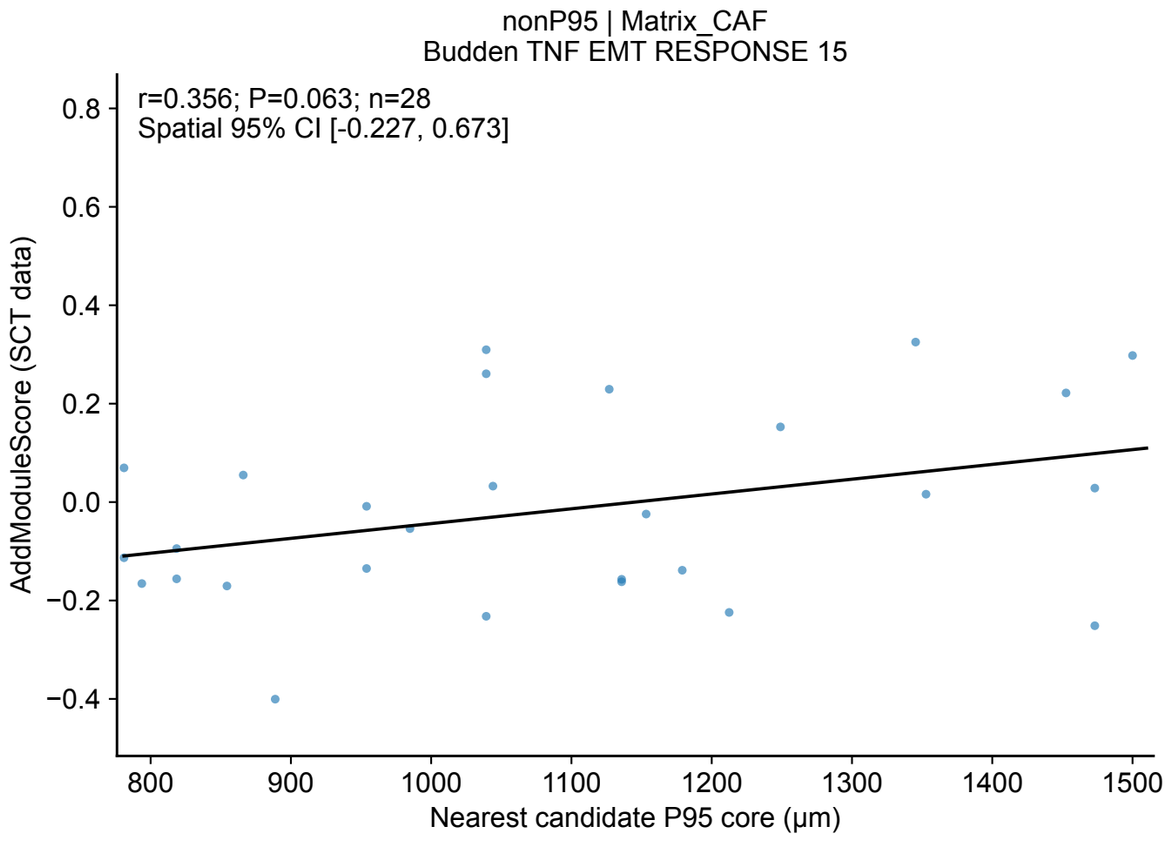


GSE238145_B18 | exploratory scatter display selected by coverage and |delta r|, not P







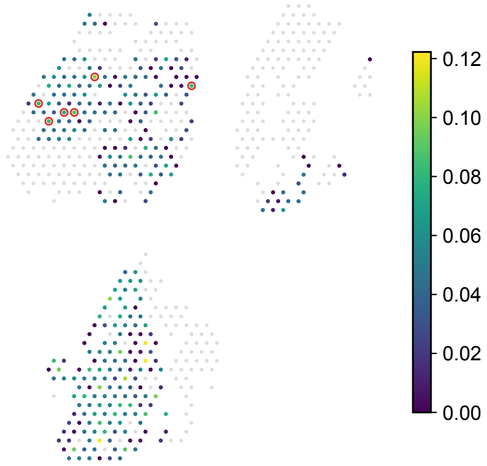


Treg / Budden S1P RESPONSE 18: delta r=-0.280 [nan, nan]; common 100–700 µm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=-0.098 [nan, nan]; common 781–1510 µm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

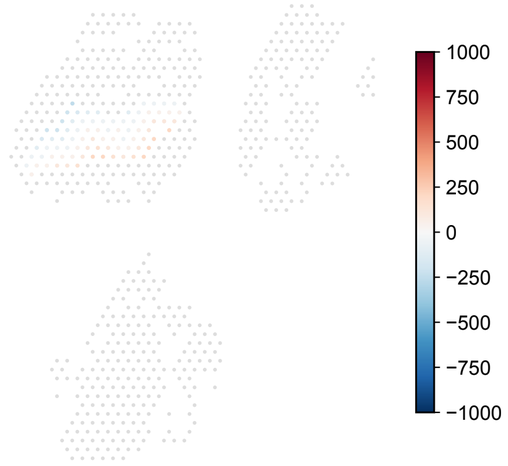
P95 nonP95 distance 0

GSE238145_B5 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

MP10 AUCell and frozen P95



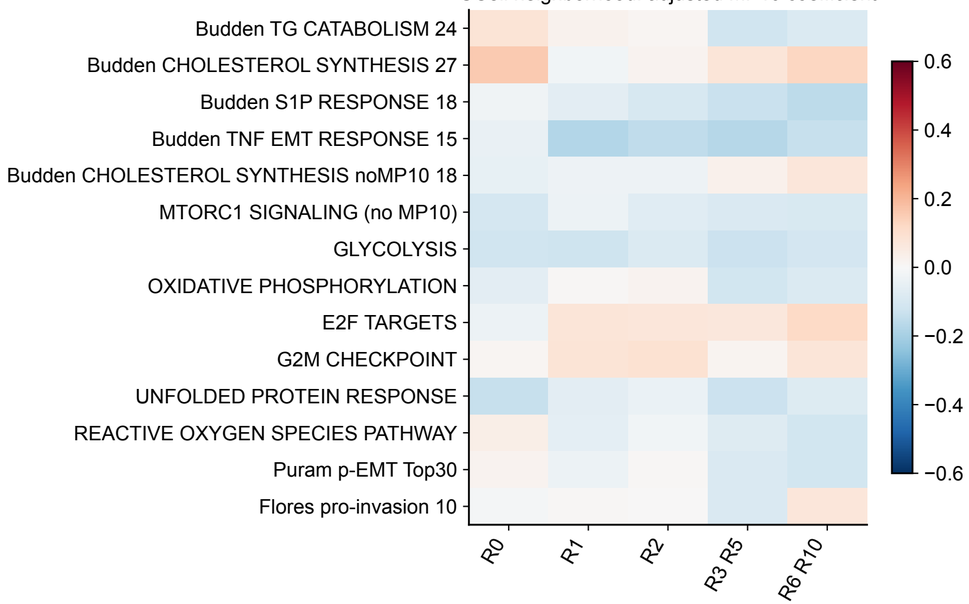
Strict v3.4 signed distance (µm)



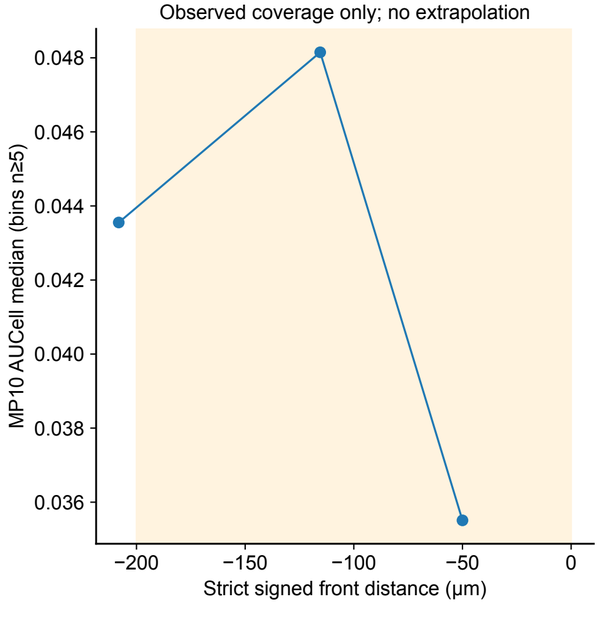
Formal R12 0–8; grey = unlabelled



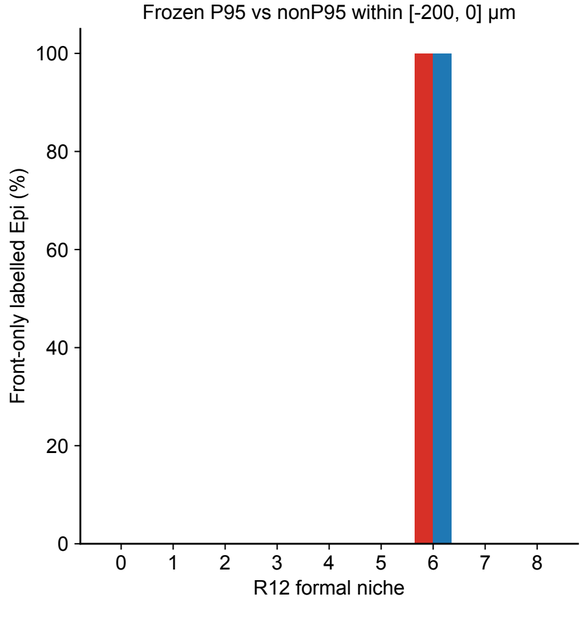
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

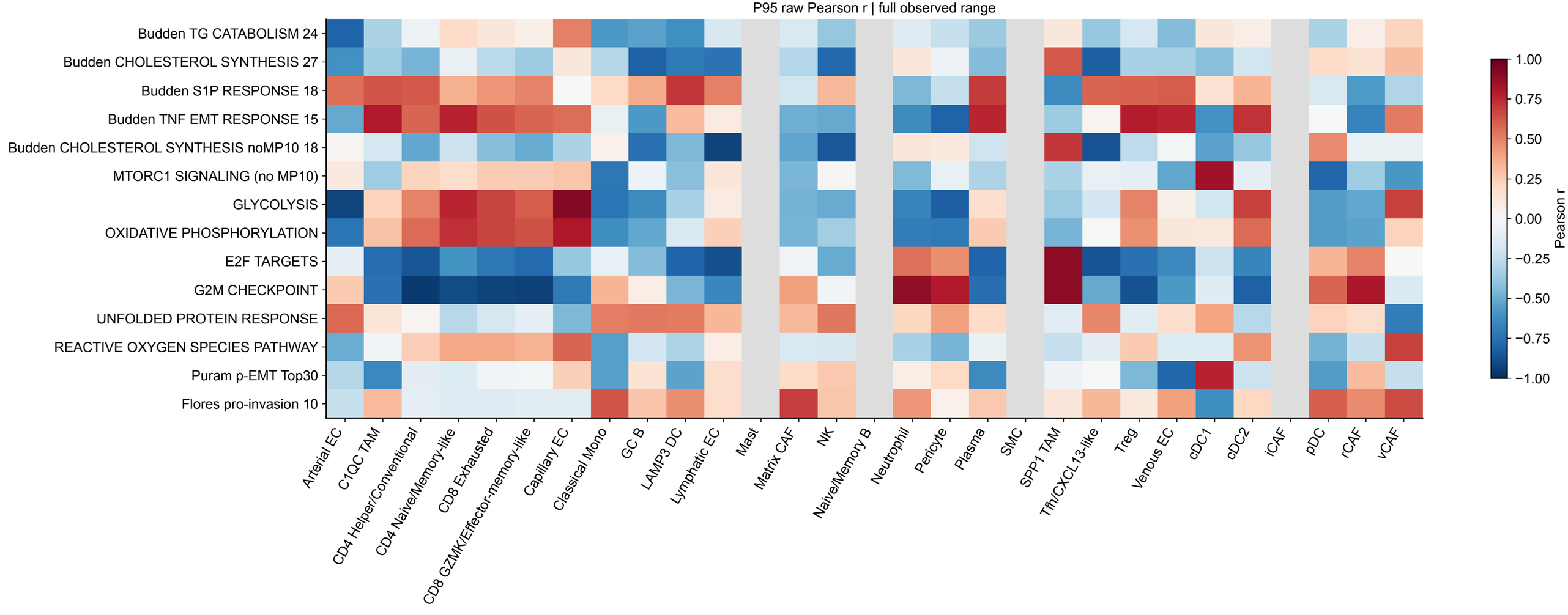


Tissue 639 | Epi 305 | internal 114 | P95 6
Exact front in Epi 40 | R12 labelled 220

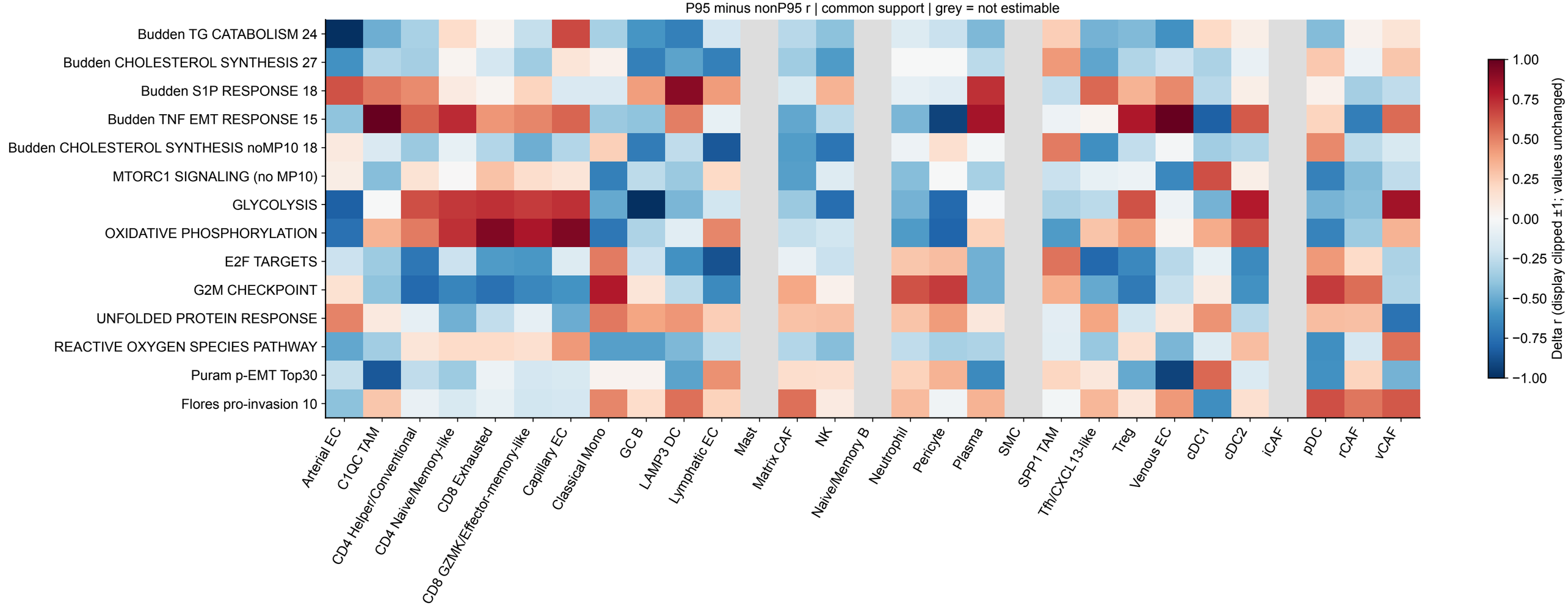


GSE238145_B5 | all 29 candidates; fixed core programs

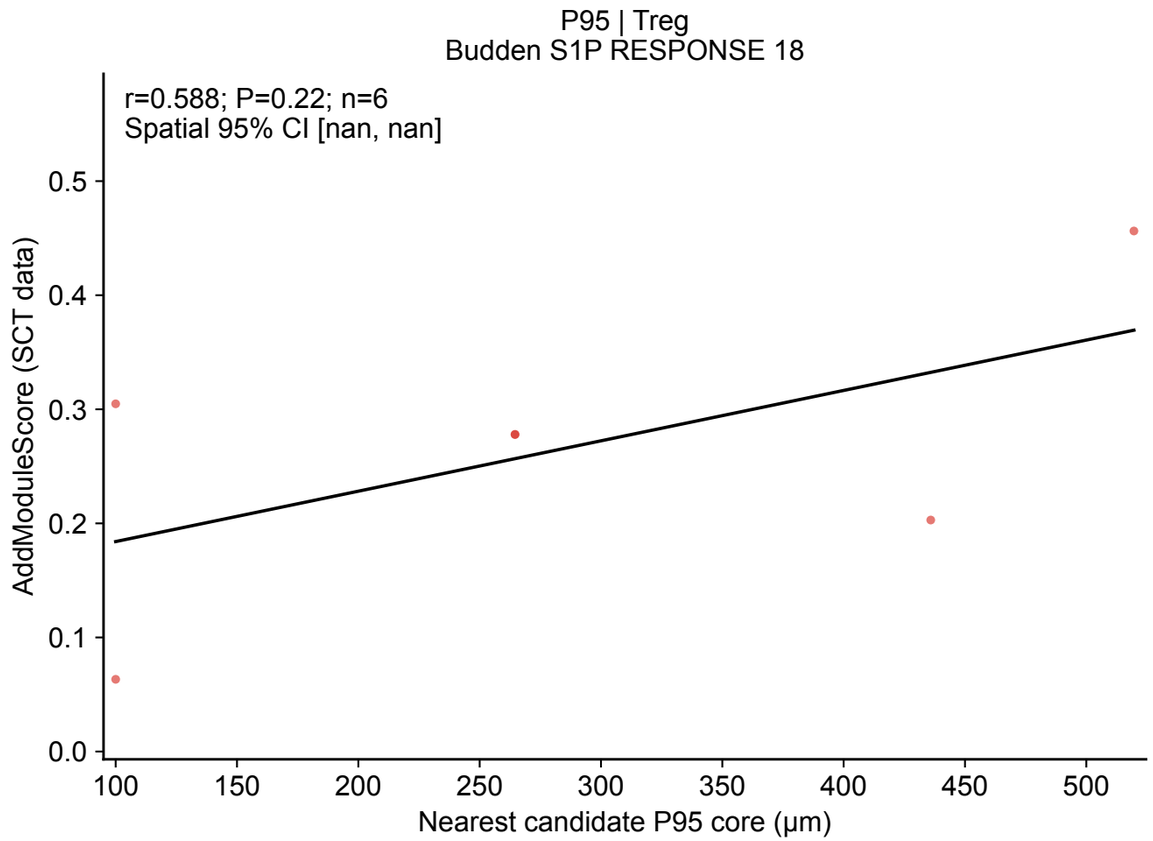
P95 raw Pearson r | full observed range

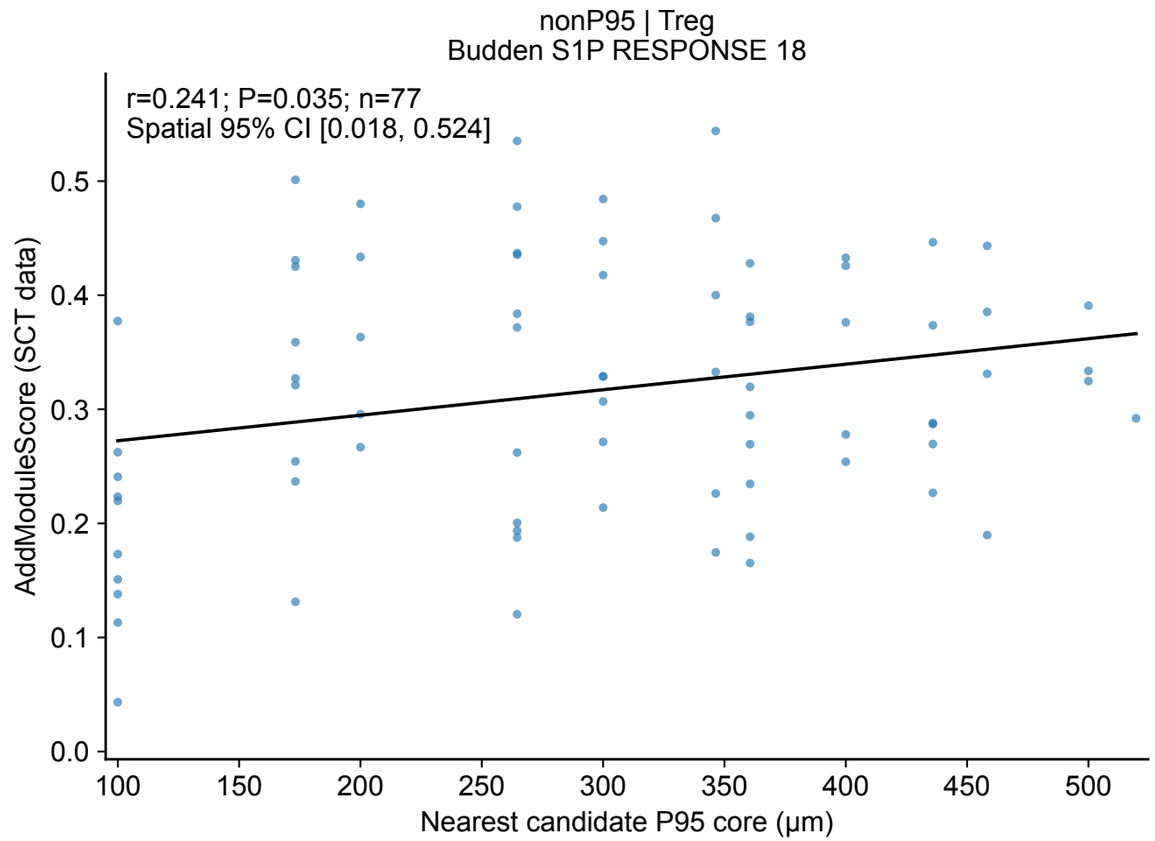


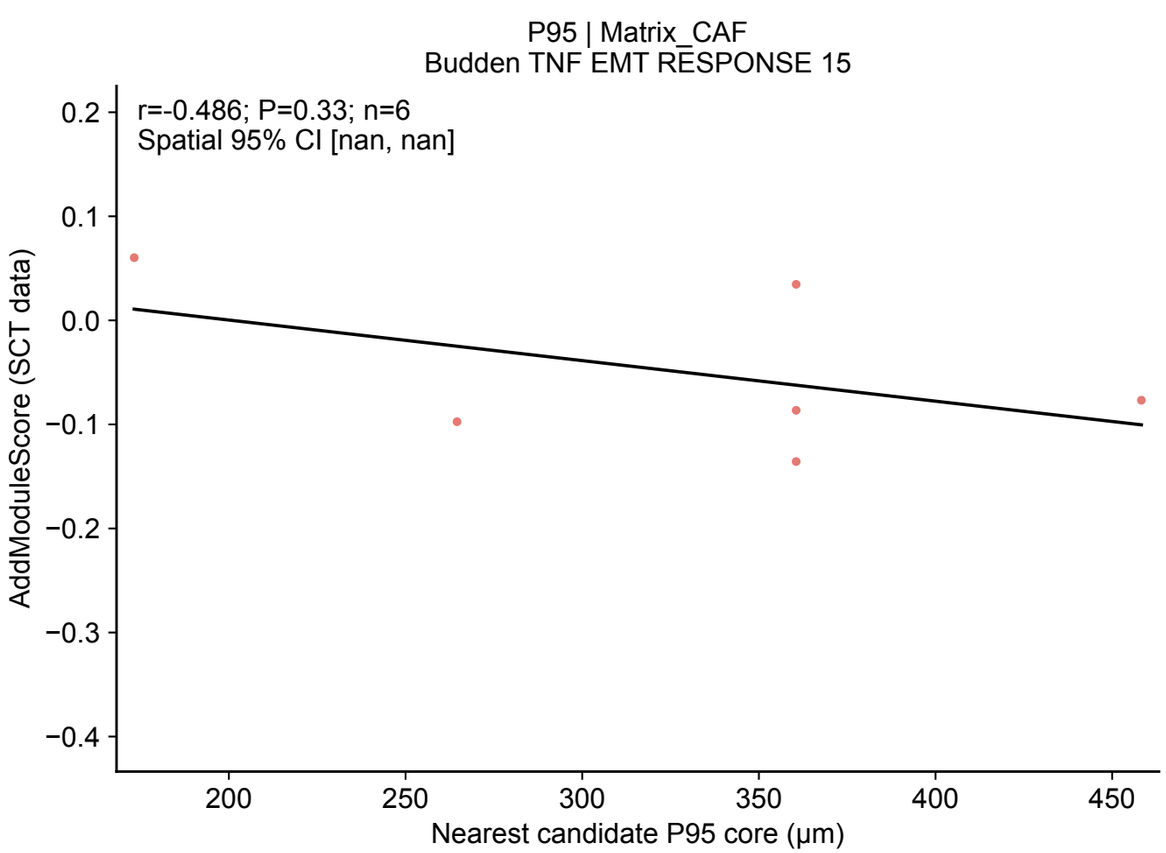
P95 minus nonP95 r | common support | grey = not estimable

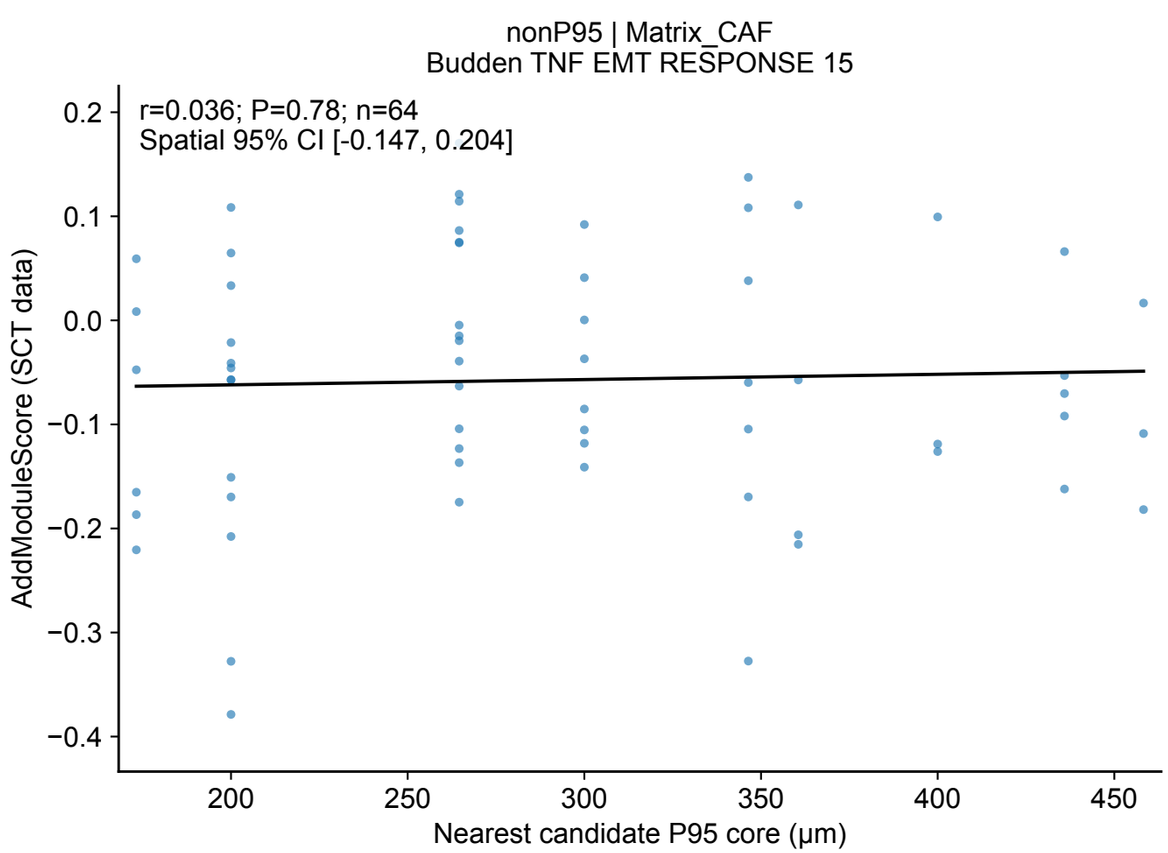


GSE238145_B5 | exploratory scatter display selected by coverage and |delta r|, not P







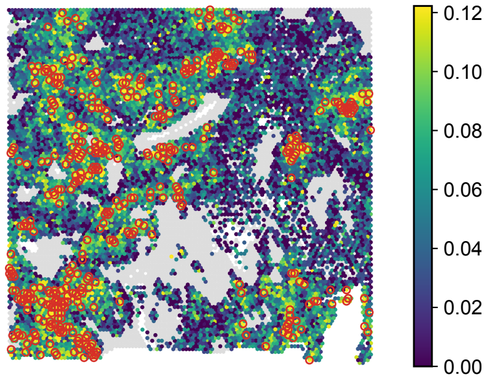


Treg / Budden S1P RESPONSE 18: delta r=0.347 [nan, nan]; common 100–520 µm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=-0.522 [nan, nan]; common 173–458 µm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

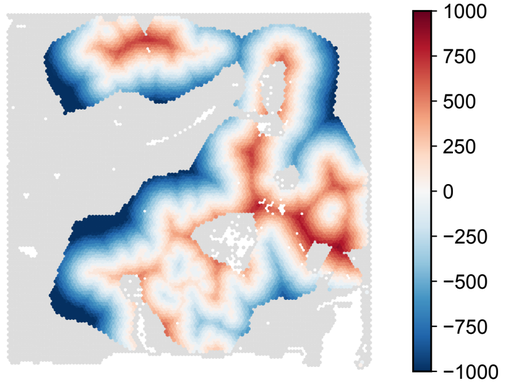
P95 nonP95 distance 0

GSE238145_B9 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

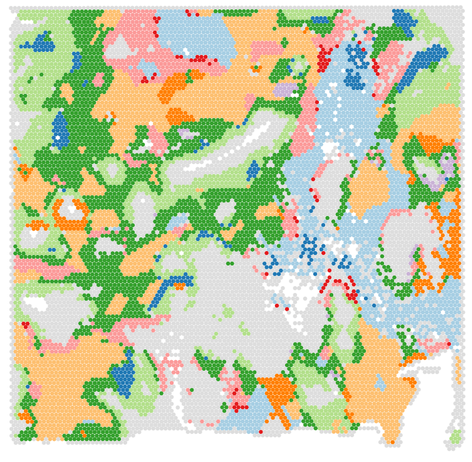
MP10 AUCell and frozen P95



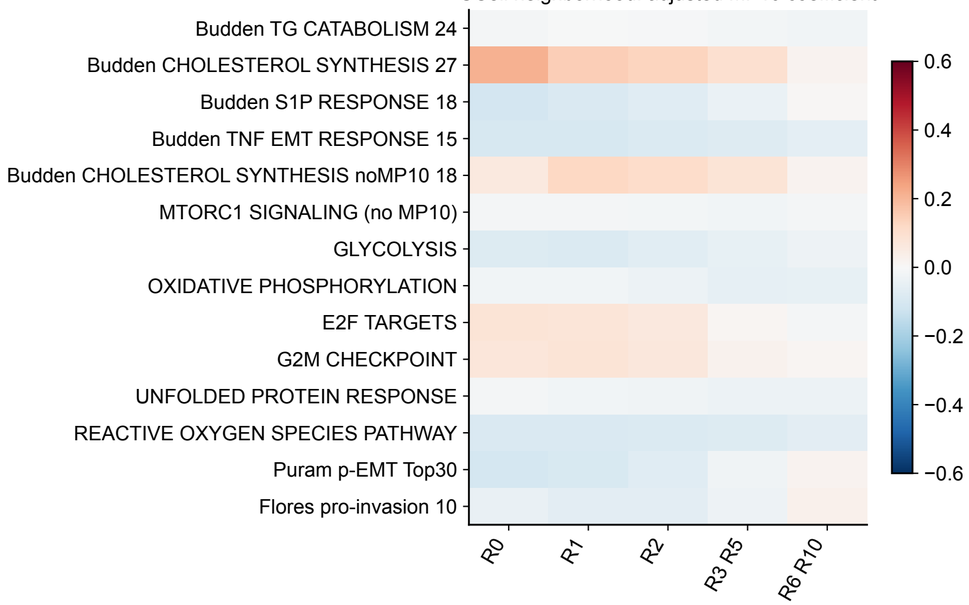
Strict v3.4 signed distance (µm)



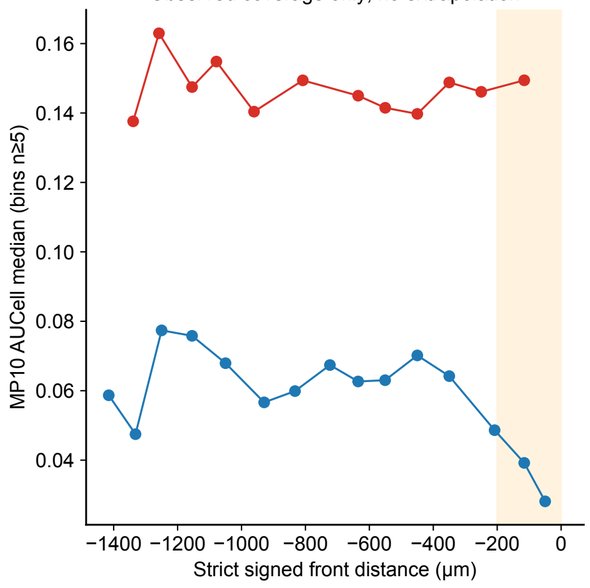
Formal R12 0–8; grey = unlabelled



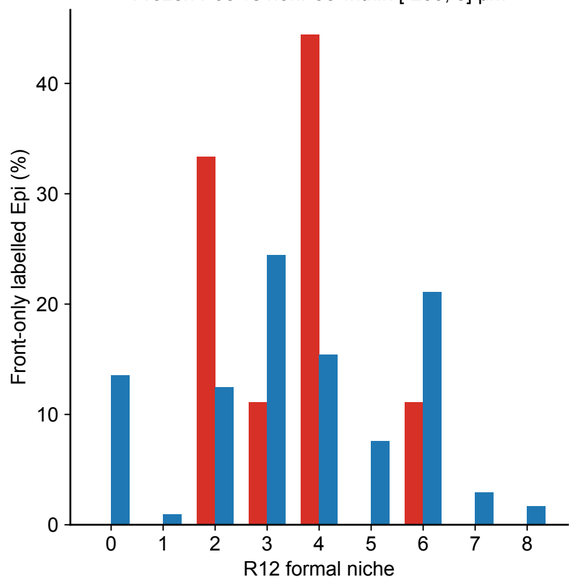
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

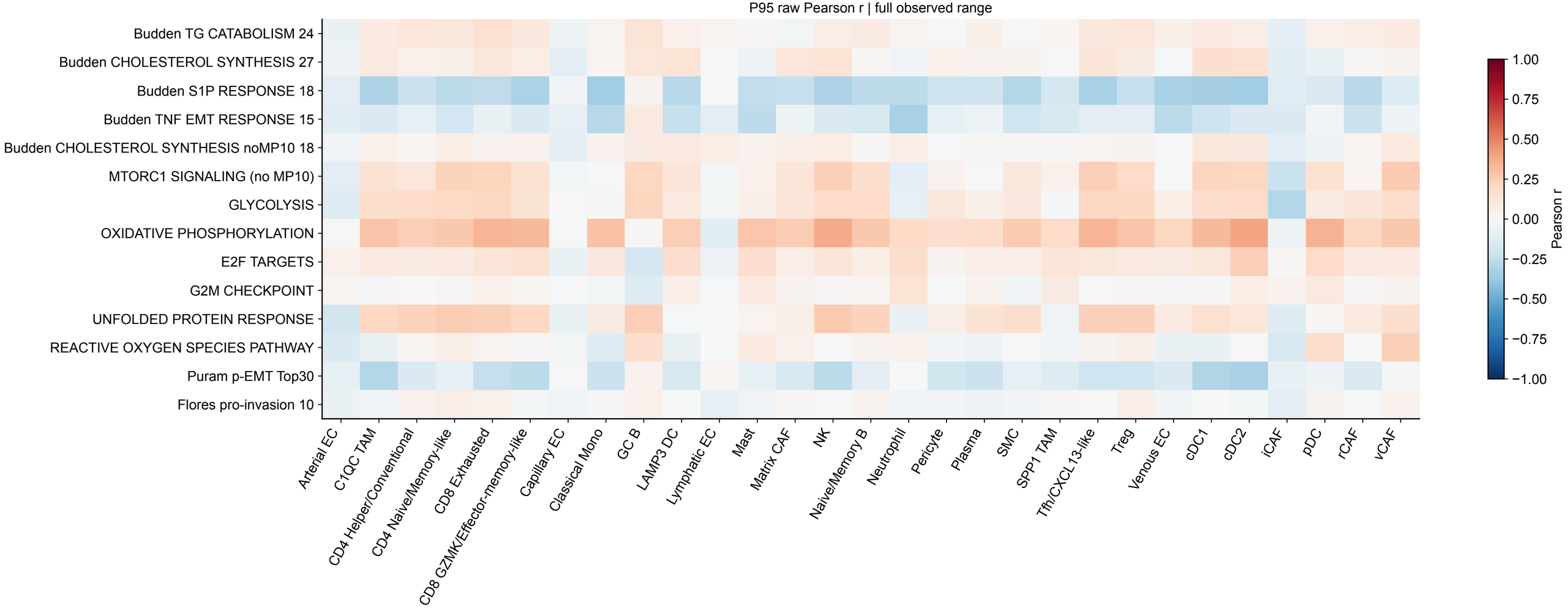


Tissue 13287 | Epi 10550 | internal 8258 | P95 414
Exact front in Epi 5435 | R12 labelled 9753

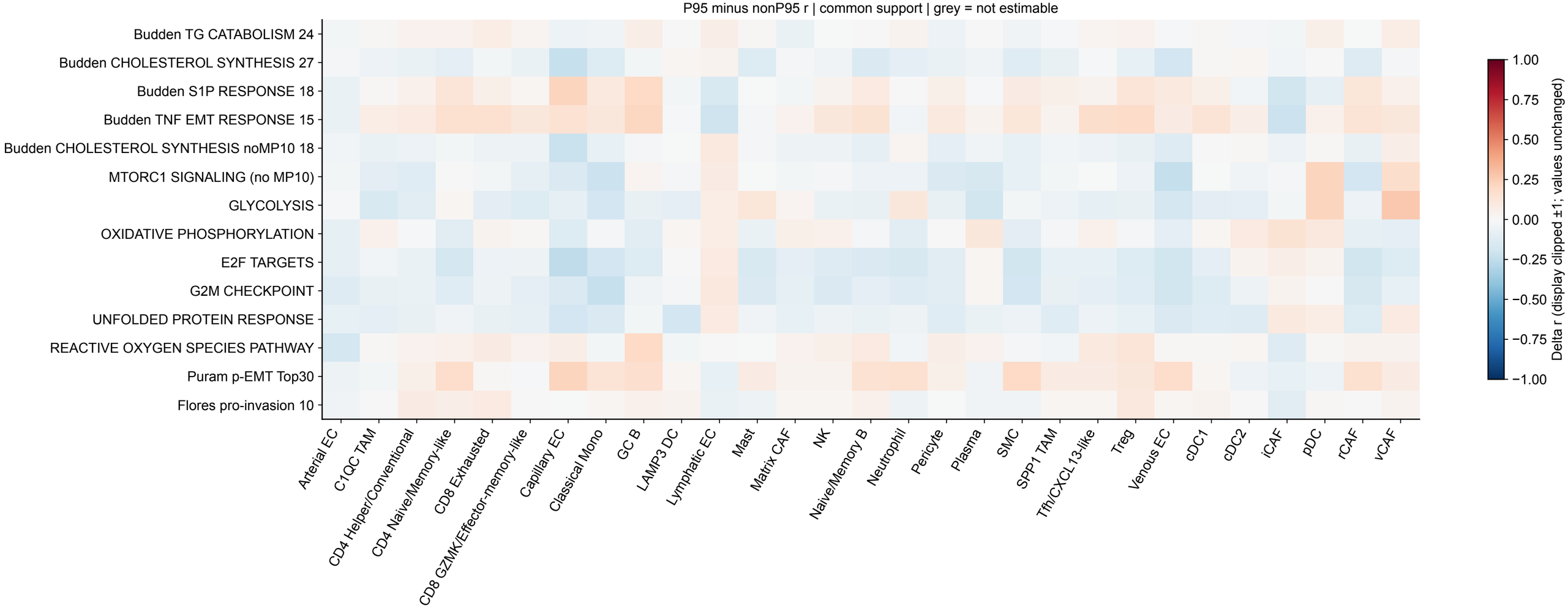
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

GSE238145_B9 | all 29 candidates; fixed core programs

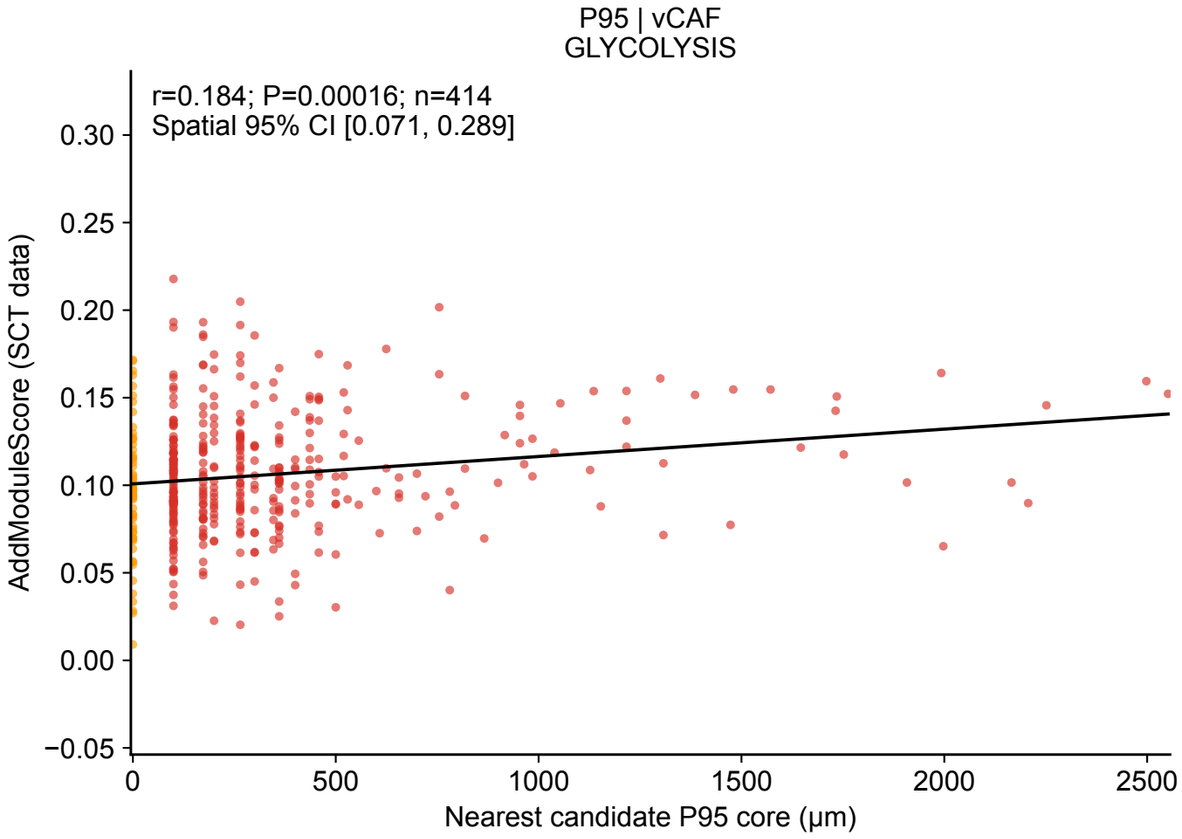
P95 raw Pearson r | full observed range

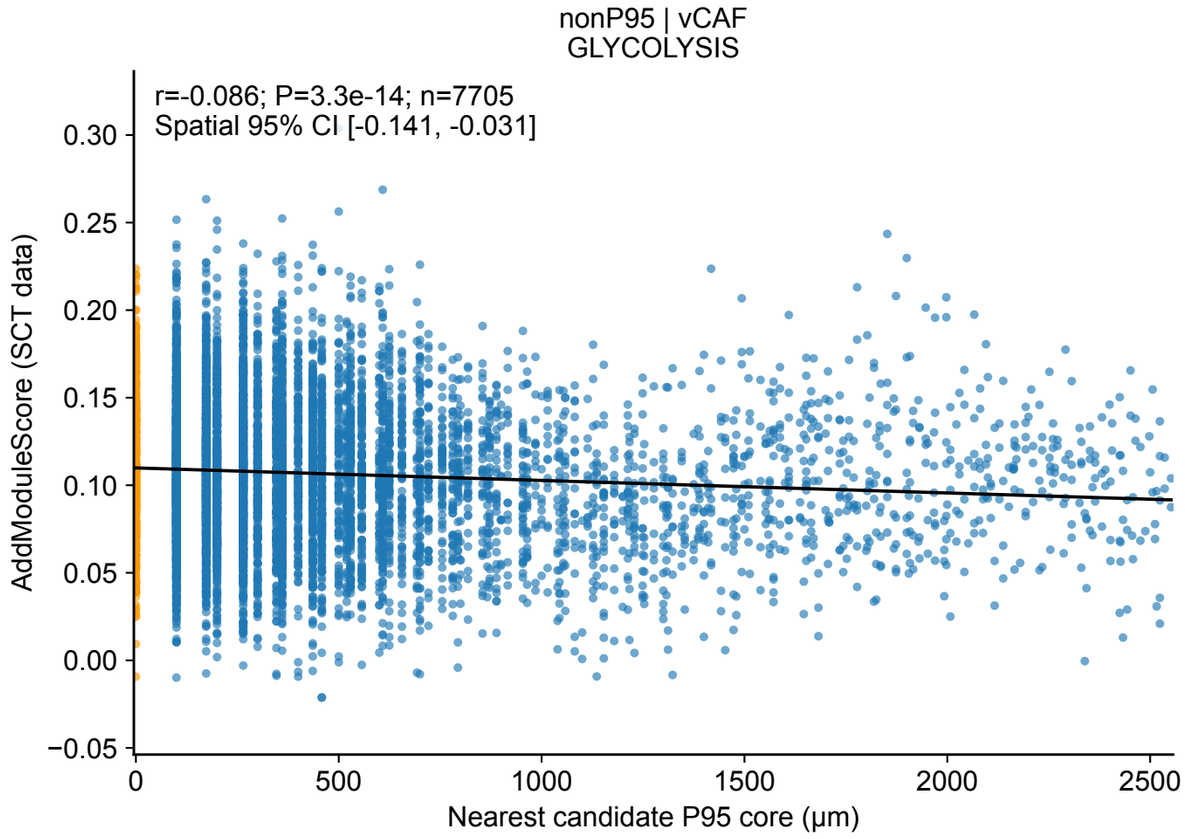


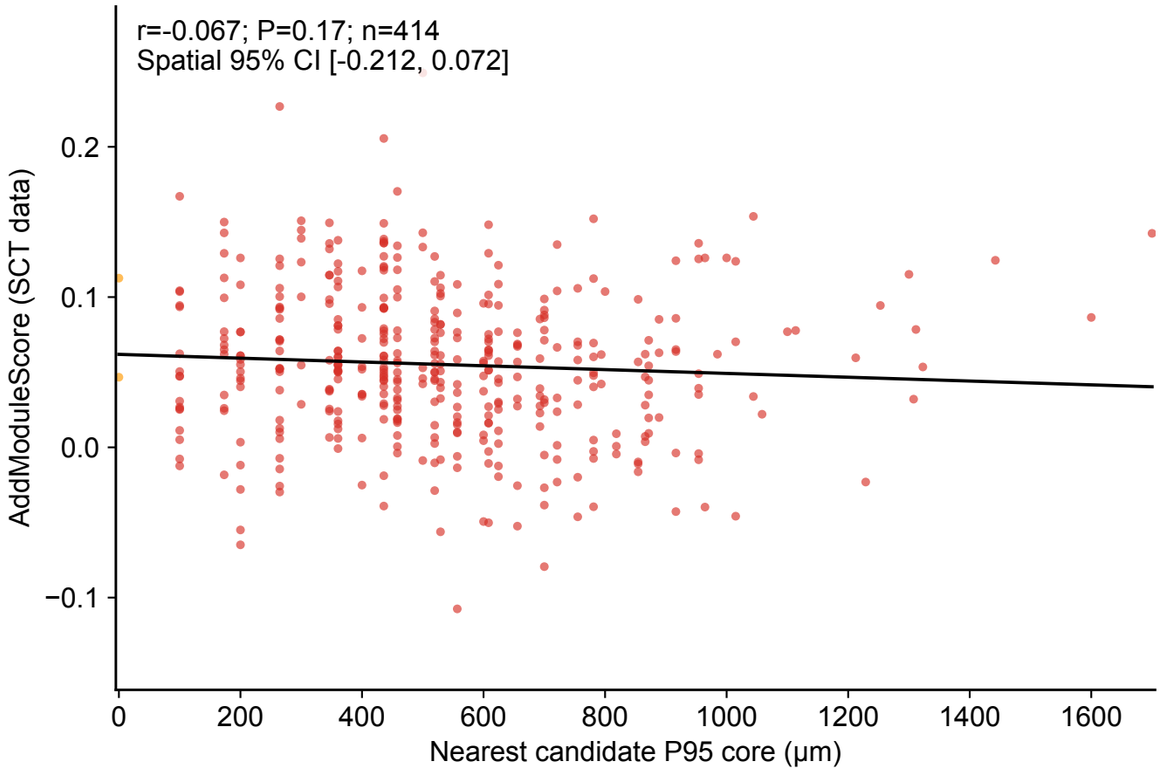
P95 minus nonP95 r | common support | grey = not estimable

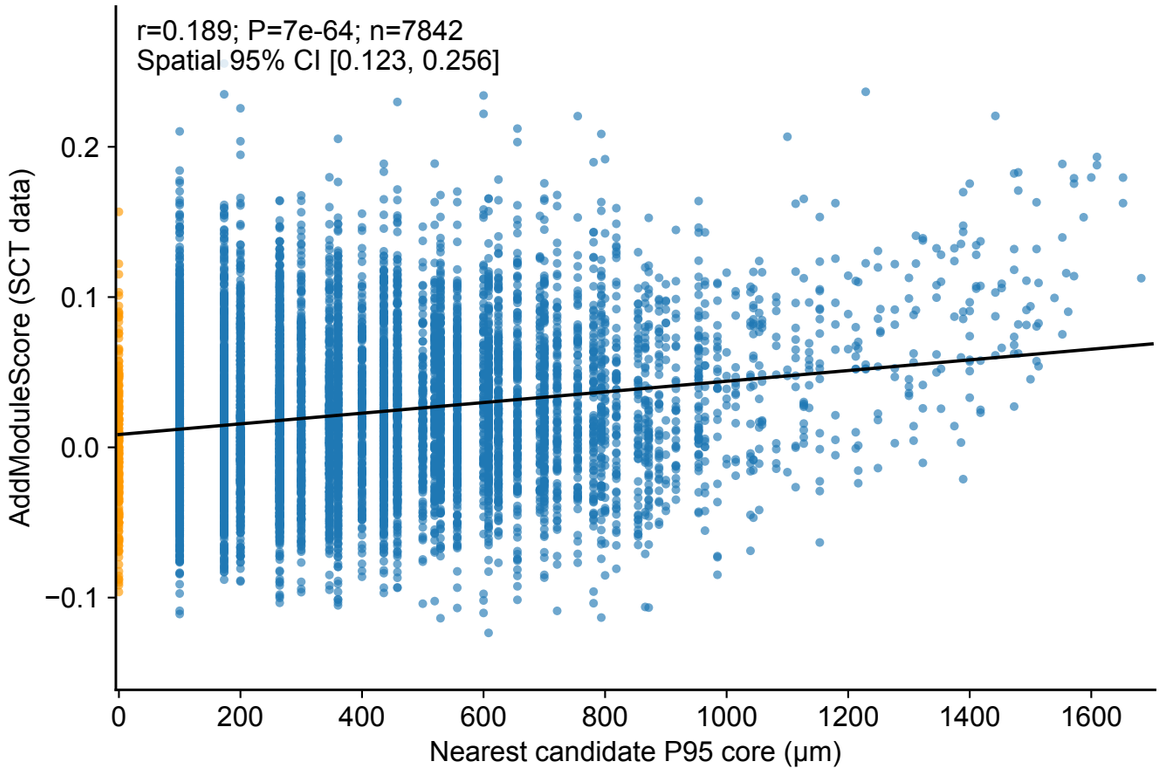


GSE238145_B9 | exploratory scatter display selected by coverage and |delta r|, not P



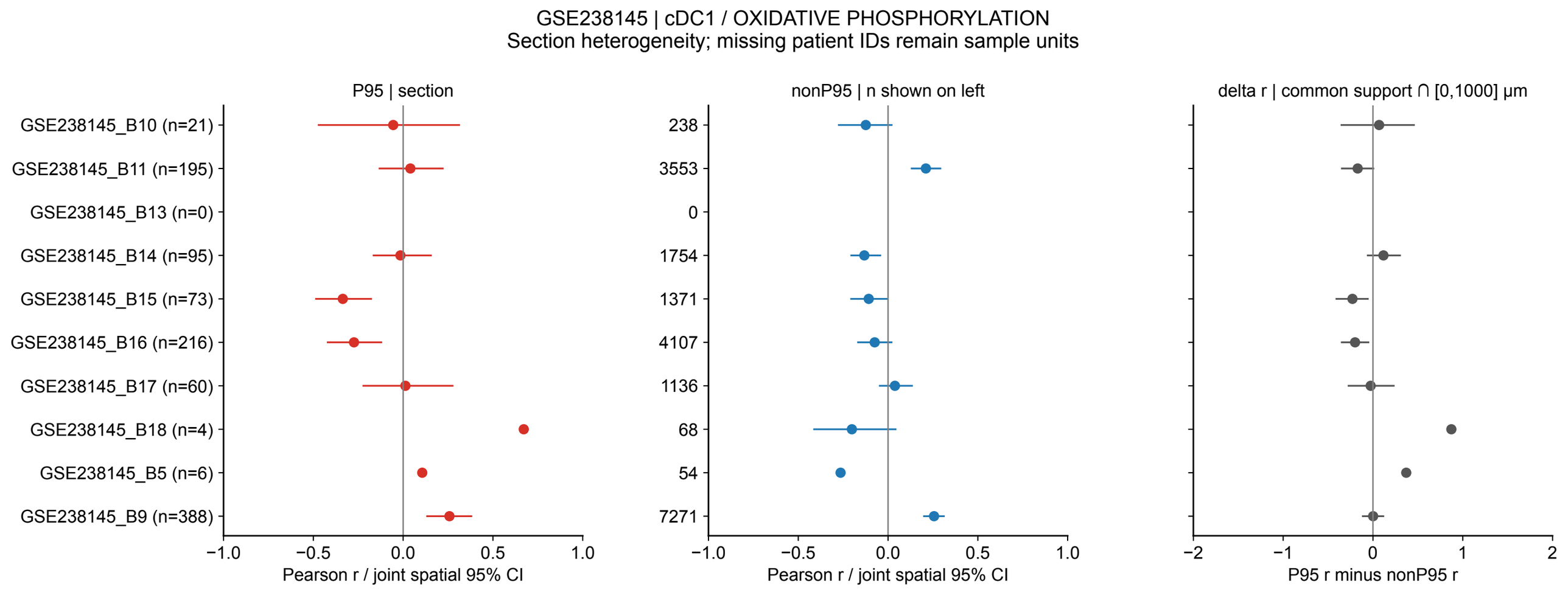


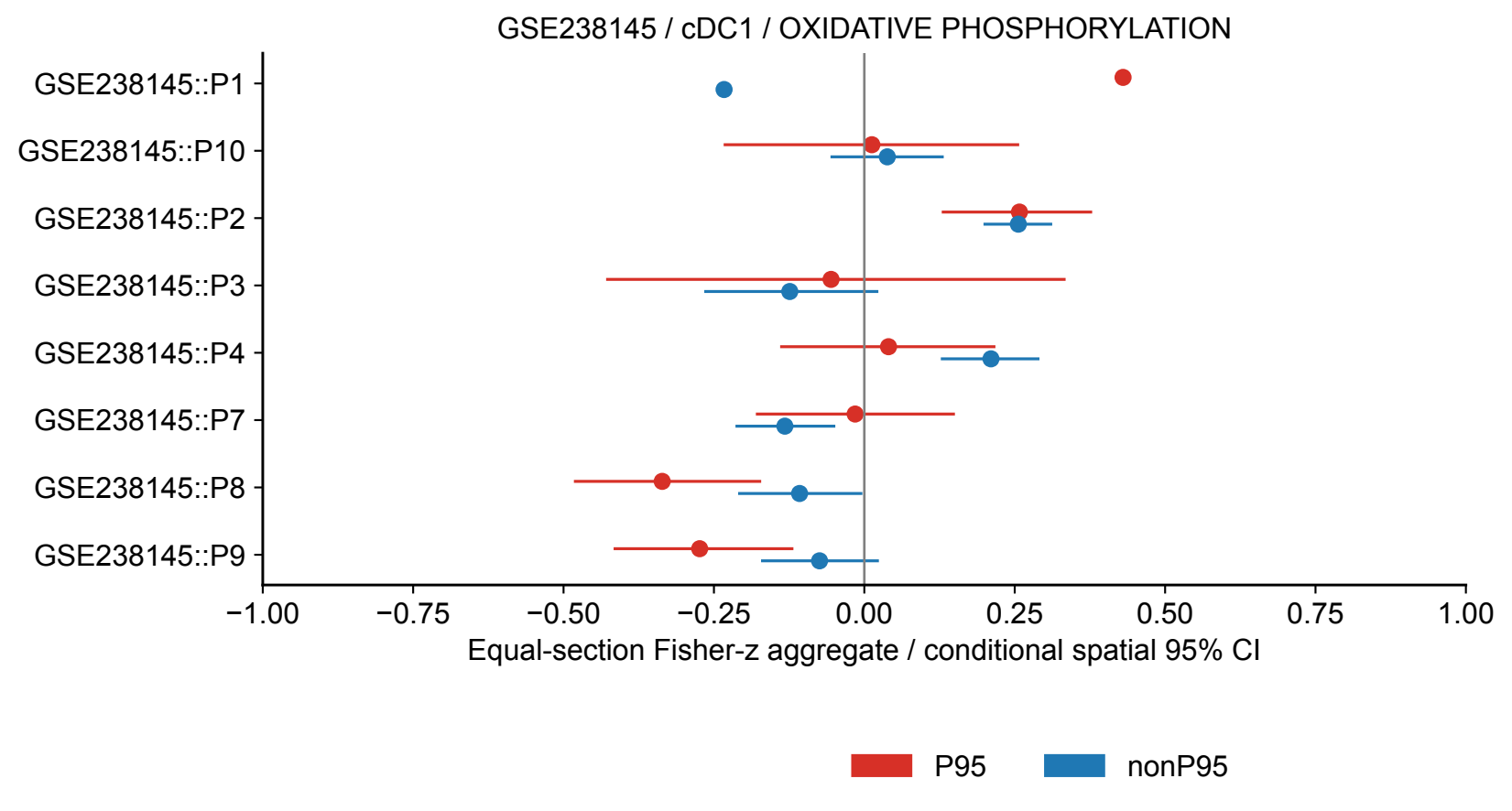


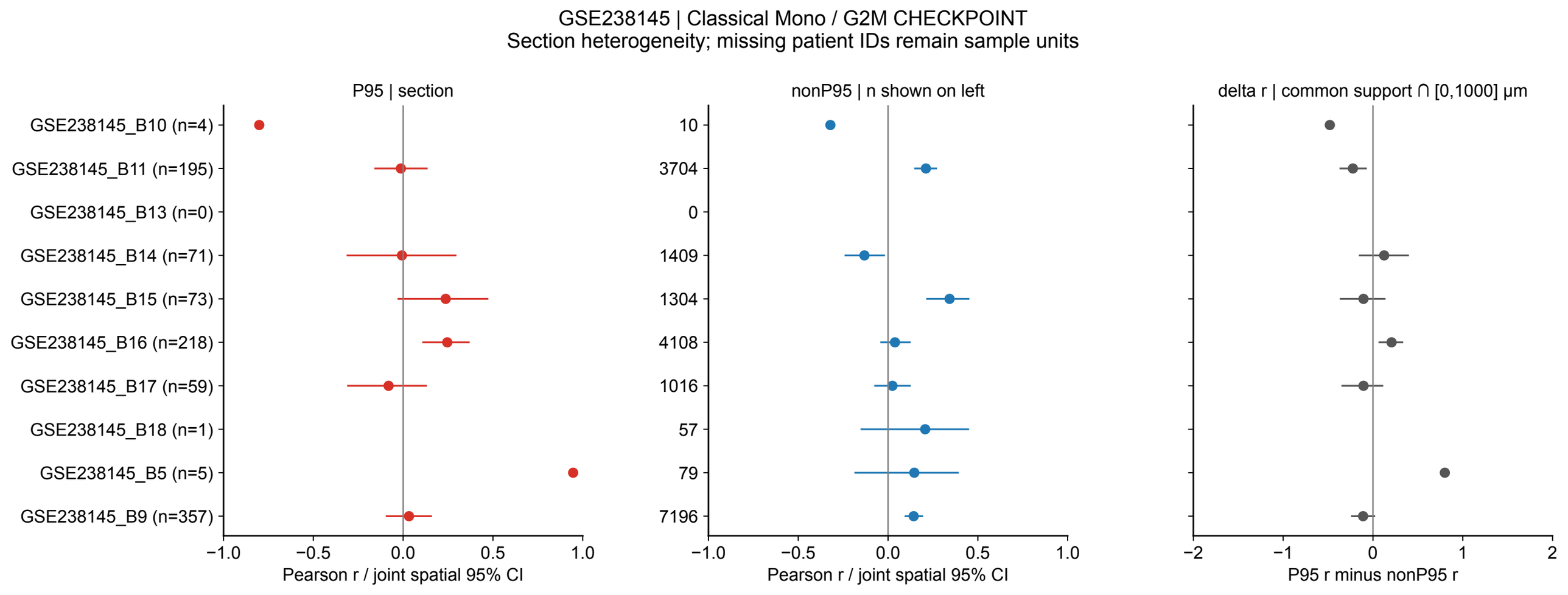


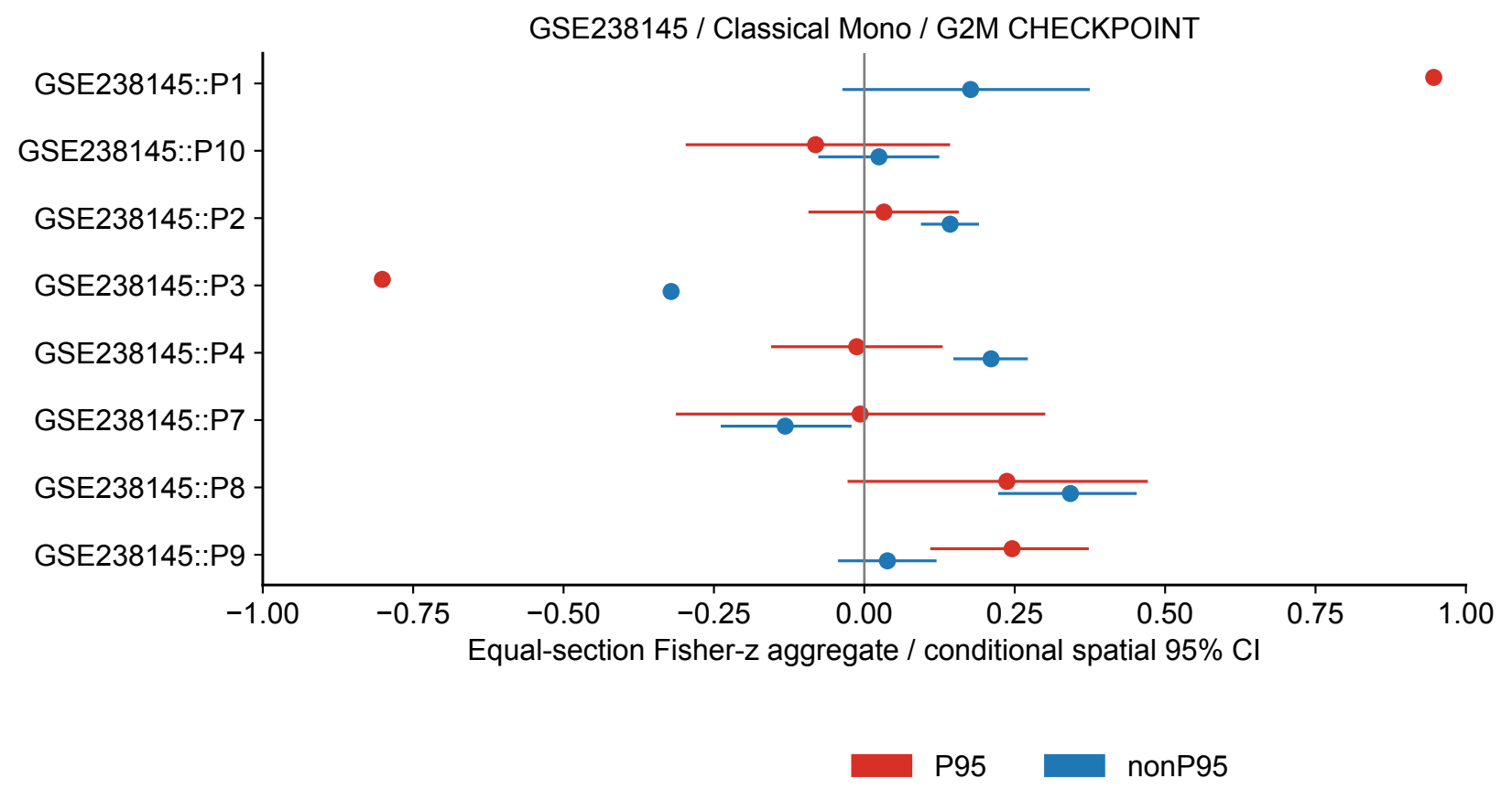
vCAF / GLYCOLYSIS: delta r=0.271 [0.161, 0.377]; common 0–2551 µm; spatial P=1.7e-06
Capillary EC / E2F TARGETS: delta r=-0.256 [-0.384, -0.124]; common 0–1700 µm; spatial P=0.00011
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

P95 nonP95 distance 0


GSE238145 | cDC1 / OXIDATIVE PHOSPHORYLATION
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
GSE238145_B10 (n=21)
GSE238145_B11 (n=195)
GSE238145_B13 (n=0)
GSE238145_B14 (n=95)
GSE238145_B15 (n=73)
GSE238145_B16 (n=216)
GSE238145_B17 (n=60)
GSE238145_B18 (n=4)
GSE238145_B5 (n=6)
GSE238145_B9 (n=388)
238
3553
0
1754
1371
4107
1136
68
54
7271
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


GSE238145 / cDC1 / OXIDATIVE PHOSPHORYLATION
GSE238145::P1
GSE238145::P10
GSE238145::P2
GSE238145::P3
GSE238145::P4
GSE238145::P7
GSE238145::P8
GSE238145::P9
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95


GSE238145 | Classical Mono / G2M CHECKPOINT
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
GSE238145_B10 (n=4)
GSE238145_B11 (n=195)
GSE238145_B13 (n=0)
GSE238145_B14 (n=71)
GSE238145_B15 (n=73)
GSE238145_B16 (n=218)
GSE238145_B17 (n=59)
GSE238145_B18 (n=1)
GSE238145_B5 (n=5)
GSE238145_B9 (n=357)
10
3704
0
1409
1304
4108
1016
57
79
7196
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


GSE238145 / Classical Mono / G2M CHECKPOINT
GSE238145::P1
GSE238145::P10
GSE238145::P2
GSE238145::P3
GSE238145::P4
GSE238145::P7
GSE238145::P8
GSE238145::P9
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95